AGR 102. Introductory Agro-meteorology & Climate Change (1+1)

Theory:

Unit - I: Climate and weather

Meteorology - Agricultural Meteorology - Importance and scope in crop production - Co-ordinates of India and Tamil Nadu - Atmosphere - Composition and vertical layers of atmosphere (stratification) - Climate - Weather - Factors affecting climate and weather - Climatic types - Different agricultural seasons of India and Tamil Nadu and climatic characteristics of India.

Unit - II: Solar radiation, RH and Wind

Solar radiation - Light intensity, quality, direction and duration - Air and Soil temperature - Diurnal variation - importance in crop production. Heat unit and its importance in agriculture. Relative Humidity and its importance - vapor pressure deficit and its importance - Wind and its effect on crops.

Unit - III: Atmospheric pressure and precipitation

Atmospheric pressure - cyclones, anticyclones, tornado, hurricane and storms - Wind systems of the world -. Clouds - types and their classification. Precipitation - forms - monsoon - Seasons of India- rainfall variability drought, flood and their effect - Cloud seeding - Evaporation - transpiration - Evapotranspiration - PET.

Unit - IV: Agroclimatic zones and remote sensing

Agro climatic Zones of India and Tamil Nadu - Agro climatic normals - Weather forecasting - synoptic chart - crop weather calendar - Remote sensing - Impact of climate and weather on crop production and pest and diseases.

Unit - V: Climate change

Climate change- climate variability - definition and causes of climate change - Impact of climate change on Agriculture.

Practical:

Observatory - Site selection and layout. Acquiring skill in use of Pyranometers - Sunshine recorder - Maximum, Minimum, Grass minimum and Soil thermometers - Thermograph, Dry and wet bulb thermometers - Hygrograph - Psychrometers - Fortein's barometer - Barograph - Altimeter; Wind vane, Anemometer - Raingauge - Ordinary and self-recording - Dew guage; Automatic weather station - Evaporimeters - Lysimeters, Automatic weather station - Preparation of synoptic charts and crop weather calendars. Rainfall probability analysis. Mapping of Agroclimatic Zones.

Theory

Lecture Schedule:

1. Meteorology - Agricultural Meteorology - Definition, their importance and scope in crop production.
2. Coordinates of India and Tamil Nadu. Atmosphere - Composition of atmosphere - Vertical layers of atmosphere based on temperature difference / lapse rate.
3. Climate and weather - Factors affecting climate and weather. Macroclimate - Meso climate - Microclimate - Definition and their importance - Different climates of India and Tamil Nadu and their characterization.
4. Solar radiation - Radiation balance - Wave length characteristics and their effect on crop production - Light - effect of intensity, quality, direction and duration on crop production.

5. Air temperature - Factors affecting temperature. Diurnal and seasonal variation in air temperature - Isotherm, Heat unit and its use - Heat and cold injuries.
6. Role of temperature in crop production. Soil temperature - Importance in crop production. Factors affecting soil temperature, diurnal and seasonal variation in soil temperature.
7. Humidity - Types - Dew point temperature - Vapour pressure deficit - Diurnal variation in Relative humidity and its effect on crop production - Wind and its role on crop production.
8. Atmospheric pressure, diurnal and seasonal variation - Isobar – cyclone, hurricane, tornado and storms.
9. **Mid Semester Examination.**
10. Wind systems of the world - wind speed in different seasons -. Clouds and their classification - Concepts of cloud seeding - present status.
11. Precipitation - Forms of precipitation - Isohyte - Monsoon - Different monsoons of India - Rainfall variability - Drought and flood - Impact on crop production.
12. Evaporation - Transpiration, evapotranspiration - Potential evapotranspiration - Definition and their importance in agricultural production. Agroclimatic zones of Tamil Nadu - Agroclimatic normals for field crops.
13. Weather forecasting - Types, importance, Agro Advisory Services - Synoptic chart - Crop weather calendar.
14. Remote sensing and its application in agriculture.
15. Effect of weather and climate on crop production, soil fertility and incidence of pest and diseases.
16. Climate change, climate variability - definition and causes of climate change.
17. Impact of climate change on Agriculture.

Practical Schedule:

1. Site selection and layout for Agromet Observatory - Calculation of local time - Time of observation of different weather elements - Reviewing agromet registers.
2. Measurements of solar radiation (pyranometers), sunshine hours (sunshine recorder) - working out weekly and monthly mean for graphical representation.
3. Measurement of air and soil temperature and grass minimum thermometers and thermographs - drawing isolines.
4. Humidity measurements - use of wet and dry bulb thermometers - Psychrometers - Hygrograph - Measurement of wind direction and wind speed and conversion (KMPH, KNOT, and M/Sec.) - Beaufort’s scale.
5. Measurement of atmospheric pressure - barograph - Fortein-s barometer - Isobars based on past data for different seasons.
6. Measurement of rainfall - Ordinary and self-recording rain gauges - Measurement of Dew - dew gauge- study of Automatic weather station.
7. Measurement of Evaporation - Open pan evaporimeter- application of evaporation data- Measurement of Evapotranspiration- Lysimeter.
8. Heat Unit concept- GDD, HTU, PTU for fixing time of sowing.
9. Probability analysis of rainfall for crop planning.
10. Drawing Synoptic charts for understanding weather.
11. Preparation of crop weather calendars and forecast based agro advisories.
12. Preparation pest weather calendar and pest forewarning.
13. Estimation of length of growing periods using weekly rainfall data.
14. Water balance studies.

15. Identification of efficient cropping zone- RYI, RSI.
16. Mapping of agro climatic Zones of India and Tamil Nadu and its characterization.
17. **Practical Examination**.

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LECT. 1. METEOROLOGY – AGRICULTURAL METEOROLOGY – DEFINITION, THEIR IMPORTANCE TO CROP PRODUCTION AND FUTURE SCOPE.

Meteorology

Greek word "Meteoro" means 'above the earth's surface' (atmosphere)

"logy" means "indicating science"

- Branch of science dealing with that of atmosphere is known as meteorology
- Lower atmosphere extending up to 20km from earth's surface where frequent physical process takes place
- Meteorology is a combination of both physics and geography

Meteorology is a combination of both physics and geography. This science utilizes the principles of Physics to study the behaviour of air. It is concerned with the analysis of individual weather elements for a shorter period over a smaller area. In other words, the physical state of the atmosphere at a given place and time is referred to as "weather". The study of weather is called 'meteorology'. It is often quoted as the "physics of atmosphere".

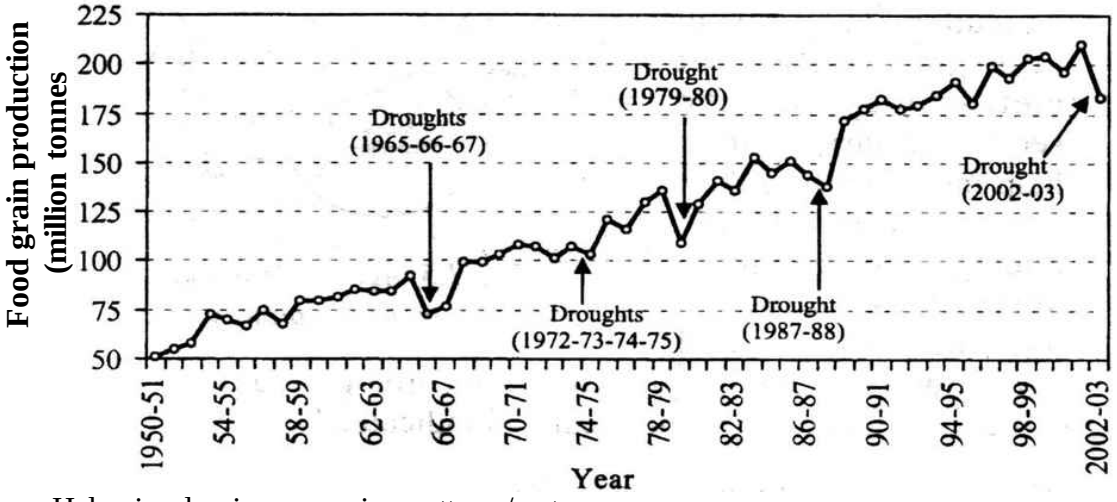
Weather: Physical state of the atmosphere at a given place and given time
Climate: Long term regime of atmospheric variables of a given place or area

Agricultural meteorology:

1. A branch of applied meteorology which investigates the physical conditions of the environment of growing plants or animal organisms
2. An applied science which deals with the relationship between weather/climatic conditions and agricultural production.
3. An applied science which deals with the relationship between weather/climatic conditions and agricultural production.
4. A science concerned with the application of meteorology to the measurement and analysis of the physical environment in agricultural systems. The word 'Agro meteorology' is the abbreviated form of agricultural meteorology.
5. To study the interaction between meteorological and hydrological factors on the one hand and agriculture in the widest sense, including horticulture, animal husbandry and forestry on the other (WMO)

Meteorology	Agricultural meteorology
Branch of atmospheric physics	Branch of applied meteorology or a branch of agriculture as it deals with agriculture
It is a weather science	It is a product of agriculture and meteorology
It is a physical science	It is a biophysical science
It aims at weather forecasting	It aims at improving quantity and quality of crop production through meteorological skills
Weather service is the concern	Agro advisory service to the farmers is the concern based on weather forecast
It is a linking science to the society	It is a linking science to the farming community

IMPORTANCE TO CROP PRODUCTION



1. Helps in planning cropping patterns/systems.
2. Selection of sowing dates for optimum crop yields.
3. Cost effective ploughing, harrowing, weeding etc.
4. Reducing losses of applied chemicals and fertilizers.
5. Judicious irrigation to crops.
6. Efficient harvesting of all crops.
7. Reducing or eliminating outbreak of pests and diseases.
8. Efficient management of soils which are formed out of weather action.
9. Managing weather abnormalities like cyclones, heavy rainfall, floods, drought etc. This can be achieved by
   (a) Protection: When rain is forecast avoid irrigation. But, when frost is forecast apply irrigation.
   (b) Avoidance: Avoid fertilizer and chemical sprays when rain is forecast
   (c) Mitigation: Use shelter belts against cold and heat waves.
10. Effective environmental protection.
11. Avoiding or minimizing losses due to forest fires.

FUTURE SCOPE

1. To study climatic resources of a given area for effective crop planning.
2. To evolve weather based effective farm operations.
3. To study crop weather relationships in all important crops and forecast crop yields based on agro climatic and spectral indices using remote sensing.
4. To study the relationship between weather factors and incidence of pests and diseases of various crops.
5. To delineate climatic/agro ecological/agro climatic zones for defining agro climatic analogues so as to make effective and fast transfer of technology for improving crop yields.
6. To prepare crop weather diagrams and crop weather calendars.
7. To develop crop growth simulation models for assessing/obtaining potential yields in different agro climatic zones.
8. To monitor agricultural droughts on crop-wise for effective drought management.
9. To develop weather based agro advisories to sustain crop production utilizing various types of weather forecast and seasonal climate forecast.
10. To investigate microclimatic aspects of crop canopy in order to modify them for increased crop growth
11. To study the influence of weather on soil environment on which the crop is grown
12. To investigate the influence of weather in protected environment (eg. Glass houses) for improving their design aiming at increasing crop production.

Lect 2. **COORDINATES OF INDIA AND TAMIL NADU, ATMOSPHERE – COMPOSITION OF ATMOSPHERE - VERTICAL LAYERS OF ATMOSPHERE BASED ON TEMPERATURE DIFFERENCE / LAPSE RATE.**

Coordinates of India

Lies between	º N and	º N latitude
	ºE and	º E longitude

Coordinates of Tamil Nadu

Lies between	08º 05' N and	13º 35' N latitude
	76º 15' E and	80º 20' E longitude

Earth is elliptical in shape and has three spheres

Hydrosphere - the water portion
Lithosphere - the solid portion
Atmosphere - the gaseous portion

Atmosphere : The atmosphere is the colourless, odourless and tasteless physical mixture of gasses which surrounds earth on all sides. It is mobile, compressible and expandable. It contains huge number of solid and liquid particles called aerosol. Some gases are permanent atmospheric constituents in fixed proportions to the total gas volume. Others very from place and time to time.

The lower atmosphere where the chemical composition of gas is uniform is called homosphere. At higher levels the chemical composition of air changes considerably and known as heterosphere.

Uses of atmosphere

1. Provides oxygen which is useful for respiration in crops
2. Provides carbon-dioxide to build biomass in photosynthesis.
3. Provides nitrogen which is essential for plant growth.
4. Acts as a medium for transportation of pollen.
5. Protects crops plants on earth from harmful U.V.rays.
6. Maintains warmth to plant life and
7. Provides rain to field crops as it is a source of water vapour, cloud, etc.

Composition of atmosphere

The following all the different gases that are present in percentage by volume approximately.

Nitrogen (N_2)	= 78.08	Oxygen (O_2)	= 20.95
Argon (Ar)	=0.93	CO_2	=0.03
Neon (Ne)	= 0.0018	Helium(He)	=0.0005
Ozone(O_3)	=0.00004	Hydrogen(H_2)	=0.00006
Methane (CH_4)	=0.00017		

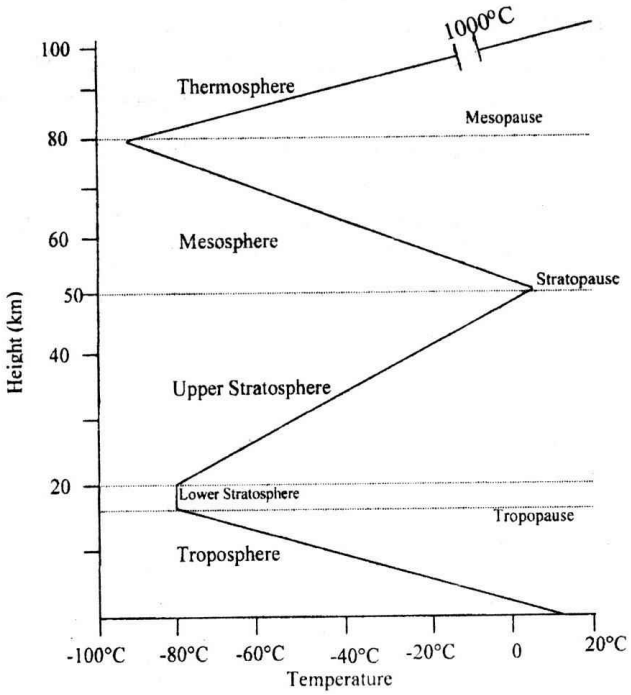
Vertical Layers of atmosphere based on temperature

On the basis of vertical temperature variation the atmosphere is divided into different spheres or layers.

A. Troposphere

1. The word "Trop" means mixing or turbulence and "sphere" means region.
2. The average height o this lower most layer of the atmosphere is about 14 km above the men sea level; at the equator it is 16 km and 7-8 km at the poles.

3. Under normal conditions the height of the troposphere changes from place to place and season to season.
4. Various types of clouds, thunderstorms, cyclone and anticyclones occur in this sphere because of the concentration of almost all the water vapour and aerosols in it. So, this layer is called as "seat of weather phenomena".
5. The wind velocities increase with height and attain the maximum at the top of this layer.
6. Another striking feature of the troposphere is that there is a decrease of temperature with increasing elevation at a mean lapse rate of about 6.5°C per km.
7. Most of the radiation received from the sun is absorbed by the earth's surface. So, the troposphere is heated from below.
8. In this layer, about 75 per cent of total gases and most of the moisture and dust particles present.
9. A the top of the troposphere there is a shallow layer separating it from the stratosphere which is known as the "Tropopasue ".
10. The tropospause layer is thin and its height changes according to the latitudes and it is a transitional zone and distinctly characterized by no major movement of air.



B). Stratosphere

1). This layer exists above the tropopause (around 20 km onwards) and extends to altitudes of about 50-55 km.This layer is called as "Seat of photochemical reactions"

2). The temperature remains practically constant at around 20 km and is characterized as isothermal because air is thin, clear, cold and dry near tropopause.

3). The temperature of this layer increases with height and also depends upon the troposphere because the troposphere is higher at the equator than at the poles.

4). In the upper parts of the stratosphere the temperatures are almost as high as those near the earth's surface, which is due to the fact that the ultra-violet radiation from the sun is absorbed by ozone in this region.

5). The air density is so much less that even limited absorption of solar radiation by the atmospheric constituents notably ozone produces a temperature increase. There is a maximum concentration of ozone between 30 and 60 km above the surface of the earth and this layer is known at the ozonosphere. A property of the ozone is that it absorbs UV rays. Had there been no layer of the ozone in the atmosphere, the ultraviolet rays might have reached the surface of the earth and no life can exist.

6). Less convection takes place in the stratosphere because it is warm at the top and cold at the bottom. There is also persistence of circulation patterns and high wind speeds. The upper boundary of the stratosphere is called the stratopause.

7). Temperature of the ozonosphere is high (warm) due to selective absorption of U.V radiation by ozone. This layer is almost free from clouds and associated weather phenomenon, making conditions most ideal for flying aeroplanes.

C). Mesosphere

1. Because of the preponderance of chemical process this sphere is called as the "chemosphere"
2. In this layer the temperature increases with height at the rate of 5°C per km.
3. According to some leading scientists the ionosphere is supposed to start at a height of 80 km above the earth's surface. The layer between 50 and 80 km is called as "Mesosphere". In this layer the temperature decreases with height. The upper boundary of this layer is called the "Mesospause".
4. Mesosphere is the coldest region in the atmosphere with temperature reaching the lowest value of nearly -95°C at the mesopause (80km)

D). Thermosphere(Ionosphere)

1) The thermosphere layer lies beyond the ozonosphere (mesosphere) at a height of about 80 km above the earth's surface and extends upto 400 km.
2) The atmosphere in the ionosphere is partly ionised enriched ion zones exist in the form of distinct ionised layers. So, this layer is called as the ionosphere.
3) Above the mesosphere the temperature increases again and is in the order of 1000°C.
4) The ionosphere reflects the radio waves because of one or multiple reflections of shortwave radio beams from the inoised shells. So, long distance radio communication is possible de to this layer.

E). Exosphere.

1) The outer most layer of the earth's atmosphere is named as the exosphere and this layer lies between 400 and 1000 km.
2) At such a greater height the density of atoms in the atmosphere is extremely low.
3) Hydrogen and Helium gases predominate in this outer most region.
4) At an altitude of about 500 to 600 km the density of the atmosphere becomes so low that collisions between the natural particles become extremely rare.

Lapse rate

The decrease in air temperature with height is known as the normal / environmental lapse rate and it is 6.5°C/km.

Adiabatic lapse rate

The rate of change of temperature in an ascending or descending air mass through adiabatic process is called as adiabatic lapse rate.

The thermodynamic transformation which occurs without exchange of heat between a system and its environment is known as adiabatic process. In adiabatic process, adiabatic cooling accompanies expansion, and adiabatic warming accompanies compression.

Lect. 3. CLIMATE AND WEATHER – FACTORS AFFECTING CLIMATE AND WEATHER. MACRO CLIMATE – MESO CLIMATE – MICRO CLIMATE – DEFINITION AND THEIR IMPORTANCE – DIFFERENT CLIMATES OF INDIA AND TAMILNADU AND THEIR CHARACTERIZATION

Weather

i) 'A state or condition of the atmosphere at a given place and at a given instant of time'.
ii) 'The daily or short term variations of different conditions of lower air in terms of temperature, pressure, wind, rainfall, etc'.
iii) State of atmosphere at a particular time as defined by the various meteorological elements. (WMO)

The aspects involved in weather include small areas and duration, expressed in numerical values, etc. The different weather elements are solar radiation, temperature, pressure, wind, humidity, rainfall evaporation, etc. is highly variable. It changes constantly sometimes from hour to hour and at other times from day to day.

Climate

i) 'The generalized weather or summation of weather conditions over a given region during comparatively longer period'.
ii) 'The sum of all statistical information of weather in a particular area during a specified interval of time, usually, a season or a year or even a decade'.
iii) Synthesis of weather conditions in a given area, characterized by long-term statistics (mean values, variances, probabilities of extreme values, etc,) of the meteorological elements in that area. (WMO)

The aspects involved are larger areas like a zone, a state, a country and is described by normal. The climatic normals are generally worked out for a period of 30 years.

Differences between weather and climate:

Weather	Climate
1. A typical physical condition of the atmosphere.	1. Generalized condition of the atmosphere which represents and describes the characteristics of a region.
2. Changes from place to place even in a small locality	2. Different in different large region
3. Changes according to time (every moment)	3. Change requires longer (years) time.
4. Similar numerical values of weather of different places usually have same weather	4. Similar numerical values of climate of different places usually have different climates.
5. Crop growth, development and yield are decided by weather in a given season.	5. Selection of crops suitable for a place is decided based on climate of the region.
6. Under abnormal weather conditions planners can adopt a short-term contingent planning.	6. Helps in long-term agricultural planning.

Factors affecting climate

i) Latitude: The distance from the equator, either south or north, largely creates variations in the climate. Based on the latitude, the climate has been classified as

i) Tropical ii) Sub-tropical iii) Temperate & iv) Polar.

ii) **Altitude(elevation):** The height from the MSL creates variation in climate. Even in the tropical regions, the high mountains have temperate climate. The temperature decreases by 6.5 ºC/Km from the sea level. Generally, there is also a decrease in pressure and increase in precipitation and wind velocity. The above factors alter the kind of vegetation, soil types and the crop production.

iii) **Precipitation**: The quantity and distribution of rainfall decides the nature of vegetation and the nature of the cultivated crops. The crop regions are classified on the basis of average rainfall which is as follow.

Rainfall(mm)	Name of the climatic region
Less than 500	Arid
500-750	Semi-arid
750-1000	Sub-arid
More than 1000	Humid

iv) **Soil type**: Soil is a product of climatic action on rocks as modified by landscape and vegetation over a long period of time. The colour of the soil surface affects the absorption, storage and re-radiation of heat. White colour reflects while black absorbs more radiation. Due to differential absorption of heat energy, variations in temperature are created at different places. In black soil areas, the climate is hot while in red soil areas, it is comparatively cooler due to lesser heat absorption.

v) **Nearness to large water bodies**: The presence of large water bodies like lakes and sea including its current affect the climate of the surrounding areas, eg. Islands and coastal areas. The movement of air from earth's surface and from water bodies to earth modify the climate. The extreme variation in temperature during summer and winter is minimized in coastal areas and island.

vi) **Topography**: The surface of landscape (leveled or uneven surface areas) produces marked change in the climate. This involves the altitude of the place, steepness of the slope and exposure of the slope to light and wind.

vii) **Vegetation**: Kinds of vegetation characterize the nature of climate. Thick vegetation is found in tropical regions where temperature and precipitation are high. General types of vegetations present in a region indicate the nature of the climate of that region.

Scales of climate and their importance

i) Microclimate: Microclimate deals with the climatic features peculiar to small areas and with the physical processes that take place in the layer of air very near to the ground. Soil-ground conditions, character of vegetation cover, aspect of slopes, and state of the soil surface, relief forms – all these may create special local conditions of temperature, humidity, wind and radiation in the layer of air near the ground which differ sharply from general climatic conditions. One of the most important tasks of agricultural meteorology is to study the properties of air near the ground and surface layer of soil, which falls under the micro climate.

ii) Meso climate: The scale of meso climate falls between micro and macro climates. It is concerned with the study of climate over relatively smaller areas between 10 & 100 km across.

iii) Macro climate: Macro climate deals with the study of atmosphere over large areas of the earth and with the large scale atmospheric motions that cause weather. The scales of air motion in different climates are given in the Table below.

Type of climate	Horizontal scale (km)	Vertical scale(km)	Time Scale(hrs)
A. Macro climate			
1. Planetary scale	2000-5000 & more	10	200 to 400
2. Synoptic scale	500-2000	10	100
B. Meso climate	1 to 100	1-10	1-10
C. Micro climate	<100m	200 m	6-12 minutes

If any weather system develops under different types of climate, it persists longer periods under the macroclimate while smaller periods under micro climates.

CLIMATES OF INDIA AND TAMILNADU AND THEIR CHARACTERIZATION

Climate classification was tried by many scientists from beginning of 19th century using many parameters. Thornthwaite during 1931 and 1948 classified the climate using precipitation and evaporation /Potential evaporation and was subsequently modified by Mathur (1955) for the Moisture Index (Im) and is give below

Im = 100 [(P-PE)/PE]

Where P = Precipitation, PE = Potential evapo-transpiration

Using the moisture Index (**Im**) the following classification was made

Im Quantity	Climate classification
100 and above	Per humid
20 to 100	Humid

0 to 20	Moist sub humid
-33.3 to 0	Dry sub humid
-66.7 to -33.3	Semi arid
-100 to -66.7	Arid

Another classification by Troll (1965) based on number of humid months, said to be of more agricultural use was modified by ICRISAT for India. Humid month is one having mean rainfall exceeding the mean Potential evapo transpiration.

Climate	Number of humid months	% geographical area of India
Arid	<2.0	17.00
Semiarid-dry	2.0-4.5	57.17
Semiarid-wet	4.5-7.0	12.31
Humid	>7.0	1.10

The ICAR under All India Coordinated Research Project on Dryland Agriculture adopted classification based Moisture Deficit Index (MDI)

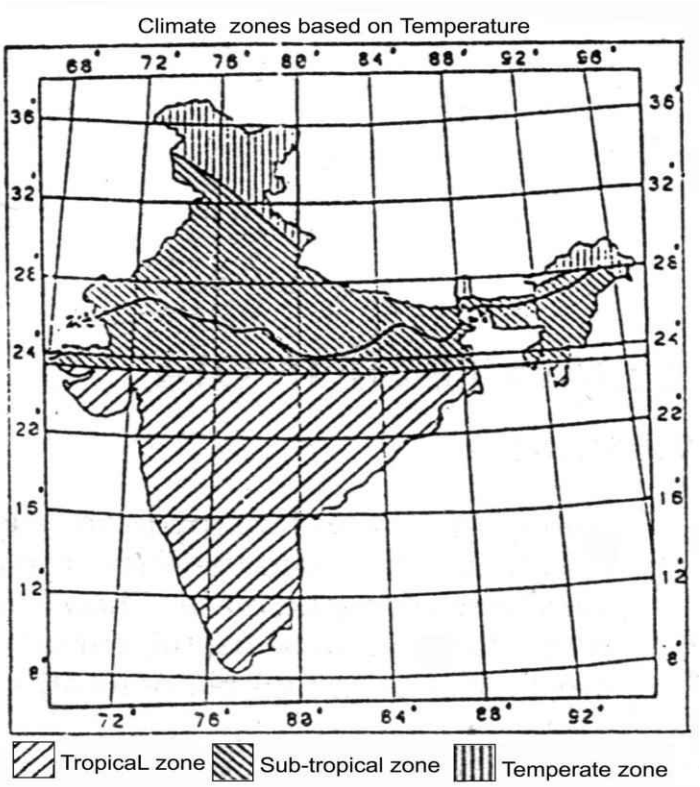
$$MDI = \frac{P - PET}{PET} \times 100$$

Where P is annual precipitation (cm) and PET is Potential Evapotranspiration. Based on MDI the climate is divided into three regions as below.

Type of climate	MDI
Subhumid	0.0 to 33.3
Semiarid	-33.3 to -66.6
Arid	> -66.6

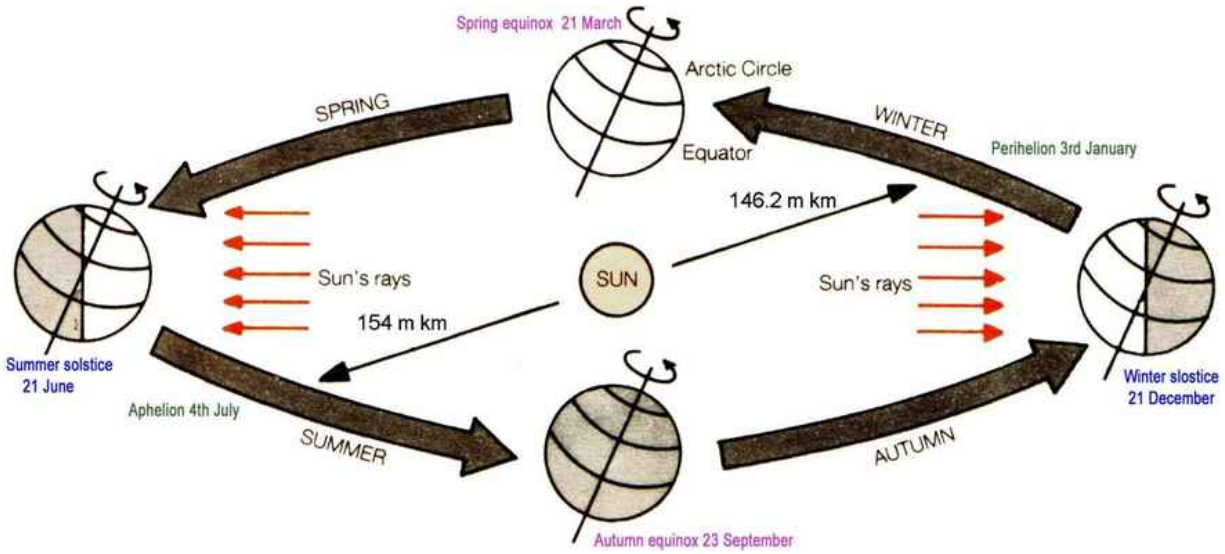
Temperature based classification

The tropic of cancer, which passes through the middle of the country, divides it into two distinct climates. The tropical climate in the South where all the 12 months of the year have mean daily temperature exceeding 20°C; and in the North where a sub-tropical climate prevails. In sub-tropics during the winter months, it is cool to cold. Frosts occur sometime during the months of December and January. Some areas in the Northern India have a temperate climate. Here it snows during the winter months and freezing temperatures may extend to two months or more during the year. Three main climatic zones of India based on temperature are shown in the map below.



Lec 4. Solar radiation - Radiation balance - Wave length characteristics and their effect on crop production - Light - effect of intensity, quality, direction and duration on crop production

- Sun is the prime source of energy
- Sun is the nearest star to the planet earth
- Diameter of the sun is 1.39 X 10^6 km
- It rotates on its axis about once every four weeks(27 days near equator & 30 days –polar)
- Sun is on an average 1.5 X 10^8 km away from the earth (149.64 M km deviation is 2.41 M km)
- Surface temperature of the sun is 5462° K
- Every minute, the sun radiates approximately 56 X 10^{26} calories of energy.
- The interior mass of the sun has a density of 80 to 100 times that of water.
- Energy is due to the fusion, Hydrogen is transformed to helium.
- 99% of the energy to biosphere is only from the sun and the rest one percent is from stars, lightning discharge, sun's radiation reflected from the moon, re-radiation from the earth etc.



Insolation

Electro magnetic energy radiated into the space by the sun

Factors affecting insolation

1. The solar constant which depends on
   a. Energy output of the sun
   b. Distance from earth to sun
2. Transparency of the atmosphere
3. Duration of daily sunlight period
4. Angle at which sun's noon rays strike the earth.

Transfer of heat

All mater, at a temperature above the absolute zero, imparts energy to the surrounding space. Three processes viz. conduction, convection and radiation are involved in heat flow or heat transfer.

Conduction

Heat transfer through matter without the actual movement of the substances or matter. Heat flows from the warmer to cooler part of the body so that the temperatures between them are equalized.

Eg. The energy transmission through an iron rod which is made warmer at one end.

Convection

Processes of transmission of heat through actual movement of molecules of the medium. This is predominant form of energy transmission on the earth as all the weather related processes involve this process. Eg. Boiling of water in a beaker

Radiation.

Transfer of energy from one body to another without the aid of the material medium (solid, liquid or gas). Radiation is not heat, only when radiation is absorbed by surface of a body heat is produced.

Eg. The energy transmission through space from the sun to the earth.

Solar radiation

The flux of radiant energy from the sun is solar radiation.

Heavenly bodies emit – short wave radiation
Near surfaces including earth emit - long wave radiation

Radiation flux

The amount of radiant energy emitted, received, transmitted across a particular area is known as radiant flux.

Radiant flux density

The radiant flux divided by the area across which the radiation is transmitted is called radiant flux density.

Emissive power

The radiant flux density emitted by a source is called its emissive power.

Energy measurement

Units	Cal cm^{-2} min^{-1}	J cm^{-2} mi^{-1}	W cm^{-2}
Cal cm^{-2} min^{-1}	1	4.1868	0.069
J cm^{-2} mi^{-1}	0.238	1	0.00165
W cm^{-2}	14.3	60.6	1

Spectrum of Radiation

Band	Spectrum	Wavelength (µ)	Importance
Ultra violet	Cosmic rays	< 0.005	Shorter wave lengths of spectrum & Chemically active, unless filtered there is danger of life on earth
	Gamma rays and X-rays	0.005 – 0.20	
	Ultraviolet rays	0.20 – 0.39	
Visible	Violet	0.39 – 0.42	Visible spectrum known as Light essential for all plant processes
	Blue	0.42 – 0.49	
	Green	0.49 – 0.54	
	Yellow	0.54 – 0.59	
	Orange	0.59 – 0.65	
	Red	0.65 – 0.76	
Infra red	Infrared rays	> 0.76	Essential for thermal energy of the plant (Source of heat)

Units of measurements of wavelength

Micron 1µ = 10^{-6} m = 10^{-4} cm
Milli micron 1 mµ = 10^{-9} m = 10^{-7} cm
Angstrom Å = 10^{-10} m = 10^{-8} cm

Solar radiation and crop plants

❖ Crop production is exploitation of solar radiation

❖ Three broad spectra

1. **Shorter than visible range**: Chemically very active
   - ➢ When plants are exposed to this radiation the effects are detrimental.
   - ➢ Atmosphere acts as regulator for this radiation and none of cosmic, Gamma and X-rays reaches to the earth.
   - ➢ The UV rays of this segment reaching to the earth are very low and it is normally tolerated by the plants.
2. **Higher than visible wavelength**
   - ⌖ Referred to IR radiation
   - ⌖ It has thermal effect on plants
   - ⌖ In the presence of water vapour, this radiation does not harm plants, rather it supplies the necessary thermal energy to the plant environment.
3. **Visible spectrum**
   - ☞ Between UV & IR radiation and also referred as light
   - ☞ All plant parts are directly or indirectly influenced by the light
   - ☞ Intensity, quality and duration are important for normal plant growth
   - ☞ Poor light leads to plant abnormalities
   - ☞ Light is indispensable to photosynthesis
   - ☞ Light affect the production of tillers, the stability, strength and length of culms
   - ☞ It affects the yield, total weight of plant structures, size of the leaves and root development.
   - ☞ Critical stages of plant growth for light
     - Radiation intensity during the third month of Maize plant
     - Rice – 25 days prior to flowering
     - Barley – flowering period

and	Wavelength(nm)	Specific effect on plant
.	Radiation within 1000 and more	No specific effect on plant activity. Radiation absorbed by plants are transformed into heat. This radiation does not interfere with bio-chemical processes.
.	1000-720	Radiation in this band helps in plant elongation, can be accepted as an adequate measure of plant elongation activity. The far red region (700-920 nm) has important role on photo-periodism, germination of seeds, flowering and colouration of fruits.
.	720-510	In this spectral region light is strongly absorbed by chlorophylls. It generates strong photosynthetic and photo-periodic activity.
.	610-510	This is green-yellow region. Absorption in this spectral region has low photosynthetic effectiveness and weak formative activity.
.	510-400	It is the strongest chlorophyll and yellow pigment absorption region. In the blue-violet range, photosynthetic activity becomes very strong. This region has very strong effect on formation of tissues.
.	400-315	Radiation in this band produces formative effects. It has dwarfing effect on plants and thickening effect on plant leaf.
.	315-280	Radiation in this band has detrimental effect on most plants
.	Less than 280	Lethal effect most of the plants get killed due to radiation in this band UV ranges have germicidal action.

Radiation balance – Solar constant – albedo – Sensible heat – Heat energy – Latent heat

A part of the incident radiation on the surface is absorbed, while a part is reflected and the remaining is transmitted.

Absorptivity

Absorptivity of a substance is defined as the ratio of the amount of radiant energy absorbed to the total amount incident upon that substance. The absorptivity of a blackbody is unity. Natural bodies like sun and earth are near perfect black bodies

Reflectivity

Reflectivity is defined as the ratio of the radiant energy reflected to the total incident radiation upon that surface. If it is expressed in percentage it becomes albedo.

Transmittivity

Transmittivity is defined as ratio of the transmitted radiation to the total incident radiation upon the surface.

Emissivity

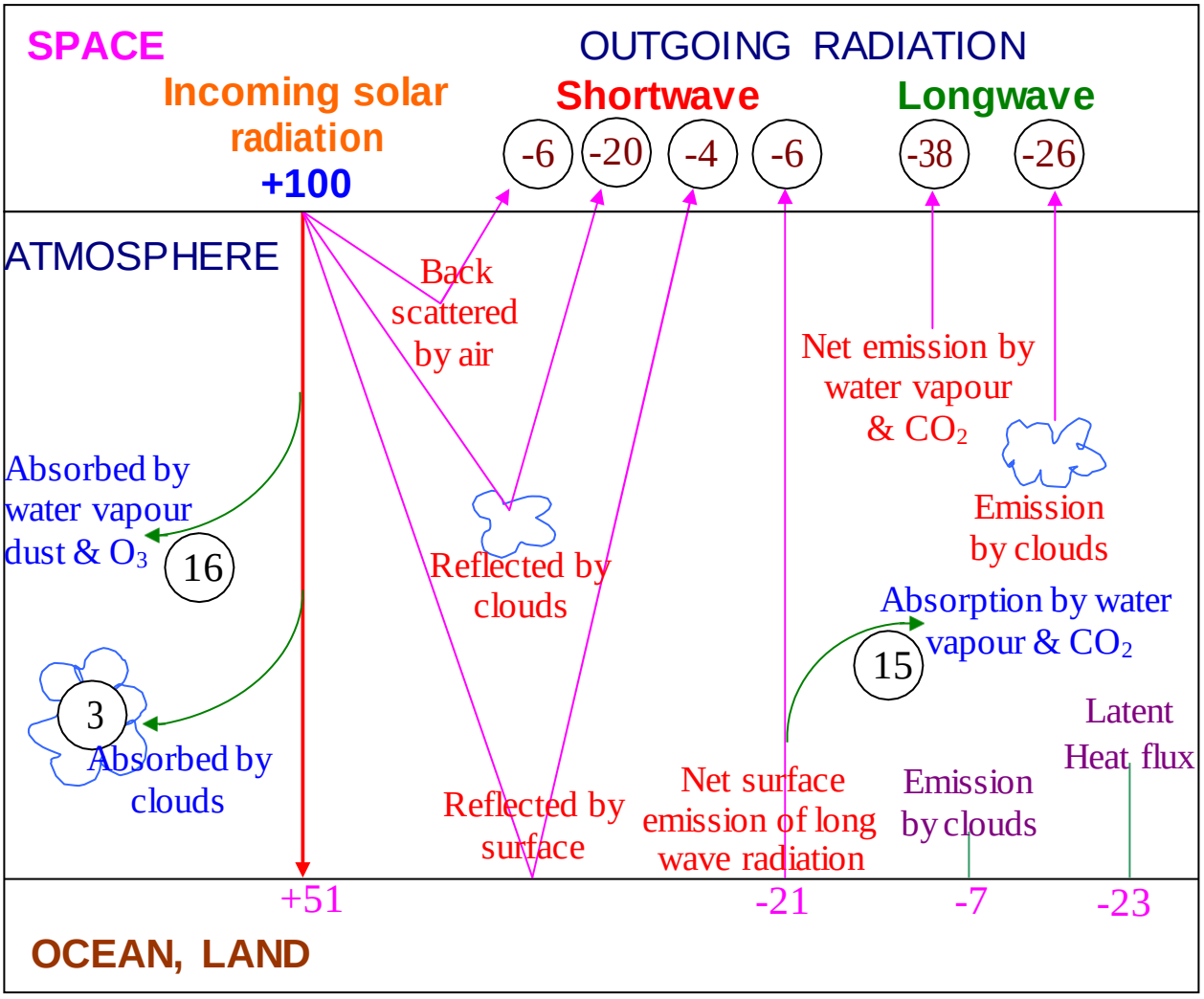
Emissivity is defined as the ratio of the radiant energy emitted by a given surface to the total heat energy emitted by a black body. The emissivity of a black body is unity.

Blackbody radiation

A Blackbody is defined as a body, which completely absorbs all the heat radiations falling on it without reflecting and transmitting any of it. It means reflectivity and transmittivity become zero. When such a black body is heated it emits radiation of all wavelengths depending upon its temp.

Radiation balance

The difference between all incoming and outgoing radiation at the earth's surface and top of the atmosphere is known as radiation balance at the earth's surface.



Solar constant

Solar constant is the energy received on a unit area at the outer most boundary of the earth (atmosphere) surface held perpendicular to the sun's direction, at the mean distance between the sun and the earth.

- ☞ Solar constant is not a true constant. It fluctuates by as much as ± 3.5 % about its mean value depending upon the distance of the earth from the sun.
- ☞ Value is 2 cal / cm^2 / min. (1.92 and 2.02) Recent measurements indicate value of 1.94 cal / cm^2 / min (133 wm^{-2}) [1 Langley = 1cal]
- ☞ 35% of the energy is contributed by U.V. and visible parts and 65% by Infra Red.

Albedo

It is the percentage of reflected radiation to the incident radiation. (Varies with colour and composition of the earth's surface, season, angle of the sun rays). Value is Highest in winter and at sunrise and sunset. Pure water – 5-20%, Vegetation 10-40%, Soils 15-50%, Earth 34-43% and clouds 55%. High albedo indicates that much of the incident solar radiation is reflected rather than absorbed.

Depends up on

1. Angle of incidence of radiation. Albedo increase with decreasing elevation of sun with minimum during noon.
2. Physical characteristics of surface
3. Season
4. Time of the day

For plant community albedo depends upon

1. Age of the crop
2. Percentage of ground cover
3. Colour and reflectivity of the foliage

Outgoing long wave radiation

- ➢ After being heated by solar radiation, the earth becomes source of radiation.
- ➢ Average temperature of the earth's surface 285º k (12º C)
- ➢ 99% of radiation is emitted in the farm of IR range (4 to 120 µ)
- ➢ About 90% of the outgoing radiation is absorbed by the atmosphere.
  Water vapour absorb in wavelengths of 5.3 to 7.7 µ and beyond 20µ.
  Ozone 9.4 to 9.8 µ.
  CO_2 – 13.1 to 16.9 µ
  Clouds – in all wavelengths
- ➢ Long wave radiation escapes to the space between 8.5 and 11 µ and this is known as the **atmospheric window.** Atmosphere for this spectrum acts as transparent medium instead of absorbing. This spectral region is used in microwave remote sensing to monitor the features of the sky in case of overcast sky.
- ➢ A large part of the radiation absorbed by the atmosphere is sent back to the earth's surface as counter – radiation. This counter radiation prevents the earth's surface from excessive cooling at night.

Radiation laws

The direct transfer of heat from the sun to the earth through the space and atmosphere indicates that radiation of heat from one place to other occurs in the form of electromagnetic waves in the same manner and with same speed of as light. The wavelength of electromagnetic radiation is given by the equation

$$\lambda = \frac{C}{V}$$

Where λ = Wavelength (The shortest distance between consecutive crests in the wave trance)
C = Velocity of light ($3x10^{10}$ cm sec^{-1})
V = Frequency means number of vibrations of cycles per second

Plank's law

Plank introduced the 'particle concept'. The electromagnetic radiation consists of a stream or flow of particles or quanta, each quantum having energy content E determined by of each quantum is proportional to the frequency given by the equation.

E= h v where
h= Plank's constant ($6.62x10^{-34}$ J sec^{-1}
V= Frequency

The law states that greater the frequency (shorter wave length) greater is the energy of quantum.

Kirchoff's law

A good absorber of radiation is a good emitter, in similar circumstances. This law states that the absorptivity 'a' of an object for radiation of a specific wavelength is equal to the emissivity 'e' for the same wavelength. The equation of the law is :

$$a(\lambda) = e(\lambda)$$

Stefan-Boltzmann's law

The intensity of radiation emitted (E) by a radiating body is directly proportional to the fourth power of the absolute temperature of that body. (Emissivity of black body = 1)

$E = \sigma T^4$

Where,

T= (273+°C) because temperature is in Kelvins
= Stefan-Boltzmann's constant which is equal to 5.673×10^{-8} W m^{-2} K^{-4}

Wein's Displacement laws

The wavelength of the maximum intensity of emission (λ_{max}) from a radiating black body is inversely proportional to its absolute temperature

$\lambda_{max} = 2897\ T^{-1}\ \mu = 2897/T\ \mu$ Where T is in °K

If the temperature of a body is high, radiation maximum is displaced towards shorter wavelengths. For the sun's surface temperature of 5793°K, the λ_{max} is 0.5μ (2897/5793). The most intense solar radiation occurs in the blue-green range of visible light. The wavelength of maximum intensity of radiation for the earth's actual surface temperature of 14°C or 287°K is about 10.0 (2897/287) microns, which is in the infrared band.

Energy balance or heat balance

The net radiation is the difference between total incoming and outgoing radiations and is a measure of the energy available at the ground surface. It is the energy available at the earth's surface to drive the processes of evaporation, air and soil heat fluxes as well as other smaller energy consuming processes such as photosynthesis and respiration. The net radiation over crop is as follows.

$R_n = G + H + LE + PS + M$

R_n is net radiation, G is surface soil heat flux, H is sensible heat flux, LE is laten heat flux, PS and M are energy fixed in plants by photosynthesis and energy involved in respiration, respectively. The PS and M are assumed negligible due to their minor contribution (about 1-2% of R_n). The net radiation is

the basic source of energy for evapotranspiration (LE), heating the air (H) and soil (S) and other miscellaneous M including photosynthesis.

Temperature

It is defined as the measure of the average speed of atoms and molecules

Kinetic energy

Energy of motion.

Heat

It is the aggregate internal energy of motion and molecules of a body. It is often defined as energy in the process of being transferred from one object to another because of the temperature between them.

Sensible heat

It is the heat that can be measured by a thermometer and thus sensed by humans. Normally measured in Celsius, Fahrenheit and Kelvin.

Latent heat

It is the energy required to change a substance to a higher state of matter. This same energy is released on the reverse process. Change of state through Evaporation and condensation is known as latent heat of evaporation and latent heat of condensation. From water to water vapour takes 600 calories and water to ice takes 80 calories.

Blue colour of the sky

If the circumference of the scattering particle is less than about 1/10 of the wavelength of the incident radiation, the scattering co-efficient is inversely proportional to the fourth power of the wavelength of the incident radiation. This is known as **Rayleigh scattering**. This is the primary cause of the blue colour of the sky.

For larger particles with circumference >30 times of wavelength of the incident radiation, scattering is independent of the wavelength (i.e) white light is scattered. This is known as **Mei scaring**

Red Colour of the sky at sunset & sunrise.

It is because of increased path length in the atmosphere. % of solar energy in the visible part decreases. With in the visible part, the ratio of the blue to the red part decreases with increased path length.

Disposition of Solar radiation

a. 25% of solar radiation is reflected back to the space by clouds (more by middle and high latitudes and less in the sub tropics)
b. 6% reflected back by air, dust and water vapour.
c. 30% scatted downwards (more in the form of shorter wavelengths able) them that in longer wave length (red).
d. 17% of solar radiation is absorbed by the atmosphere. (Mostly by Oxygen, O_3, CO_2 & H_2O vapour).

O_2	– absorb the extreme UV wavelengths (0.12 to 0.6 µ)
O_3	– UV (0.2 to 0.32 µ) and Visible part of radiation (0.44 to 0.7 µ)
H_2O vapour	– Near infra red (0.93, 1.13, 1.42 µ)
CO_2	- IR band 2.7 µ.

e. About 50% of solar radiation reaches earth's surface, after reflection, scattering and absorption.

Light – Effect of Light Intensity, Quality, Direction And Duration On CropProduction – Air Temperature – Factors Affecting Temperature.

Light:

Light is the visible portion of the solar spectrum with wavelength range is from 0.39 to 0.76μ.

- ❖ Light is one of the important climatic factors for many vital functions of the plant.
- ❖ It is essential for the synthesis of the most important pigment ie. Chlorophyll, Chlorophyll absorbs the radiant energy and converts it into potential energy of carbohydrate.
- ❖ The carbohydrate thus formed is the connecting link between solar energy and living world.
- ❖ In addition, it regulates the important physiological functions.
- ❖ The cahrateristics of light viz. intensity, quality, duration and direction are important for crops.

Light intensity

The intensity of light is measured by comparing with a standard candle. The amount of light received at a distance of one metre from a standard candle is known as "Metre candle or Lux". The light intensity at one foot from a standard candle is called 'foot candle' or 10.764 luxes and the instrument used is called as lux metre.

- ➢ About one percent of the light energy is converted into biochemical energy.
- ➢ Very low light intensity reduces the rate of photosynthesis resulting in reduced growth.
- ➢ Similarly, very high intensity is detrimental to plant in many ways as below.
  - ▪ It increases the rate of respiration.
  - ▪ It also causes rapid loss of water (ie) increases the transpiration rate of water from the plants.
  - ▪ The most harmful effect of high intensity light is that it oxidises the cell contents which is termed as 'Solarisation'. This oxidation is different from respiration and is called as photo-oxidation.
- ➢ Under field conditions the light is not spread evenly over the crop canopy but commonly passed by reflection and transmission through several layers of leaves.
- ➢ The intensity of light falls at exponential rate with path length through absorbing layers according to Beer's law. ie the relative radiation intensity decreases exponentially with increasing leaf area.
- ➢ At ground level the light intensity is below the light compensation point (The light intensity at which the gas exchange resulting from photosynthesis is equal to that resulting from respiration)

Based on the response to light intensities the plants are classified as follows.

(i) Sciophytes (shade loving plants): The plants grow better under partially shaded conditions. (eg) Betel vine, buck wheat etc.

(ii) Hetrophytes (Sun loving): Many species of plants produce maximum dry matter under high light intensities when the moisture is available at the optimum level. (eg) Maize, sorghum, rice etc.

Quality of Light

When a beam of white light is passed through a prism, it is dispersed into wavelengths of different colours. This is called the visible part of the solar spectrum. The different colours and their wave length are as follows:

Violet	400 – 435 m μ	Blue	435 – 490 m μ
Green	490 – 574 m μ	Yellow	574 – 595 m μ
Orange	595 – 626 m μ	Red	626 – 750 m μ

- The principal wavelength absorbed and used in photosynthesis are in the violet – blue and the orange - red regions.
- Among this, short rays beyond violet such as X rays, gamma rays and larger rays beyond red such as infrared, are detrimental to plant growth.

- Red light is the most favourable light for growth followed by violet – blue.
- Ultra – violet and shorter wave lengths kill bacteria and many fungi.

c) Duration of light:

- The duration of light has greater influence than the intensity for canopy development and final yield.
- It has a considerable importance in the selection of crop varieties.
- The response of plants to the relative length of the day and night is known as phtoperiodism. The plants are classified based on the extent of response to day length which is as follows.
  - **(i) Long day plants:** The plants which develop and produce normally when the photoperiod is greater than the critical minimum (greater than 12 hours). eg. Potato, Sugarbeet, Wheat, Barley etc.
  - **(ii) Short day plants:** The plants which develop normally when the photoperiod is less than the critical maximum (less than 12 hours). Rice, Sorghum, cotton, Sunflower
  - **(iii) Day neutral plants / Indeterminate:** Those plants which are not affected by photoperiod. (eg) Tomato, Maize

The photoperiodism influences the plant character such as floral initiation or development, bulb and rhizome production etc. In long day plant, during periods of short days, the growth of internodes are shortened and flowering is delayed till the long days come in the season. Similarly when short day plants are subjected to long day periods, there will be abnormal vegetative growth and there may not be any floral initiation.

Direction of light

- The direction of sunlight has a greater effect on the orientation of roots and leaves.
- In temperate regions, the southern slopes plants produce better growth than the northern slopes due to higher contribution of sunlight in the southern side.
- The change of position or orientation of organs of plants caused by light is usually called as phototropism ie the leaves are oriented at right angles to incidence of light to receive maximum light radiation.

Photomorphogenesis: Change in the morphology of plants due to light. This is mainly due to U.V and violet ray of the sun.

5. Air temperature - Factors affecting temperature. Diurnal and seasonal variation in air temperature - Isotherm, Heat unit and its use - Heat and cold injuries

AIR TEMPERATURE

Temperature is defined as, "The measure of speed per molecule of all the molecules of a body". Where as heat is, "the energy arising from random motion of all the molecules of a body'. (Degree of molecular activity). It is the intensity aspect of heat energy.

Conduction: Heat transfer when two bodies of unequal temperatures come into contact. Heat passes from point to point by means of adjacent molecules.

Convection: Transfer through movement of particles (part of mass) in fluids and gasses. These ar able to circulate internally and distribute heated part of the mass.

Radiation: It is the process of transmission of energy by electromagnetic waves between two bodies without the necessary aid of an intervening material medium.

Factors affecting air temperature

i. Latitude
ii. Altitude
iii. Distribution of land and water
iv. Ocean currents
v. Prevailing winds
vi. Cloudiness
vii. Mountain barriers
viii. Nature of surface
ix. Relief
x. Convection and turbulence etc.

1. Latitude:

The time of occurrence of maximum monthly mean temperature and minimum monthly mean temperature also depends on latitude of a place. (eg.) The coldest month is January in northern regions of India while December in the south. Similarly, the warmest month is May in the south while June in the north across the country.

2. Altitude:

The surface air temperature decreases with increasing altitude from the mean sea level as the density of air decreases. Since the density of air is less at higher altitudes, the absorbing capacity of air is relatively less with reference to earth's longwave radiation.

3. Distribution of land and water:

Land and water surfaces react differently to the insolation. Because of the great contrasts between land and water surfaces their capacity for heating the atmosphere varies. Variations in air temperature are much greater over the land than over the water. The differential heating process between land and sea surfaces are due to their properties. It is one of the reasons for Indian monsoon.

4. Ocean currents:

The energy received over the ocean surface carried away by the ocean currents from the warm areas to cool areas. This results in temperature contrast between the equator and poles. The occurrence of El-Nino is due to change in sea surface temperature between two oceanic regions over the globe.

5. Prevailing winds:

Winds can moderate the surface temperature of the continents and oceans. In the absence of

winds, we feel warm in hot climates. At the same time, the weather is pleasant if wind blows.

6. Cloudiness:

The amount of cloudiness affects the temperature of the earth's surface and the atmosphere. A thick cloud reduces the amount of insolation received at a particular place and thus the day time temperature is low. At the same time, the lower layers in the atmosphere absorb earth's radiation. This results in increasing atmospheric temperature during night. That is why, cloudy nights are warmer. This is common in the humid tropical climates.

7. Mountain barriers:

Air at the top of the mountain makes little contact with the ground and is therefore cold while in the valley at the foothills makes a great deal of contact and is therefore warm. That is, the lower region of the earth's atmosphere is relatively warmer when compared to hillocks.

DIURNAL AND SEASONAL VARIATION IN AIR TEMPERATURE– ISOTHERM. HEAT UNIT DEFINITION AND ITS USE – HEAT AND COLD WAVES – ROLE OF TEMPERATURE IN CROP PRODUCTION.

Diurnal and seasonal variation of air temperature

- The minimum air temperature occurs at about sunrise, after which there is a constant rise till it reaches to maximum.
- The maximum air temperature is recorded between 1300 hrs and 1400 hrs although the maximum solar radiation is reaches at the moon.
- A steady fall in temperature till sunrise is noticed after is attains maximum. Thus the daily March displays one maximum and one minimum. The difference between the two is called the diurnal range of air temperature.
- The diurnal range of air temperature is more on clear days while cloudy weather sharply reduces daily amplitudes.
- The diurnal range of temperature is also influenced by soils and their coverage in addition to seasons.
- Addition of daily maximum and minimum temperature divided by two is nothing but daily mean / average temperature.
- In northern hemisphere winter minimum occurs in January and summer maximum in July.

Horizontal air temperature distribution

The lines connecting points of equal temperature is called as **isotherm**

- ✓ It is largely depends latitude. A general decrease in temperature from equator towards poles is one of the most fundamental factors of climatology.
- ✓ Irregular distribution of land and water on earth's surface breaks the latitudinal variation in temperature.
- ✓ Land areas warm and cool rapidly than water bodies
- ✓ Mountain barriers influence horizontal distribution of temperature by restricting movement of air masses.
- ✓ On local scale topographic relief exerts an influence on temperature distribution.

Vertical air temperature distribution

Decrease in temperature with increase in height

Temperature inversion

Occasionally at some altitude the temperature abruptly increases instead of decreasing. This condition in which this abrupt rise instead of fall in temperature occurs in the air is known as the temperature inversion. This may occurs under the following conditions.

- When the air near the ground cools off faster than the overlying layer, because of heat loss during cooling nights.
- When an actual warm layer passing over a lower cold layer
- Cold air from hill tops and slopes tend to flow downward and replaced by warm air.

Significance of Temperature inversion

- Cloud formation, precipitation and atmospheric visibility are greatly influenced by inversion phenomenon
- Fog formation may take place near the ground which may affect the visibility to both human beings and animals. Affects air navigation.
- Diurnal temperature is affected by temperature inversions.

The incoming solar radiation and its conversion in to heat is affected.

Heat Units

It is a measure of relative warmth of growing season of a given length. Normally it is indicated as Growing Degree Days (GDD). A heat unit is the departure from the mean daily temperature above the minimum threshold temperature.

The minimum threshold temperature is the temperature below which no growth takes place. Usually ranges from 4.5 to 12.5 ºC for different crops (Most commonly used value is 6.0ºC)

Degree Day

A degree day is obtained by subtracting the threshold temperature from daily mean temperature. Summation of the daily values over the growth period gives degree days of the crops.

$$GDD = \Sigma \left\{ \frac{T_{max} + T_{min}}{2} \right\} - T_b$$

Where

T_{max} – Maximum air temperature of the day
T_{min} – Minimum air temperature of the day
T_b - Base temperature of the crop

The base temperature is the threshold temperature.

Advantages / Importance of growing degree Day Concept

1. In guiding the agricultural operations and planting land use.
2. To forecast crop harvest dates, yield and quality
3. In forecasting labour required for agricultural operations
4. Introduction of new crops and new varieties in new areas
5. In predicting the likelihood of successful growth of a crop in an area.

HEAR INJURIES

'Thermal death point" – the temperature at which the plant cell gets killed when the temperature ranges from 50-60°C. This varies with plant species. The aquatic and shade loving plants are killed at comparatively lower temperature (40°C).

High temperature

- results in desiccation of plants
- disturbs the physiological activities like photosynthesis and respiration
- increases respiration leading to rapid depletion of reserve food.

Sun clad

Injury caused on the barks of stem by high temperature during day time and low temperature during the night time.

Stem griddle

The stem at ground level scorches around due to high soil temperature. It causes death of plant by destroying conductive tissues. Eg. This type of injury is very common in young seedlings of cotton in sandy soil when soil temperature exceeds 60°C.

COLD INJURY

(i) Chilling injury: Plants which are adapted to hot climate, if exposed to low temperature for sometime, are found to be killed or severely injured or development of chloratic condition (yellowing) (eg.) cholratic bands on the leaves of sugarcane, sorghum and maize in winter months when the night temperature is below 20°C.

(ii) Freezing injury: This type of injury is commonly observed in plants of temperate regions. When the plants are exposed to very low temperature, water freezes into ice crystals in the intercellular spaces of plants. The protoplasm of cell is dehydrated resulting in the death of cells.

(eg.) Frost damage in potato, tea etc.

(iii) Suffocation: In temperature regions, usually during the winter season, the ice or snow forms a thick cover on the soil surface. As a result, the entry of oxygen is prevented and crop suffers for want of oxygen. Ice coming in contact with the root prevents the diffusion of CO_2 outside the root zone. This prevents the respiratory activities of roots leading to accumulation of harmful substances.

(iv) Heaving: This is a kind of injury caused by lifting up of the plants along with soil from its normal position. This type of injury is commonly seen in temperate regions. The presence of ice crystals increases the volume of soil. This causes mechanical lifting of the soil.

Role of temperature in crop production:

1. Temperature influences distribution of crop plants and vegetation.
2. The surface air temperature is one of the important variables, which influences all stages of crop during its growth development and reproductive phase.
3. Air temperature affects leaf production, expansion and flowering.
4. The diffusion rates of gases and liquid changes with temperature.
5. Solubility of different substances is dependent on temperature.

6. Biochemical reactions in crops (double or more with each 10°C rise) are influenced by air temperature.
7. Equilibrium of various systems and compounds is a function of temperature.
8. Temperature affects the stability of enzymatic systems in the plants.
9. Most of the higher plants grow between 0°C – 60°C and crop plants are further restricted from 10 – 40°C, however, maximum dry matter is produced between 20 and 30°C
10. At high temperature and high humidity, most of the crop plants are affected by pests and diseases.
11. High night temperature increases respiration and metabolism.
12. A short duration crop becomes medium duration or long duration crop depending upon its environmental temperature under which it is grown.
13. Most of the crops have upper and lower limits of temperature below or above which, they may not come up and an optimum temperature when the crop growth is maximum. These are known as cardinal temperatures and different crops have different temperatures.

Sl No	Crop	Minimum	Optimum	Mximum
1	Wheat and Barley	0 – 5	25 – 31	31 – 37
2	Sorghum	15 – 18	31 – 36	40 – 42

Thermo periodic response

Response of living organism to regular changes in temperature either day or night or seasonal is called thermoperiodism.

Lect. 6. SOIL TEMPERATURE – THEIR IMPORTANCE IN CROP PRODUCTION. FACTORS AFFECTING SOIL TEMPERATURE. DIURNAL AND SEASONAL VARIATION IN SOIL TEMPERATURE.

Soil temperature:

The soil temperature is one of the most important factors that influence the crop growth. The sown seeds, plant roots and micro organisms live in the soil. The physio-chemical as well as life processes are directly affected by the temperature of the soil. Under the low soil temperature conditions signification is inhibited and the intake of water by roof is reduced. In a similar way extreme soil temperatures injures plant and its growth is effected.

Eg. On the sunny side, plants are likely to develop faster near a wall that stores and radiates heat. If shaded by the wall, however, the same variety may mature later. In such cases soil temperature is an important factor.

Importance of soil temperature on crop plants:

The soil temperature influences many process.

1. Governs uptake of water, nutrients etc needed for photosynthesis.
2. Controls soil microbial activities and the optimum range is 18-30°C.
3. Influences the germination of seeds and development of roots.
4. Plays a vital role in mineralization of organic forms of nitrogen.(inc with inc in temp)
5. Influences the presence of organic matter in the soil.(more under low soil temperature)
6. Affects the speed of reactions and consequently weathering of minerals.
7. Influences the soil structure (types of clay formed, the exchangeable ions present, etc.)

Factors affecting soil temperature:

Heat at ground surface is propagated downward in the form of waves. The amplitude deceases with depth. Both meteorological and soil factors contribute in bringing about changes of soil temperature.

I) Meteorological factors:

1. Solar radiation:

a) The amount of solar radiation available at any given location and point of time is directly proportional to soil temperature.
b) Even though a part of total net radiation available is utilised in evapotranspiration and heating the air by radiation (latent and sensible heat fluxes) a relatively substantial amount of solar radiation is utilized in heating up of soil (ground heat flux) depending up on the nature of surface.
c) Radiation from the sky contributes a large amount of heat to the soil in areas where the sun's rays have to penetrate the earth's atmosphere very obliquely.

2. Wind:

Air convection or wind is necessary to heat up the soil by conduction from the atmosphere. (eg.) The mountain and valley winds influence the soil temperature.

3. Evaporation and condensation:

a) The greater the rate of evaporation the more the soil is cooled. This is the reason for coolness of moist soil in windy conditions.
b) On the other hand whenever water vapour from the atmosphere or from other soil depths condenses in the soil it heats up noticeably. Freezing of water generates heat.

4. Rainfall (Precipitation):

Depending on its temperature, precipitation can either cool or warm the soil.

II. Soil factors:

1. Aspect and slope:

a) In the middle and high latitudes of the northern hemisphere, the southern slopes receive more insolation per unit area than the northern exposure.
b) The south west slopes, are usually warmer than the south east slopes. The reason is that the direct beam of sunshine on the south east slope occur shortly after prolonged cooling at night, but the evaporation of dew in the morning also requires energy.

2. Soil texture

a) Because of lower heat capacity and poor thermal conductivity, sandy soils warm up more rapidly than clay soils. The energy received by it is concentrated mainly in a thin layer resulting in extraordinary rise in temperature.
b) Radiational cooling at night is greater in light soils than in heavy soils. In the top layer, sand has the greatest temperature range, followed by loam and clay.
c) The decrease of range with depth is more rapid in light soils than heavy soils when they are dry but slower when they are wet.
d) A soil with rough surface absorbs more solar radiation than one with a smooth surface.

3. Tillage and Tilth:

a) By loosening the top soil and creating a mulch, tillage reduces the heat flow between the surface and the sub soil.
b) Since, the soil mulch has a greater exposure surface than the undisturbed soil and no capillary connection with moist layers below, the cultivated soil dries up quickly by evaporation, but the moisture in the sub-soil underneath the dry mulch is conserved.
c) In general soil warms up faster than air. The diurnal temperature wave of the cultivated soil has a much larger amplitude than that of the uncultivated soil.
d) The air 2-3 cm above the tilled soil is often hotter (10°C or above) than that over an untilled soil.
e) At night loosened ground is colder and more liable to frost than the uncultivated soil.

4. Organic matter:

a) The addition of organic matter to a soil reduces the heat capacity and thermal conductivity. But, the water holding capacity increases.
b) The absorbtivity of the soil increases because of the dark colour of the organic matter.
c) At night, the rapid flow of heat from sub-soil by radiation is reduced with the addition of organic matter because of its low thermal conductivity.
d) The darker the colour, the smaller the fraction of reflected radiation.
e) The dark soils and moist soils reflect less than the light coloured and dry soils.

5. Soil moisture:

a) Moisture has an effect on heat capacity and heat conductivity.
b) Moisture at the soil surface cools the soil through evaporation.
c) Therefore, a moist soil will not heat up as much as a dry one.
d) Moist soil is more uniform in temperature throughout its depth as it is a better conductor of heat than the dry soil.

Variations in soil temperature:

There are two types of soil temperature variations; daily and seasonal variation of soil temperature

1. Daily variations of soil temperature:

a) These variations occur at the surface of the soil.
b) At 5 cm depth the change exceeds 10°C. At 20 cm the change is less and at 80 cm diurnal changes are practically nil.
c) On cooler days the changes are smaller due to increased heat capacity as the soils become wetter on these days.
d) On a clear sunny day a bare soil surface is hotter than the air temperature.
e) The time of the peak temperature of the soil reaches earlier than the air temperature due to the lag of the air temperature.
f) At around 20 cm in the soil the temperature in the ground reaches peak after the surface reaches its maximum due to more time the heat takes to penetrate the soil. The rate of penetration of heat wave within the soil takes around 3 hours to reach 10 cm depth.
g) The cooling period of the daily cycle of the soil surface temperature is almost double than the warming period.
h) Undesirable daily temperature variations can be minimised by scheduling irrigation.

2. Seasonal variations of soil temperature:

a) Seasonal variations occur much deeper into the soil.
b) When the plant canopy is fully developed the seasonal variations are smaller.
c) In winter, the depth to which the soil freezes depends on the duration and severeness of the winter.
d) In summer the soil temperature variations are much more than winter in tropics and sub tropics.

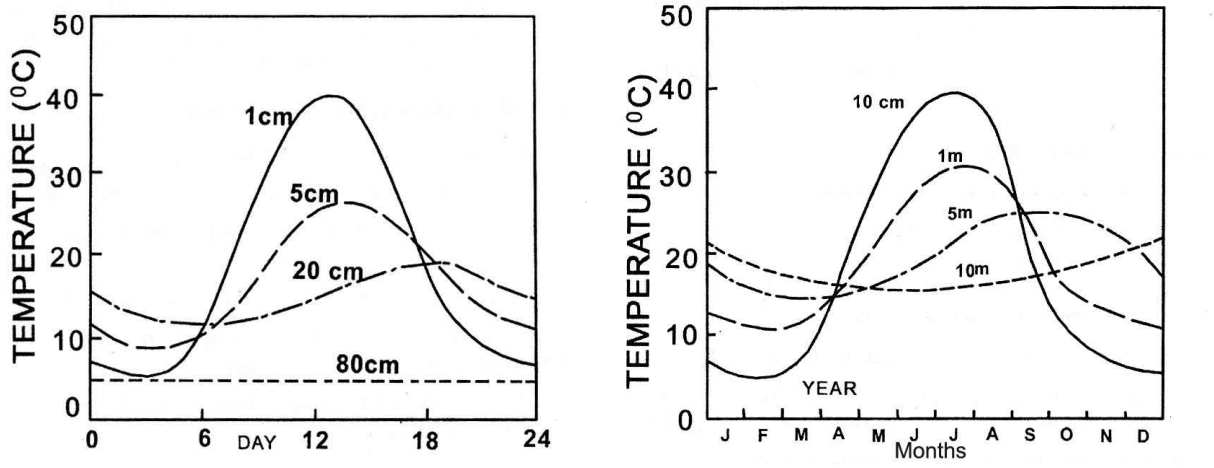


Fig. Daily and monthly variation of soil temperature

Lect 7. Humidity – Types - Dew point temperature - Vapour pressure deficit - Diurnal variation in Relative humidity and its effect on crop production – Wind and its role on crop production

Humidity: The amount of water vapour that is present in atmosphere is known as atmospheric moisture or humidity.

Absolute humidity: The actual mass of water vapour present in a given volume of moist air. It is expressed as grams of water vapour per cubic meter or cubic feet.

Specific humidity: Weight of water vapour per unit weight of moist air. It is expressed as grams of water vapour per kilogram of air (g/kg).

Mixing ratio: The ratio of the mass of water vapour contained in a sample of moist air to the mass of dry air. It is expressed as gram of water vapour per kilogram dry air.

Relative Humidity: The ratio between the amount of water vapour present in a given volume of air and the amount of water vapour required for saturation under fixed temperature and pressure. There are no units and this is expressed as percentage. In other terms it is the ratio of the air's water vapour content to its maximum water vapour capacity at a given temperature expressed in percentage. The relative humidity gives only the degree of saturation of air. The relative humidity of saturated air is 100 per cent.

Dew Point temperature: The temperature to which a given parcel of air must be cooled in order to become saturation at constant pressure and water vapour content. In this case, the invisible water vapour begins to condense into visible form like water droplets.

Vapour Pressure deficit: The difference between the saturated vapour pressure (SVP) and actual vapour pressure (AVP) at a given temperature. This is an another measure of moisture in the atmosphere which is useful in crop growth studies. When air contains all the moisture that it can hold to its maximum limit, it is called as saturated air, otherwise it is unsaturated air, at that temperature. The vapour pressure created at this temperature under saturated conditions is vapour pressure or saturated vapour pressure (SVP).

Importance of Humidity on crop plants

The humidity is not an independent factor. It is closely related to rainfall, wind and temperature. It plays a significant role in crop production.

1. The humidity determines the crops grown in a given region.
2. It affects the internal water potential of plants.
3. It influences certain physiological phenomena in crop plants including transpiration
4. The humidity is a major determinant of potential evapotranspiration. So, it determines the water requirement of crops.
5. High humidity reduces irrigation water requirement of crops as the evapotranspiration losses from crops depends on atmospheric humidity.
6. High humidity can prolong the survival of crops under moisture stress. However, very high or very low relative humidity is not conducive to higher yields of crops.
7. There are harmful effects of high humidity. It enhances the growth of some saprophytic and parasitic fungi, bacteria and pests, the growth of which causes extensive damage to crop plants.
8. Examples: a. The blight disease on potato.
   b. The damage caused by thrips and jassids on several crops.

9. High humidity at grain filling reduces the crop yields.
10. A very high relative humidity is beneficial to maize, sorghum, sugarcane etc, while it is harmful to crops like sunflower and tobacco.
11. For almost all the crops, it is always safe to have a moderate relative humidity of above 40%

Varitation In Humidity:

1. Absolute humidity is highest at the equator and minimum at the poles.
2. Absolute humidity is minimum at sunrise and maximum in afternoon from 2 to 3 p.m. The diurnal variations are small in desert regions.
3. The relative humidity is maximum at about the sunrise and minimum between 2 to 3 p.m.
4. The behaviour of relative humidity differs a lot from absolute humidity. At the equator it is at a maximum of 80 per cent and around 85 per cent at the poles. But, near horse latitudes it is around 70 per cent.

Wind: Air in horizontal motion is known as wind. Vertical movement is noticed but negligibly small compared to horizontal movement as the height of the atmosphere is only for few kilometers. However vertical movement or uplift of air only causes significant weather changes in cloud formation and rain.

Effect of wind on crop plants

1) Transports heat in either sensible or latent heat form from lower to higher altitudes
2) Wind affects the plant directly by increasing transpiration and the intake of CO_2 and also causes several types of mechanical damage.
3) Wind helps in pollination and dispersal of seeds.
4) Light and gentle winds are helpful for cleaning the agricultural produce.
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6) Wind has powerful effect on humidity.
7) Long, continued warm, dry winds injured blossoms by evaporating the secretion of the stigma.
8) Provides moisture which is necessary for precipitation
9) Wind prevents frost by disrupting atmospheric inversion
10) Causes soil erosion

Wind speed in different season

Winds represent air in motion. The primary cause of all winds is regional differences in temperature, producing regional differences in pressure. When these pressure differences persist for several hours, the rotation of the earth modifies the direction of motion, till the winds blow along lines of equal pressure. Wind direction and speed are modified frequently due to seasonal variation in solar radiation and differential heating of the earth's surface.

Wind Speed The winds are generally measured over level, open terrain at 3 meters above ground. Yet, a general idea of the distribution of the mean daily wind speed, on an annual basis as well as on a monthly basis, would be useful. The mean daily wind speed is the value obtained by averaging the wind speed (irrespective of direction) for a whole day. This averaged for all the days of a month is the mean daily wind speed for that month. The daily values averaged for all the 365 days of the year is the annual mean daily wind speed.

Wind Direction Winds are always named after the direction they come from. Thus, a wind from the south, blowing towards north is called south wind. The wind vane is an instrument used to find out the direction of the wind. Windward refers to the direction wind comes from, and leeward refers to the direction it blows to. When a wind blows more frequently from one

direction than from any other, it is called a prevailing wind.

South West Monsoon wind direction: During South West Monsoon period of June to September, the westerly winds prevail on the west of Kerala and south winds on the west of northern Circars, Orissa and Bengal. During April and May the region of high temperature is shifted to north viz., upper Sind, lower Punjab and Western Rajasthan. This area becomes the minimum barometric pressure area to which monsoon winds are directed.

North East Monsoon wind direction: During North East Monsoon period of October to December, on account of the increase in barometric pressure in Northern India, there is a shift in the barometric pressure to the South East and North Easterly winds begin to flow on the eastern coast, by the end of September. These changes bring on heavy and continue rainfall to the Southern and South Eastern India.

Average wind speed for different seasons at Coimbatore. Month wise wind speed (KMPH)

Month	Wind speed (KMPH)	
	Monthly	Season
January	5.4	Winter - 5.5
February	5.5	
March	7.6	
April	4.6	Summer - 5.3
May	3.7	
June	13.4	
July	12.7	SWM - 10.8
August	11.2	
September	6.0	
October	2.8	NEM - 2.7
November	2.5	
December	2.8	

LECT.8 ATMOSPHERIC PRESSURE – DIURNAL AND SEASONAL VARIATION – PRESSURE SYSTEMS OF THE WORLD – CAUSES OF VARIATION – ISOBAR – LOW, DEPRESSION, ANTICYCLONE, TORNADO, HURRICANE AND STORMS.

Atmospheric pressure

The atmospheric pressure is the weight of the air, which lies vertically above a unit area centered at a point. The weight of the air presses down the earth with the pressure of 1.034 gm / cm^2. It is expressed in millibar (mb) equal to 100 N/m^2 or 1000 dynes/cm^2. Unequal heating of the earth and its atmosphere by the sun and rotation of the earth bring about differences in atmospheric pressure.

Isobars: The distribution of pressure is represented on maps by 'isobars'. Isobars are defined as the imaginary lines drawn on a map to join places having the same atmospheric pressure.

Diurnal and seasonal variation

(a) Diurnal pressure variation

a. There is a definite rhythm in the rise and fall of the pressure in a day.
b. Radiational heating (air expansion) and radiational cooling (air contraction) are the main reasons for diurnal variation in the air pressure.
c. Diurnal variation is more prominent near the equator than at the mid latitudes.
d. The areas closer to sea level record relatively larger amount of variation than in land areas.
e. Equatorial regions absorbs more heat than it loses while the polar region gives up more heat than they receive

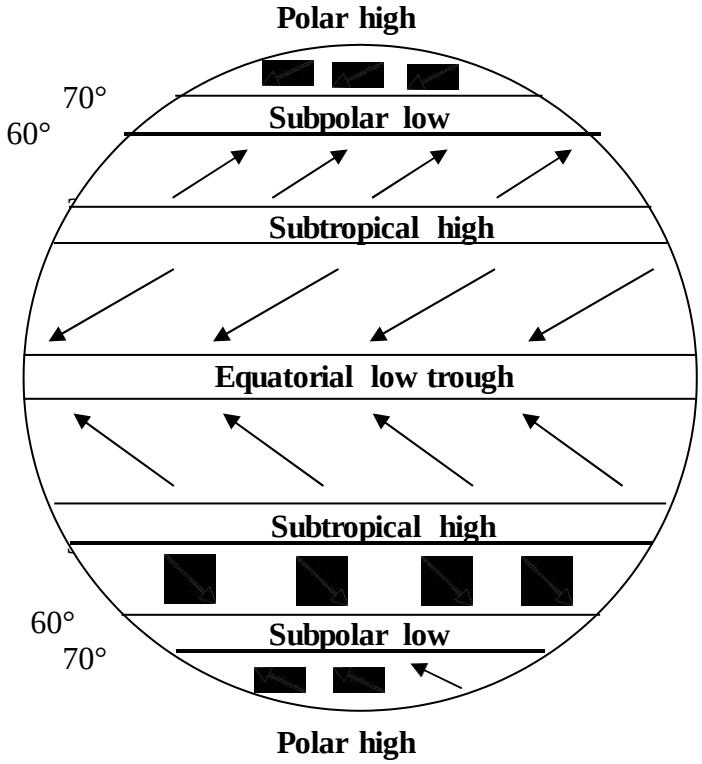
b) Seasonal pressure variation

a. Due to the effect caused by annual variation in the amount of insolation, distinct seasonal pressure variations occur.
b. These variations are larger in the tropical region than the mid latitude and polar regions.
c. Usually, high pressures are recorded over the continents during the cold season and over the oceans during the warm season.

Pressure systems of the world

The shape of the earth is not uniform and subjected to uneven distribution of solar radiation, when it revolves around the sun. The uneven distribution of solar radiation over different regions of the globe leads to contrast in surface air temperature. This results in variations of surface atmospheric pressure systems, which are known as standard atmospheric pressure systems / belts. There are altogether seven alternating low and high pressure belts on the earth's surface. They are as follows:

i. Equatorial trough of low pressure (between 5°N and 5°S)
ii. Subtropical high pressure belt (Northern hemisphere) (30° and 35°N)
iii. Subtropical high pressure belt (Southern hemisphere) (30° and 35°S)
iv. Subpolar low pressure belt (Northern hemisphere) (60° and 70°N)
v. Subpolar low pressure belt (Southern hemisphere) (60° and 70°S)
vi. Polar high (Northern hemisphere)
vii. Polar high (Southern hemisphere)



The equatorial low pressure belt or the ITCZ (0° to 5° N and 0° to 5°S):

1. This belt of low pressure extends from around 5° N to 5°S latitude. In the equatorial region, the rays of Sun fall vertically on the Earth's surface, therefore the level of insolation is high.
2. The air gets heated and rises, leading to the creation of low pressure. This zone is also known as the doldrums as little or no surface wind blows in this region.
3. This low pressure zone is also referred to as the inter-tropical convergence zone (ITCZ), as air converges here from surrounding areas, drawn by the low pressure.
4. The air is then carried up by powerful convection currents that cause severe thunderstorms in this region. Places in this zone receive precipitation over 200 days of the year.

The sub-tropical high pressure belt (30° N to 35° N and 30° S to 35° S):

1. This belts extend along the Tropics of Cancer and Capricorn, around 30° N and S latitudes.
2. Air rises above the equator, and when it reaches the troposphere, it starts flowing towards the poles.
3. This air starts converging (or coming together) between 25° and 35° N and S latitudes. The air also loses heat through radiation.
4. Upper air convergence and cooling, together, causes the air to subside over the sub-tropics. This descending air creates the sub-tropical high pressure belts.

The sub-polar low pressure belt (60° N to 70° N and 60° S to 70° S):

1. The sub-polar low pressure belts occur around 60° N and S.
2. This is a zone where two wind systems converge-the Prevailing Westerlies and the Polar Easterlies.
3. The Westerlies being warmer, is forced to rise over the cold polar air. This rising air creates this area of low pressure. This is a zone of storms caused by the powerful uplift of air.

The polar high pressure belt (85° N to 90° N and 85° S to 90° S):

1. The polar high pressure belts are found over the north and south poles.
2. In these regions the solar energy intercepted by the surface of the Earth is very low, leading to low terrestrial radiation.

3. The air over the poles is therefore always cold, dense and heavy, thereby creating belts of high pressure.

Causes of variation

The atmospheric pressure changes continuously due to several factors. The most important factors are changes with temperature, altitude, water vapour content and rotation of earth.

a) Temperature

Hot air expands and exerts low pressure. Cold air contracts and exerts high pressure. So the equator has a low pressure due to prevalence of high temperature but poles have a high pressure.

b) Altitude

At sea level, the air column exerts its full pressure, but when we stand on a hill or when we go in the upper layers of atmosphere, we leave a portion of air which cannot exert its full pressure. At sea level, a coastal town enjoys high pressure but on high altitude one will register a low pressure. For every 10 m of ascend, the pressure get reduced by 1 hPa.

c) Water vapour

The water vapour content is lighter in cold area than in air which is dry with the result that moist air ofa high temperature exerts a less pressure when compared to cold air.

d) Rotation of the earth

On account of rotation of the earth, the pressure at 60-70°N and S becomes low. The rotation of the earth near sub-polar belts, makes the air to escape from these belts which move towards the horse latitude (30° - 35°N and 30 – 35°S). These latitudes absorb the air from sub polar belts making the pressure high. G.D.Coriolis (1844) a French Mathematician indicated that air is deflected towards right in the Northern Hemisphere and Left in the Southern hemisphere due to rotation of earth and this was termed after him as Coriolis force. Coriolis force is not actually a force but it is effect created by rotation of earth.

Weather abberations

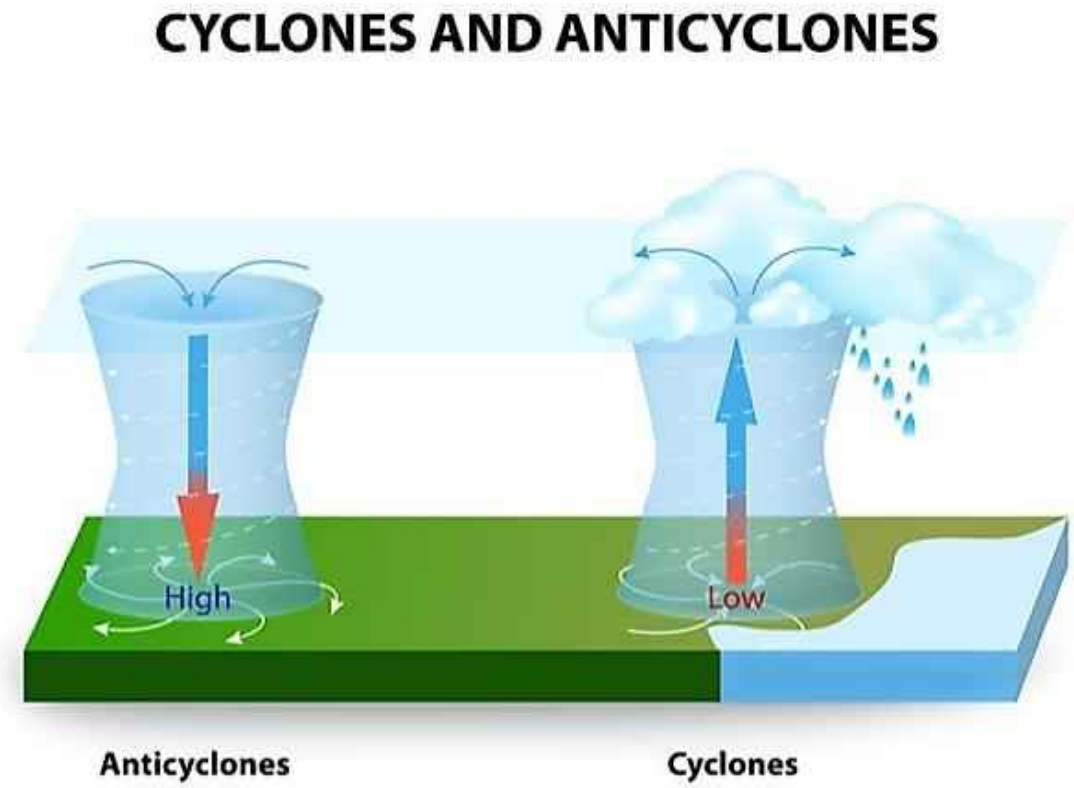
Low / Depression: When the isobars are circular or elliptical in shape, and the pressure is lowest at the centre, such a pressure system is called 'Low' or 'Depression' or 'Cyclone'. The movement will be anti-clockwise in the Northern hemisphere while it is clockwise in the southern hemisphere. Wind spped hardly exceeds 40 km per hour.

Anticyclone: When isobars are circular, elliptical in shape and the pressure is highest at the centre such a pressure system is called 'High' or 'Anticyclone'. When the isobars are elliptical rather than circular the system is called as 'Ridge' or 'Wedge'. The movement will be clockwise in the Northern hemisphere while it is anti-clockwise in the southern hemisphere.

Storm: Low pressure centre surrounded by winds having their velocities in the range of 40 to 120 km/hour. A more favourable atmosphere condition for their occurrence exists during the summer season. The Bay of Bengal and Arabian sea offer ideal condition for origin and growth of the storms. These storms produce heavy precipitation and bring about a change in the existing weather. It occurs very rarely. It causes wide spread damage.

Hurricane: A severe tropical cyclone with wind speed exceeding 120 km per hour. The name hurricane is given to the tropical cyclones in the North Atlantic and the eastern North Pacific Ocean. The tropical cyclones of Hurricane force in the western North pacific are known as typhoons. In Australia this type of storm is given the name willy-willy, whereas in the Indian Ocean they are called as Cyclones. Hurricanes are fueled by water vapour (i.e.) pushed up from

the warm ocean surface, so they can last longer and sometimes move much further over water than over land. A combination of heat and moisture along with the right wind conditions can create a new hurricane.



Hurricane



Thunderstorms: Storms produced by cumulonimbus clouds and always accompanied by lightening and thunder. They are usually of short duration, seldom over 2 hours. They are also accompanied by strong wind gusts, heavy rain and sometimes hail.

Tornadoes: Defined as a violently rotating column of air attended by a funnel-shaped or tubular cloud extending downward from the base of cumulonimbus cloud. Tornadoes are the most violent storms of lower troposphere. They are very small in size and of short duration. They mostly occur during spring and early summer. They have been reported at widely scattered locations in the mid latitudes and tropics. Crop losses are heavy due to this event. Unknown in other parts of the world.

Waterspouts: It is column of violently rotating air over water having a similarity to a dust devil of tornado. In other words, tornadoes are weak visible vortices occurring over water are called waterspouts. They are formed over tropical and subtropical oceans.

Tornadoes



Water spouts



Lect. 10. WIND – WIND SYSTEMS OF THE WORLD – INTER TROPICAL CONVERGENCE ZONES (ITCZ) – WIND SPEED IN DIFFERENT SEASONS – EFFECT OF WIND ON CROP PRODUCTION.

Wind: Air in horizontal motion is known as wind. Vertical movement is noticed but negligibly small compared to horizontal movement as the height of the atmosphere is only for few kilometers. However vertical movement or uplift of air only causes significant weather changes in cloud formation and rain.

Wind systems of the world

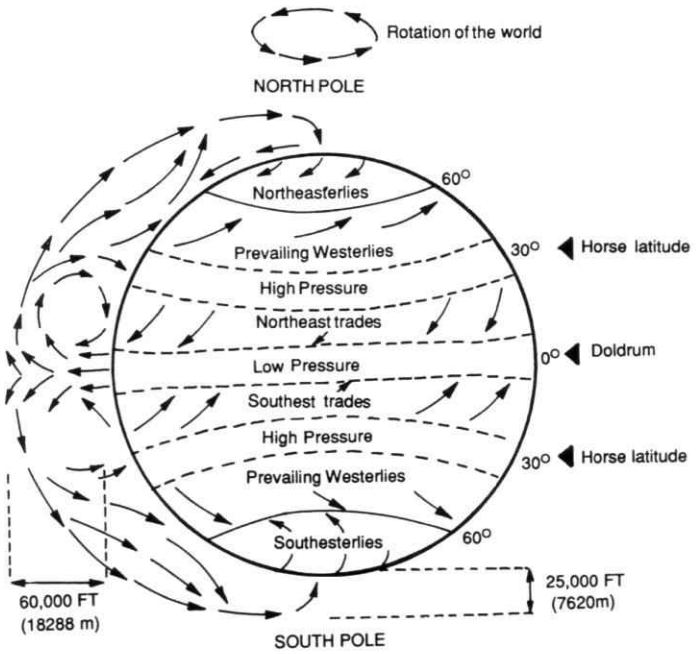
The wind belts found on earth's surface in each hemisphere are:

a. Doldrums
b. Trade wind belt
c. Prevailing westerlies
d. Polar easterlies

1. Doldrums

Owing to continuous heating of the earth by insolation, pressures are low and winds converge and rise near the equator. This intertropical convergent zone is known as 'Doldrums'.

a) These are the equatorial belts of calms and variable winds.
b) The location is 5°S and 5°N latitudes.
c) Wind is light due to negligible pressure gradient.
d) Mostly, there are vertical movements in the atmosphere.
e) The atmosphere is hot and sticky.



GLOBAL WIND CIRCULATION

2. Trade winds (Tropical Easterlies)

a) The regular high temperature at the equator results in a high pressure forming in the upper levels of the equator.
b) Then, the air is transferred to the northward and southward directions until 35° North and South in both the hemisphere.
c) Due to this reduction in surface pressure on the equator (doldrums) there is an increase in pressure at 35°N and 35°S which are known as horse latitude (sub-tropical high).
d) As a result, the winds flow from the horse latitude to the equatorial region.
e) While moving, these winds are deflected by Coriolis force to the right in northern hemisphere and to the left in southern hemisphere.
f) These winds flow from 35°N to the equator in NE direction in the northern hemisphere and from 35°S to the equator in SE direction in the southern hemisphere. These are known as 'Trade winds'. These are known as 'Tropical easterlies'.
g) These are most constant winds in force and direction and flow over nearly half the globe.

3. Anti-trade winds

a) This is a supplementary wind system of the earth which is effective at higher levels.
b) This system works in opposite direction to the surface winds.
c) The anti-trade winds mostly flow from land to ocean and brings no rain.

4. Prevailing Westerlies

a) The winds that flow from sub-tropical high to the low-pressure area about 60-70° latitudes in both the hemispheres are known as 'Prevailing westerlies'.
b) In the northern hemisphere the direction of Prevailing westerlies is SW and in southern hemisphere NW.

c) These winds are forceful and are irregular as compare to the trade winds in the tropical regions.
d) High precipitation zone

5. Polar Easterlies / Polar winds

b) A permanent high pressure exists on the poles.
c) From these high pressure polar regions, cold winds flow to areas at about 60-65° latitudes in both the hemispheres.
d) The winds flow in NE direction in the northern hemisphere and in SE direction in the southern hemisphere.

Mountain winds

a) Blows from mountain up slope to base
b) Occurs during night time
c) Cooling of air close to slope takes place
d) Adiabatic heating decreases this phenomenon
e) Also known as 'Katabalic winds'

Valley winds

a) Blow from valley base to up slope.
b) Occurs during day time
c) Over heating of air adjacent to slope takes place
d) Adiabatic cooling decreases this phenomenon
e) Also known as 'Anabatic winds'

Sea breeze

During the daytime, more so in summer, land is heated more than the adjacent body of water. As a result warmed air over the land expands producing an area of low pressure. The isobaric surfaces bend upward as a result of which the cooler air starts moving across the coast line from sea to land. This is the 'Sea breeze; or 'On shore breeze'.

Land breeze

At night because of nocturnal radiation land is colder than adjacent sea and the pressure gradient is directed from land to sea. There is a gentle flow of wind from land to sea. This 'off-share' wind is called 'Land breeze'.

Sl.	Sea Breeze	Land Breeze
1.	Occurs in day time	Occurs in night time
2.	Flows from sea	Flows from land
3.	Have more moisture than land breeze	Do not have more moisture
4.	Occurrence depends on topography of coast to grater extent	Occurrence depends on topography of land to little extent
5.	Modifies weather on hot summer afternoon	Produces cooler winters and warmer summers
6.	Stronger than land breeze	Weaker than sea breeze

Effect of wind on crop plants

1) Transports heat in either sensible or latent heat form from lower to higher altitudes
2) Wind affects the plant directly by increasing transpiration and the intake of CO_2 and also causes several types of mechanical damage.
3) Wind helps in pollination and dispersal of seeds.
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Wind speed in different season

Winds represent air in motion. The primary cause of all winds is regional differences in temperature, producing regional differences in pressure. When these pressure differences persist for several hours, the rotation of the earth modifies the direction of motion, till the winds blow along lines of equal pressure. Wind direction and speed are modified frequently due to seasonal variation in solar radiation and differential heating of the earth's surface.

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South West Monsoon wind direction: During South West Monsoon period of June to September, the westerly winds prevail on the west of Kerala and south winds on the west of northern Circars, Orissa and Bengal. During April and May the region of high temperature is shifted to north viz., upper Sind, lower Punjab and Western Rajasthan. This area becomes the minimum barometric pressure area to which monsoon winds are directed.

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Average wind speed for different seasons at Coimbatore.
Month wise wind speed (KMPH)

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CLOUDS AND THEIR CLASSIFICATION – PRECIPITATION – FORMS OF PRECIPITATION – ISOHYETS – MONSOON RAINFALL VARIABILITY – DROUGHT AND FLOOD - CONCEPTS OF CLOUD SEEDING.

Clouds

"An aggregation of minute drops of water suspended in the air at higher altitudes". The rising air currents tend to keep the clouds from falling to the ground.

WMO cloud classification

The World Meteorological Organisation (WMO) classified the clouds according to their height and appearance into 10 categories. From the height, clouds are grouped into 4 categories (viz., family A, B, C and D) as stated below and there are sub- categories in each of these main categories.

Family A

The clouds in this category are high. The mean lower level is 7 kilometers and the mean upper level is 12 kilometers in tropics and sub-tropics. In this family there are three sub-categories.

1. Cirrus (Ci)
   - ❖ In these clouns ice crystals are present.
   - ❖ Looks like wispy and feathery. Delicate, desist, white fibrous, and silky appearance.
   - ❖ Sun rays pass through these clouds and sunshine without shadow.
   - ❖ Does not produce precipitation.

2. Cirrocumulus (Cc)
   - ❖ Like cirrus clouds ice crystals are present in these clouds also.
   - ❖ Looks like rippled sand or waves of the sea shore.
   - ❖ White globular masses, transparent with no shading effect.
   - ❖ Meckerel sky.

3. Cirrostraturs (Cs)
   - ❖ Like the above two clouds ice crystals are present in these clouds also.
   - ❖ Looks like whitish veil and covers the entire sky with milky white appearance.
   - ❖ Produces "Halo".

Family B

The clouds in this category are middle clouds. The mean lower level is 2.5 kilometers and the mean upper level is 7 kilometers in tropics and sub-tropics. In this family there are 2 sub-categories as details below:

1. Altocumulus (Ac)
   - ❖ In these clouds ice water is present.
   - ❖ Greyish or bluish globular masses.
   - ❖ Looks like sheep back and also known as flock clouds or wool packed clouds.

2. Alto-stratus (As)
   - ❖ In these clouds water and ice are present separately.
   - ❖ Looks like fibrous veil or sheet and grey or bluish in colour.
   - ❖ Produces coronos and cast shadow.
   - ❖ Rain occurs in middle and high latitudes.

Family C

The clouds in this category are lower clouds. The height of these clouds extends from ground to upper level of 2.5 kilometers in tropics and sub-tropics. In this family, like high clouds, there are 3 sub-categorises.

1. Strato cumulus (Sc)

- These clouds are composed of water.
- Looks soft and grey, large globular masses and darker than altocumuls.
- Long parallel rolls pushed together or broken masses.
- The air is smooth above these clouds but strong updrafts occur below.

2. Stratus (St)

- These clouds are also composed of water.
- Looks like for as these clouds resemble grayish white sheet covering the entire portion of the sky (cloud near the ground).
- Mainly seen in winter season and occasional drizzle occurs.

3. Nimbostratus (Ns)

- These clouds are composed of water or ice crystals.
- Looks thick dark, grey and uniform layer which reduces the day light effectively.
- Gives steady precipitation.
- Sometimes looks like irregular, broken and shapeless sheet like.

Family D

These clouds form due to vertical development i.e., due to convection. The mean low level is 0.5 and means upper level goes up to 16 kilometers.

In this family two sub-categories are present.

1. Cumulus (Cu)

These clouds are composed of water with white majestic appearance with flat base.

Irregular dome shaped and looks like cauliflower with wool pack and dark appearance below due to shadow.

These clouds usually develop into cumuolo-nimbus clouds with flat base.

2. Cumulonimbus (Cb)

- The upper levels of these clouds possess ice and water is present at the lower levels.
- These clouds have thunder head with towering envil top and develop vertically.
- These clouds produces violent winds, thunder storms, hails and lightening, during summer.

Forms of Precipitation

Rain: It is precipitation of liquid water particles either in the form of drops having diameter greater than 0.5 mm or in the form of smaller widely scattered drops. When the precipitation process is very active, the lower air is moist and the clouds are very deep, rainfall is in the form of heavy downpour. On occasions, falling raindrops completely evaporate before reaching the ground.

Drizzle: It is fairly uniform precipitation composed of fine drops of water having diameter less than 0.5 mm small and uniform size and seems to be floated in the air, it is referred as drizzle. If the drops in a drizzle completely evaporates before reaching the ground, the condition is referred to as ‘mist’.

Snow: It is the precipitation of white and opaque grains of ice. Snow is the precipitation of solid water mainly in the form of branched hexagonal crystals of stars. In winter, when temperatures are below freezing in the whole atmosphere, the ice crystals falling from the Altostratus do not melt and reach the ground as snow.

Sleet: It refers to precipitation in the form of a mixture of rain and snow. It consists of small pellets of transparent ice, 5 mm or less in diameter. It refers to a frozen rain that forms when rain falling to the earth passing through a layer of cold air and freezes. This happens when temperature is very low. It is not commonly seen in India expect high ranges, that too in winter, in extreme north and northeast India.

Hail: Precipitation of small pieces of ice with diameter ranging from 5 to 50 mm or something more is known as hail. Hailstorms are frequent in tropics. In India, the period from March to May offers the ideal condition for hailstorms. It is the most dreaded and destructive form of precipitation produced in thunderstorms or cumulonimbus clouds.

Isohyets: Isohyets are the lines connecting various locations, having an equal amount of precipitation.

Monsoon Rainfall Variability

Indian continent receives its annual rainfall by the peculiar phenomenon known as monsoon. It consists of series of cyclones that arise in India Ocean. These travel in northeast direction and enter the Peninsular India along its west coast. The most important of these cyclones usually occur from June to September resulting in summer monsoon or southwest monsoon. This is followed by a second rainy season from October to December. A third and fourth rainy seasons occur from January to February and from March to May respectively. Of the four rainy seasons, southwest monsoon is the most important as it contribute 80 – 95% of the total rainfall of the country.

Two types of monsoon systems are a) South West Monsoon, b) North East Monsoon.

(a) South West Monsoon

Beginning of the year temperature of the Indian Peninsular rapidly rises under the increasing heat of the sun. A minimum barometric pressure is established in the interior parts of the Peninsular by the month of March. Westerly winds prevail on the west Kerala and south winds on the west of northern Circars, Orissa and Bengal. During April and May the region of high temperature is shifted to north viz., upper Sind, lower Punjab and Western Rajasthan. This area becomes the minimum barometric pressure area to which monsoon winds are directed.

The western branch of South West monsoon touches North Karnataka, Southern Maharastra and then it make its way to Gujarat. When the South West Monsoon is fully operating on the Western India, another branch of the same is acting in the Bay of Bengal. It carries rains to Burma, Northern portions of the east coast of India, Bengal, Assam and the whole of North India in general.

b) North east Monsoon

During September end, the South West Monsoon penetrates to North Western India but stays on for a full month in Bengal. On account of the increase in barometric pressure in Northern India, there is a shift of the barometric pressure to the South East and North Easterly winds begin to flow on the eastern coast. These changes bring on heavy and continuous rainfall to the Southern and South Eastern India.

c) Winter Rainfall

It is restricted more to Northern India and is received in the form of snow on the hills and as rains in the plains of Punjab, Rajasthan and central India. Western disturbance is a dominant factor for rainfall during these months in northwestern India.

d) Summer Rainfall:

The summer Rainfall is received from March to May as local storms. It is mostly received in the South East of Peninsular and in Bengal. Western India does not generally receive these rains.

Rainfall distribution in Tamilnadu in different seasons

Rainfall season	Quantity (mm)	Percentage share
South West monsoon	311.7	32
North East Monsoon	457.8	47
Cold weather period	48.7	5
Hot weather period	155.9	16

Drought: The term drought can be defined by several ways.

1. The condition under which crops fail to mature because of insufficient supply of water through rains.
2. The situation in which the amount of water required for transpiration and evaporation by crop plants in a defined area exceeds the amount of available moisture in the soil.
3. A situation of no precipitation in a rainy season for more than 15 days continuously. Such length of non-rainy days can also be called as dry spells.

Classification of Drought

Droughts are broadly divided into 3 categories based on the nature of impact and spatial extent.

i. Meteorological Drought

If annual rainfall is significantly short of certain level (75 per cent) of the climatologically expected normal rainfall over a wide area, then the situation is called meteorological drought. In every state each region receives certain amount of normal rainfall. This is the basis for planning the cropping pattern of that region or area.

ii. Hydrological drought

This is a situation in which the hydrological resources like streams, rivers, reservoirs, lakes, wells etc dry up because of marked depletion of surface water. The ground water table also depletes. The industry, power generation and other income generating major sources are affected. If Meteorological drought is significantly prolonged, the hydrological drought sets in.

iii. Agricultural Drought

This is a situation, which is a result of inadequate rainfall and followed by soil moisture deficit. As a result, the soil moisture falls short to meet the demands of the crops during its growth. Since, the soil moisture available to a crops insufficient, it affects growth and finally results in the reduction of yield. This is further classified as early season drought, mid season drought and late season drought.

Flood: Years in which actual rainfall is 'above' the normal by twice the mean deviation or more is defined as years of floods or excessive rainfall. Like droughts, the definition of floods also varies one situation to another and form one region to other. Some of the flood years characterized based on the spatial damage due to high and intense rainfall in India are as follows.

India: 1878,1872,1917,1933,1942,1956,1959,1961,1970,1975,1983,1988.

Principles of rainmaking: Clouds are classified into warm and cold clouds based on cloud top temperature. If the cloud temperature is positive these clouds are called warm clouds and if it is negative they are called as cold clouds. The nucleus needed for precipitation differs with type of clouds. Hygroscopic materials are necessary as nucleus for warm clouds

Cloud seeding: Cloud seeding is one of the tools to mitigate the effects of drought. It is defined as a process in which the precipitation is encouraged by injecting artificial condensation nuclei through aircrafts or suitable mechanism to induce rain from rain bearing cloud. The rain drops are several times heavier than cloud droplets. These mechanisms are different for cold and warm clouds.

Seeding of cold clouds

This can be achieved by two ways (1. Dry ice seeding and 2. Silver Iodide seeding).

1. Dry ice seeding

- Dry ice (solid carbon-dioxide) has certain specific features. It remains as it is at –80°C and evaporates, but does not melt. Dry ice is heavy and falls rapidly from top of cloud and has no persistent effects due to cloud seeding.
- Aircrafts are commonly used for cloud seeding with dry ice.
- Aircraft flies across the top of a cloud and 0.5 – 1.0 cm dry ice pellets are released in a steady stream.
- While falling through the cloud a sheet of ice crystals is formed.
- From these ice crystals rain occurs.

This method is not economical as 250 kg of dry ice is required for seeding one cloud. To carry the heavy dry ice over the top of clouds special aircrafts are required, which is an expensive process.

2. Silver Iodide seeding

Minute crystals of silver iodide produced in the form of smoke acts as efficient ice-farming nuclei at temperatures below –5°C. When these nuclei are produced from the ground generators, these particles are fine enough to diffuse with air currents. Silver iodide is the most effective nucleating substance because; its atomic arrangement is similar to that of ice. The time for silver iodide smoke released from ground generator to reach the super cooled clouds was offer some hours, during which it would draft a long way and decay under the sun light. The appropriate procedure for seeding cold clouds would be to release silver iodide smoke into super cooled cloud from an aircraft. In seeding cold clouds silver iodide technique is more useful than dry ice techniques, because, very much less of silver iodide is required per cloud. There is no necessity to fly to the top of the cloud, if area to be covered is large.

Seeding of warm clouds

1) Water drop Technique

Coalescence process is mainly responsible for growth of rain drops in warm cloud. The basic assumption is that the presence of comparatively large water droplets is necessary to initiate the coalescence process. So, water droplets or large hygroscopic nuclei are introduced in to the cloud. Water drops of 25 mm are sprayed from aircraft at the rate of 30 gallons per seeding on warm clouds.

2) Common salt technique

Common salt is a suitable seeding material for seeding warm clouds. It is used either in the form of 10 per cent solution or solid. A mixture of salt and soap avoid practical problems. The spraying is done by power sprayers and air compressors or even from ground generators. The balloon burst technique is also beneficial. In this case gun powder and sodium chloride are arranged to explode near cloud base dispersing salt particles.

Lect. 12. EVAPORATION – TRANSPIRATION, EVAPOTRANSPIRATION – POTENTIAL EVAPOTRANSPIRATION – DEFINITION AND THEIR IMPORTANCE IN AGRICULTURAL PRODUCTION.

Evaporation: A physical process in which liquid water is converted into its vapour. Evaporation is the most important water loss term in water balance equation.

Importance of Evaporation in crop plants

1) Evaporation is an important process of hydrologic cycle.
2) The evaporation from the soil is an important factor deciding the irrigation water requirements of a crop
3) In modifying the microclimate of a crop the evaporation from the soils is an important factor for consideration.
4) Evaporation is the most important of all the factors in the heat budget, after radiation.
5) The evaporation is also one of the most important factors in the water economy.
6) Since, a certain amount of evaporation also demands a definite amount of heat, it provides a link between water budget and heat budget.

Factors affecting evaporation

The evaporation from a fully exposed water surface is the function of several environmental factors

1. Environmental factors
   a. Water temperature
      With an increase of temperature the kinetic energy of water molecules increases and surface tension decreases which increases evaporation.
   b. Wind
      The evaporation from fully exposed surface is directly proportional to the velocity of wind and vice-versa, because dry wind replaces the moist air near water. The process of evaporation takes place continuously when there is a supply of energy to provide latent heat of evaporation (540 calories / gram of water).
   c. Relative humidity
      The evaporation is greater at low RH than at high RH.
   d. Pressure
      The evaporation is more at low pressure and less at high pressure.

2. Water factors
   a. Composition of water
      The dissolved salts and other impurities decreases the rate of evaporation. The evaporation is inversely proportional to the salinity of water.
   b. Area of evaporation
      The larger the area of exposure, greater will be the evaporation.

Transpiration: This is a physiological phenomenon, which takes place only in living plants. The loss of water from living parts of the plant is known as Transpiration. The loss of water through stomatal openings of the leaves is termed as stomatal transpiration. The loss of water through cuticle is known as cuticular transpiration and from lenticels is known as lenticular transpiration.

Importance of transpiration on crop plants

1. Dissipation of radiant energy by plant parts
2. Translocation of water in the plants
3. Translocation of minerals in the plant

Factors affecting transpiration

I. Environmental factors

1. Light: By directly opening and closing the of stomata there is periodicity in the transpiration rate. Indirectly by increasing the temperature of leaf cells the transpiration is increased.

2. Atmospheric humidity: The rate of transpiration is almost inversely proportional to atmospheric humidity.
3. Air Temperature: Increase in Temperature results in opening of stomata which in turn increases transpiration.
4. Wind velocity: The higher the wind speed higher the transpiration

II. Plant factors

1. Plant height: Water need of the crop varies with height.
2. Leaf characteristics: Reduction in leaf area brings reduction in transpiration.
3. Availability of water to the plant: If there is little water in the soil the tendency for dehydration of leaf causes stomatal closure and a consequent fall in transpiration.

Differences between evaporation and transpiration

Sl	Evaporation	Transpiration
1.	Controlled by meteorological factors	Controlled by both meteorological and plant factors
2.	Diffusive resistant is absent	Diffusive resistance occurs due to internal leaf geometry and presence of stomata.
3.	Also occurs in night due to advective heat transportation.	Reduced in the night due to closure of stomata.
4.	This is purely a physical phenomenon which takes place from any exposed surface with moisture.	This is physiological phenomenon which takes place only in living plants.
5.	This takes place through any openings or pores	This takes place through guard cells of stomata, cuticle and lenticules etc.

Evapotranspiration: Evapotranspiration denotes the quantity of water transpired by plants or retained in the plant tissue plus the moisture evaporated from the surface of the soil. As long as the rate of root uptake of soil moisture balances the water losses from the canopy, evapotranspiration continues to occur at its potential rate. When the rate of root water uptake falls below the transpiration demand, actual transpiration begins to fall below the potential rate. This is either because the soil cannot supply water to roots quickly or the plant can no longer extract water to meet the evaporational demand.

Reference Evapotranspiration (ET_0): This represents the maximum rate of evapotranspiration from an extended surface of 8 to 10 centimeters tall green grass cover, actually growing and completely shading the ground under limited supply of water.

Potential Evapotranspiration (PET): Potential evapotranspiration (PET) for any crop is obtained from reference evapotranspiration and crop factors (Kc) when water supply is unlimited.

$$PET = Kc \times ET_0$$

Importance of Evapotranspiration and Potential Evapotranspiration for crop plants

1. Estimation of the soil moisture there by planning irrigation schedule of crops.
2. Understanding relationship between the crop yield and irrigation water.
3. Guiding for the production of a crop with a fully developed canopy.
4. The evapotranspiration can also help to demarcate soil climatic zones including the drought prone areas.
5. These will form the base for developing suitable soil and crop management practices, crop varieties, water conservation techniques, cropping pattern and ways to improve productivity rainfed crops.

AGROCLIMATIC ZONES OF TAMIL NADU

Based on rainfall distribution, irrigation pattern, soil characteristics, cropping pattern and other ecological, social and physical status, the State of Tamil Nadu could be classified into seven distinct agro-climatic zones listed below.

1. North Eastern Zone
2. North Western Zone
3. Western Zone
4. Cauvery Delta Zone
5. Southern Zone
6. High Rainfall Zone
7. Hilly Zone

1. North Eastern Zone

This zone covers the districts of Thiruvallur, Vellore, Chinglepattu, Thiruvannamalai, Viluppuram, Cuddalore (excluding Chidambaram and Kattumannarkoil taluks), some parts of Perambalur including Ariyalur taluks and also Chennai. The mean annual rainfall of this region is 1054 mm received in 53 rainy days and is benefited by both the monsoons. The mean monthly maximum temperature ranges between 28.2 to 38.9 °C and theminimum ranges from 19.5 to 24.8 °C.

2. North Western Zone

This zone comprises of Dharmapuri district (excluding hilly areas), Salem, Namakkal district (excluding Tiruchengode taluk) and Perambalur taluk of Perambulur district. The climate prevailing in this region is dry and sub humid. This region has been identified as moderately drought prone area. The elevation varies from 330 to 1070 metres above mean sea level. The mean annual rainfall of this region is appreciably lower than in North Eastern Zone and is 825 mm received in 47 rainy days. The region is benefited by both south-west and north-east monsoon rains but unlike the NEZ, the former contributed more to the total rainfall. The mean monthly maximum temperature ranges between 30 to 37 °C with minimum temperature ranging between 19 to 25.5 °C. The annual PET of this region is 1727 mm compared to the annual precipitation of 825 mm.

3. Western Zone

This zone comprises of Erode, Coimbatore, Dindugal, Theni districts, Tiruchengode taluk of Namakkal district, Karur taluk of Karur district and some western part of Madurai district. The mean annual rainfall is 718 mm in 45 rainy days. The monthly mean maximum temperature is 35°C in April and 30°C in January and November. The monthly mean minimum temperature is 19°C in January and 24°C in May.

4. Cauvery Delta Zone

This zone comprises the Cauvery Delta area in Thanjavur, Thiruvarur, Nagapattinam districts and Musiri, Tiruchirapalli, Lalgudi, Thuraiyur and Kulithalai taluks of Tiruchirapalli district, Aranthangi taluk of Pudukottai district and Chidambaram and Kattumannarkoil taluks of Cuddalore district. The mean annual rainfall of the zone is 1078 mm out of which 40.0mm is received during winter, 69.2mm during summer, 295.4mm during South West Monsoon and 673.8mm during North East Monsoon.

5. Southern Zone

This is the biggest among the seven zones of Tamil Nadu. It is typical zone surrounded by coastal areas on the East and mountains in the West. This zone comprises Sivagangai, Ramanathapuram, Virudunagar, Tuticorin and Tirunelveli districts and Natham and Dindigul taluks of Dindigul district, Melur, Tirumangalam, Madurai South and Madurai North taluks of Madurai district and Pudukkottai district excluding Aranthangi taluk. This zone lies on the Southern part of the State under rain shadow area. Because of this, the area is prone to drought very often. The climate is semi-arid tropics. The elevation varies from mean sea level to 300 metres. The mean annual rainfall is 776 mm received in 43 rainy days. The monthly mean maximum temperature in this region ranges from 28.5° C in December to 38.5 ° C in June. The monthly mean minimum temperature varies from 21.0° C in January to 27.5° C in June.

6. High Rainfall Zone

This zone consists of Kanayakumari district. This district situated in the southern most part of the

Peninsular India, with its high rainfall having a climate which is entirely different from the rest of the state. The climate is monsoon tropics and there is seasonal in shores flow of moist air. The elevation ranges from sea level to about 600 m. The mean annual rainfall of the district is 1469 mm received in 64 rainy days. There is not much fluctuation in the mean monthly air temperature. The monthly mean maximum temperature varies from 28.0°C in December to 33.5°C in May. The monthly mean minimum temperature varies from 22°C in December to 26.5°C in May.

7. High Altitude and Hilly Zone

This zone comprises the hilly regions, namely the Nilgiris, Shevroys, Elagiri-Javvadhu, Kollimalai, Patchaimalai, Anamalais, Palanis and Podhigaimalais. The rainfall varies from 850 mm in Kalrayan hills to about 4500 mm in Anamalai hills.

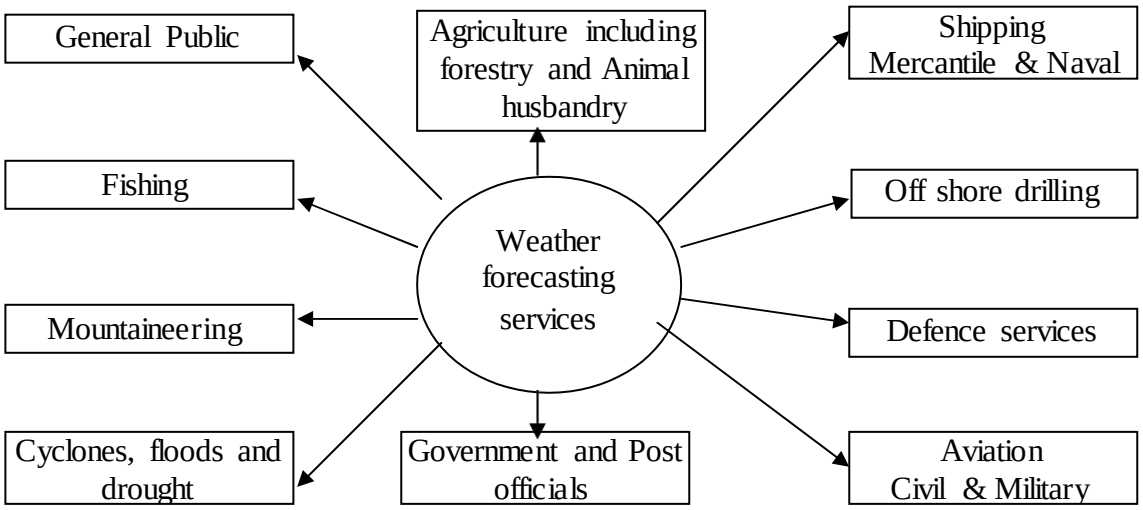
LEC . 13. WEATHER FORECASTING–TYPES, IMPORTANCE – SYNOPTIC CHART – CROP WEATHER CALENDAR, AGROCLIMATIC NORMALS,

Climatic normals

The climatic normals are the average value of 30 years of a particular weather element. The period may be week, month and year. The crop distribution, production and productivity depend on the climatic normals of a place. If the crops are selected for cultivation based on the optimum climatic requirements it is likely that the crop production can be maximized.

Weather forecast

The prediction of weather for the next few days to follow. The Figure below depicts different weather forecasting services normally practiced in a country.



NEED / IMPORTANCE OF FORECAST

- Basically weather has many social and economic impacts in a place.
- Among different factors that influence crop production, weather plays a decisive role as aberrations in it alone explains up to 50 per cent variations in crop production
- The rainfall is the most important among the required forecast, which decides the crop production in a region and ultimately the country's economy.
- The planning for moisture conservation under weak monsoon condition and for flood relief under strong monsoon condition is important in a region.
- A reliable weather forecasting when disseminated appropriately will pave way for the effective sustainability.
- One can minimize the damage, which may be caused directly or indirectly by unfavourable weather.
- The recurring crop losses can be minimized if reliable forecast on incidence of pest and diseases is given timely based on weather variables.

- ❖ Help in holding the food grain prices in check through buffer stock operations. This means that in good monsoon years when prices fall, the government may step in and buy and in bad years when price tend to rise, it may unload a part of what it had purchased.
- ❖ Judicious use of water can be planned in a region depending up on the forecast.

Type of weather forecast

	Types of forecast	Validity period	Main users	Predictions
1	Short range a) Now casting	Up to 72 hours 0-2 hours	Farmers marine agencies, general public	Rainfall distribution, heavy rainfall, heat and cold wave conditions, thunder storms etc.
	b) Very short range	0-12 hours		
2	Medium range	Beyond 3 days and upto 10 days.	Farmers	Occurrence of rainfall, Temperature.
3	Long range	Beyond 10 days upto a month and a season.	Planners	This forecasting is provided for Indian monsoon rainfall. The out looks are usually expressed in the form of expected deviation from normal condition.

Synoptic charts

An enormous volume of meteorological data is being collected from all over the world continuously round the clock through various telecommunication channels. To assess, assimilate and analyse the vast data, they have to be suitably presented. For this purpose, the observations are plotted on maps in standard weather codes. These maps are called 'Synoptic maps or charts'.

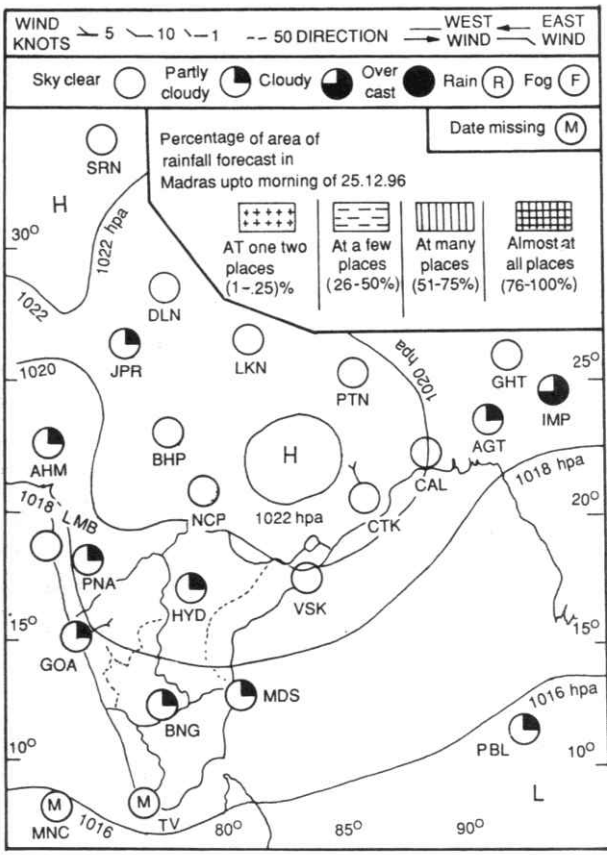
Synoptic charts display the weather conditions at a specified time over a large geographical area. The surface synoptic charts plotted for different synoptic hours (00, 03, 06, 09, 12, 15, 18, 21 UTC) depict the distribution of pressure, temperature, dew point, clouds, winds, present and past weather. In place of GMT, UTC (Universal Time Co-ordinate) is used. The upper air charts are also prepared at the standard pressure levels of the atmosphere (different heights) of the atmosphere wherein the pressure, wind and temperature are plotted. The surface charts together with the upper air charts provide a composite three-dimensional weather picture pertaining to a given time. Thus it gives a birds eye view of the state of atmosphere at a time over a large area and is a important tool used by operational meteorologists and scientists.

The surface synoptic charts are the most used charts. It contains the maximum number of observations with the largest number of parameters plotted and often forms the base on which the pressure level charts are built up. The pattern of the pressure distribution is brought out by drawing isobars, troughs, ridges, lows, highs, depressions, cyclones, cols, fronts and discontinuities. These systems are clearly marked and labeled using appropriate symbols and colours.

In synoptic charts different weather phenomena and atmospheric characters are marked with different symbols as mentioned below.

S.No	Symbols	Weather element/character/phenomenon
1.	Narrow black lines	Isobars
2.	Numbers at ends of isobars	Pressure values in hPa
3.	Shading	Precipitation
4.	Arrows	Wind direction
5.	Feathers in the arrows	Wind velocity
6.	Small circles with shading	Amount of clouds

In addition to the above, different symbols are used for recording weather phenomena.



Sample daily weather report

	SQUALL		DEW		SHOWERS OF LIGHT SNOW
	GALE		FROST		CONTINUOUS HEAVY SNOW
	DUST STORM		SOLAR HALO		SOFT HAIL
	DUST DEVIL		LUNAR HALO		HAIL
	LIGHTNING		RAINBOW		MOIST HAZE
	THUNDERSTORM		LIGHT THUNDERSTORM WITH RAIN		SEVERE DUST STORM
	SHOWERS		SHOWER OF LIGHT RAIN		INDUSTRIAL SMOKE
,	DRIZZLE		INTERMITTENT MODERATE DRIZZLE		SOLAR CORONA
,,	CONTINUOUS LIGHT DRIZZLE		CONTINUOUS MODERATE DRIZZLE		CLEAR SKY
•	RAIN		CONTINUOUS HEAVY RAIN		FOG
••	CONTINUOUS LIGHT RAIN		CONTINUOUS LIGHT SNOW		MIST

SYMBOLS FOR RECORDING WEATHER PHENOMENA

Weather calendar

In order to provide the farmers with an efficient weather service, it is essential that the weather forecaster should be familiar with the crops that are grown in a particular agroclimatic zone. The type of forewarnings to be given depend on the stages of the crop. In case of farmers, they should become familiar with weather bulletins and learn how to interpret. To meet the above requirement, the detailed information collected from the agricultural departments has been condensed by the IMD and presented in a pictorial form known as crop weather calendar. This calendar has three parts as follows.

a) Bottom part
b) Middle part
c) Top part

a) Bottom part provides the activities related to crop or information related to phenological stages of the crop and the months.

b) Middle part gives information regarding normal weather condition required for active crop growth. It is divided into different sections according to rainfall, rainy days, minimum temperature, maximum temperature, pan evaporation and sunshine hours.

c) Top part gives information related to the weather abnormalities or to take precautionary measures. Top part is divided into different sections according to dry spell length, high wind, heavy rainfall and cloudy weather.

CROP WEATHER CALENDAR

STATE : TAMILNADU **CROP** : COTTON **VARIETY** : LRA-5166, MCU-7 **IRRIGATED**

DISTRICTS : SOUTH-ARCOT **DURATION** : 150 DAYS **SOIL** : RED LOAM

Weather warnings	*Rain*	> 100 MM/DAY		
	Duration of wet spell	> 30 MM FOR 10 DAYS		
	Cloudy weather			
	Drought	40 DAYS	35 DAYS	35 DAYS
	High winds			> 30 KM/HOUR
	Temperature			
	Hail storm			
Weather conditions favourable for incidence of pests and diseases	*Pests*	STEM WEEVIL, BOLL WORMS	STEM WEEVILS, BOLL WORMS	STEM WEEVILS, BOLL WORMS
	Weather			
	Diseases	WILT	WILT	
	Weather			

| Normal phase wise water requirement (mm) | | | |

WEATHER NORMALS FOR AGRICULTURAL CROPS

Sl. No.	Crops	Optimum Temperature ° C		Day length	Rainfall (mm)	Altitude above MSL (m)
		Germination	Growth stage			
1	Rice	Min 10 ° C	22-25 (flowering) 20-21(grain formn) 20-25(ripening)		1500	<3000
2	Maize	35-44 ° C				
3	Sorghum	7-10	25-30	Short day		
4	Pearl millet		28-32		400-750	
5	Finger millet				500-1000	
6	Kodo millet				400-500	
7	Wheat	20-22	16-22		250-1800	<3500
8	Barley		12-15 (growth) 30(reproduction)	Long day	400-500	
9	Oats	15-25			380-1140	
10	Ground nut	27-30	24-27		500-1250	
11	Sesame	25-27		Short day	500-650	<1250
12	Castor	20-26		Long day	500-600	<3000
13	Sunflower	20-25			500-700	<2500
14	Rape seed and Mustard	18-25		Long day	300-400	
15	Safflower	15-16	25-30	Day neutral	600-900	
16	Soybean	15-32	30-33		600-650	1200-2000
17	Pigeon pea	20-30				
18	Green gram	15	20-40	Short day	600-1000	
19	Black gram					1500
20	Cow pea	12-15	21-35	Short day	600	
21	Bengal gram		15-25		600-1000	
22	Cotton	18	21-27	Day neutral	500	
23	Jute		27-40	Short day	1500	
24	Tobacco	28	25-35		500-1000	
25	Sugar cane		24-30	Long day	2000-2500	
26	Sugar beet	12-15	22-30	Long day		
27	Potato	18-20	18-20			

Lect. 14. **DEFINITION AND USES OF REMOTE SENSING AND CROP WEATHER MODELING – CLIMATE CHANGE AND VARIABILITY – EL- NINO, LA- NINA**

Definition: Remote sensing is defined as the art and science of gathering information about objects or areas from a distance without having physical contact with objects area being investigated.

Uses: Remote sensing techniques are used in agricultural and allied fields.

1. Collection of basic data for monitoring of crop growth
2. Estimating the cropped area
3. Forecasting the crop production
4. Mapping of wastelands
5. Drought monitoring and its assessment
6. Flood mapping and damage assessment
7. Land use/cover mapping and area under forest coverage
8. Soil mapping
9. Assessing soil moisture condition, irrigation, drainage
10. Assessing outbreak of pest and disease
11. Ground water exploration

Remote Sensing platforms:

Three platforms are generally used for remote sensing techniques. They are ground based, air based and satellite based. Infrared thermometer, Spectral radiometer, Pilot-Balloons and Radars are some of the ground based remote sensing tools while aircrafts air based remote sensing tools. Since the ground based and air based platforms are very costly and have limited use, space based satellite technology has become handy for wider application of remote sensing techniques. The digital image processing, using powerful computers, is the key tool for analyzing and interpretation of remotely sensed data. The advantages of satellite remote sensing are:

- Synoptic view – Wide area can be covered by a single image/photo (One scene of Indian Remote Sensing Satellite IRS series cover about 148 x 178 sq.km area).
- Receptivity – Can get the data of any area repeatedly (IRS series cover the same area every 16-22 days).

Coverage – Inaccessible areas like mountains, swampy areas and thick forests are easily covered.

Space based remote sensing is the process of obtaining information about the earth from the instruments mounted on the Earth Observation Satellites. The satellites are subdivided into two classes and the types of satellite are as follows:

Polar orbiting satellites: These satellites operate at an altitude between 550 and 1,600 km along an inclined circular plane over the poles. These satellites are used for remote sensing purposes. LANDSAT (USA), SPOT (FRANCE), and IRS (INDIA) are some of the Remote Sensing Satellites.

Geostationary satellite: These have orbits around the equator at an altitude of 36,000 km and move with the same speed as the earth so as to view the same area on the earth continuously. They are used for telecommunication and weather forecasting purposes. INSAT series are launched from India for the above purposes. All these satellites have sensors on board operating in the visible and near infrared regions of the electromagnetic spectrum. INSAT-3A was launched on 10th April, 2003.

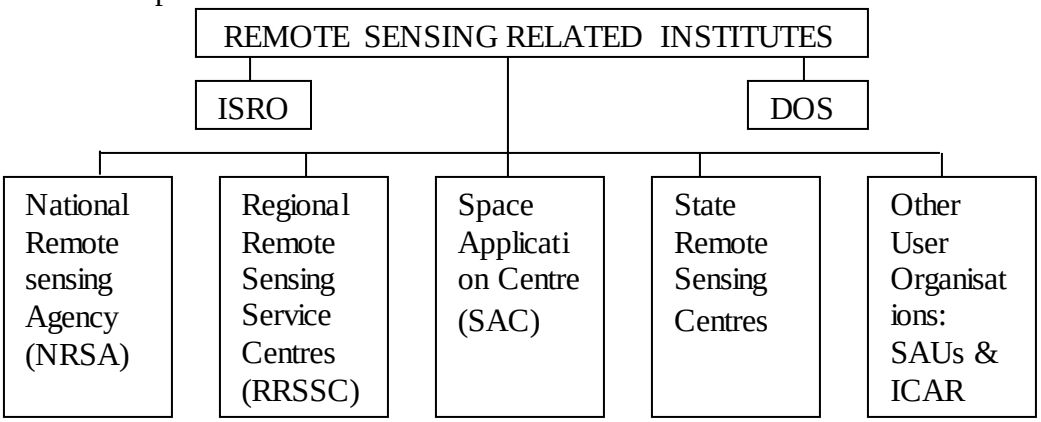
Role of Remote Sensing in agriculture

Agricultural resources are important renewable dynamic natural resources. In India, agriculture sector alone sustains the livelihood of around 70 percent of the population and contributes nearly 35 percent of the net national product. Increasing agricultural productivity has been the main concern since scope for increasing area under agriculture is rather limited. This demands judicious and optimal management of both land and water resources. Hence, comprehensive and reliable information on land use/cover, forest area,

soils, geological information, extent of wastelands, agricultural crops, water resources both surface and underground and hazards/natural calamities like drought and floods is required. Season-wise information on crops, their acreage, vigour and production enables the country to adopt suitable measures to meet shortages, if any, and implement proper support and procurement policies. Remote Sensing systems, having capability of providing regular, synoptic, multi-temporal and multi-spectral coverage of the country, are playing an important role in providing such information. A large number of experiments have been carried out in developing techniques for extracting agriculture related information from ground borne, air borne and space borne data.

Indian Remote Sensing programme:

India, with the experience gained from its experimental remote sensing satellite missions BHASKARA-I and II, has now established satellite based operational remote sensing system in the country with the launch of Indian Remote Sensing Satellite IRS-IA in 1988, followed by IRS-IB (1992), IRS-IC (1995) and IRS-ID (1997). The Department of Space (DOS) / Indian Space Research Organisation (ISRO) as the nodal agency for establishing an operation remote sensing system in the country initiated efforts in the early 1970s for assessing the potentials of remotely sensed data through several means. In order to meet the user requirement of remote sensing data analysis and interpretation, ISRO/DOS has set up a system to launch remote sensing satellites once in three or four years to maintain the continuity in data collection. The remote sensing and some of its related institutes are depicted.



Crop weather modeling

Crop model: It is a representation of a crop through mathematical equations explaining the crops interaction with both above ground and below ground environment.

The increase in dry matter of the crop is referred to as growth. The rate of growth of a healthy crop depends on the rate at which radiation is intercepted by foliage and / or on the rate at which water and nutrients are captured by root systems and therefore on the distribution of water and nutrients in the soil profile. The crop development is described in terms of various phenophases through which the crop completes its lifecycle. That is the progress of the crop from seeding or primordial initiation to maturity. Finally the yield of crop stand is expresses as a product of three components, viz., the period over which dry matter is accumulated (the length of the growing period), the mean rate at which dry matter is accumulated and the fraction of dry matter treated as yield when the crop is harvested.

It is understood that the crop growth, development and yield depend upon the mean daily temperature (DTT), the length of the day and the amount of solar radiation (PAR) received by the crop.

$$DTT = \frac{\text{Max daily temperature} + \text{Min daily temperature}}{2} - \text{base temperature}$$

Where, DTT = Daily thermal time accumulation.

The time needed for the crop to reach a development stage depends upon temperature measured above a base value (DTT) and for photo periodically sensitive phases such as flowering, the

day length above a fixed base. In the absence of stress, the harvest index does not vary much from year to year for a specified cultivar / variety. Therefore, crop weather modeling is based on the principles that govern the development of crop and its growing period based on temperature and / or day length. They are used to quantify the rate of crop growth in terms of radiation interception, water use and nutrient supply which moderate harvest index when the crops experience stress condition.

The basic information required to be generated for crop weather modeling includes.

a) Crop phonology in relation to the temperature and day length
b) Water use by the crop during different phenophases of crop growth
c) The relationship between radiation interception, crop water use and total dry matter production
d) Partitioning of dry matter into various plant components as influenced by water and nutrient availability, and
e) The effect of weather parameters on biotic interference to crop growth.

Types of models

a) Statistical models
These models express the relationship between the yield or yield components and the weather parameters. The relationships are measured in a system using statistical techniques. Simple regression techniques explaining weather crop relationships are also considered as models.

b) Mechanistic model
These models explain not only the relationships between the weather parameters and the yield, but explain the relationship of influencing dependent variables.

c) Deterministic models
These models estimate the exact value of the yield or dependent variable. These models also have defined co-efficient.

d) Stochastic models
A probability element is attached to each output. For each set of inputs different outputs are given along with probabilities. These models define the yield or state of dependent variable at a given rate.

e) Dynamic models
Time is included as a variable. Both dependent and independent variables are having values which remain constant over a given period of time. Over a period of time these variables are changing due to change in rate of increment.

f) Static models
Time is not included as a variable. The dependent and independent variables having values remain constant over a given period of time.

g) Simulation models
Computer models in general, are a mathematical representation of a real world system. One of the main goals of crop simulation models is to estimate agricultural production as a function of weather and soil conditions as well as crop management. These models use one or more sets of differential equations over time, normally from planting until harvest maturity or final harvest.

h) Descriptive models
A descriptive model defines the behaviour of a system in a simple manner. The model reflects little or none of the mechanisms that are the causes of phenomena but consists of one or more mathematical equations. An example of such an equation is the one derived from successively measured weights quickly the weight of the crop where no observation was made.

i) Explanatory models
This model consists of quantitative description of the mechanisms and process that cause the behaviour of the system. To create this model, a system is analyzed and its process and mechanisms are quantified separately. The model is built by integrating these descriptions for the entire system. It

contains descriptions of distinct processes such as leaf area expansion, tiller production etc. Crop growth is a consequence of these processes.

Lect. 16.– EFFECT OF WEATHER AND CLIMATE ON CROP PRODUCTION, SOIL FERTILITY AND INCIDENCE OF PEST AND DISEASE.

Different season of India

Based on the rainfall pattern and temperature distribution, the India Meteorological Department, Government of India has divided the whole year into four seasons viz.

1. South west monsoon (June – September)
2. Post Monsoon (October – November)
3. Winter (December – February)
4. Summer or Pre monsoon (March / April – May / June)

The above seasons coincide with the agricultural seasons viz.

1) Kharif
2) Rabi and
3) Summer

The Kharif season is nothing but the Southwest monsoon or autumn. The Rabi coincides with post monsoon and winter seasons. In summer, the cultivable land under seasonal crops is kept in many regions of the country. Wherever water is plenty, vegetables are grown in some parts of the country during summer. Spring falls between January and March. Based on temperatures, ranges, there are three distinct crops seasons in India. They are

1) Hot weather (Mid February – Mid June)
2) Kharif or rainy season (Mid June – Mid October)
3) Rabi (Mid October – Mid February)

In Tamil Nadu there is a slight variation in the seasons based on rainfall duration as

2) Winter – January and February
3) Summer – March to May
4) Rainy seasons
   a) South west monsoon (June – September)
   b) North east monsoon (October – December)

The criteria for division of growing seasons are broadly based on monthly precipitation and temperature. The pattern followed as

a) Hot month – if the average temperature is above 20°C
b) Cold month – if the mean temperature is between 0-10°C
c) Warm month – if the mean temperature is 15-20°C

Effect of weather and climate on Incidence of pest and disease

Considerable crop losses caused due to pests and diseases in the humid and sub humid tropics. Many of the restrictions on productivity and geographical distribution of plants and animals are imposed by pests and diseases. The geographical distribution of pests is mainly based on climatic factors. The climatic conditions show a gradient from place to place and there is a related gradient in the abundance of a particular pest / disease. The periodic or seasonal nature of incidence and out breaks of several pests and diseases of many crops can be ascribed to weather conditions as the triggering factors. These epidemics of diseases are principally weather dependent, either in terms of local weather conditions being favourable for growth and development of the casual organisms or the prevailing winds helping to disseminate airborne pathogens or spores of diseases such as mildew, rusts, scabs and blights.

The migration and dispersion of insect pests depend on the wind speed and direction besides the nature of air currents. Some plant pathogenic viruses suitable for the development of these vectors favour the transmission of such diseases. A surfeit of pests and diseases, which infest plants are kept in chock by seasonal fluctuations in atmospheric temperature or relative humidity and other weather factors. Insect pest outbreaks occur as a result of congenial weather conditions, which facilitate their un-interrupted multiplication. The weather and climate greatly influence the quantity and quality of food provided by the host crops to the associated species of pests. The abundance or otherwise of the pestiferous species is thus dependent on climatic conditions, indirectly also.

The surface air temperature, relative humidity, dew fall, sunshine, cloud amount, wind, rainfall and their pattern and distribution are the primary weather factors influencing the incidence or

outbreaks of pests and diseases of crops. In the humid tropics, the weather variables namely air temperature, intermittent rainfall, cloudy weather and dewfall may play a crucial role in the outbreaks of pests and diseases. The impact of various weather components on pests and diseases is experienced in a location and crop specific manner.

Among the major pests associated with crops, insect, mite and nematode species are of a serious nature in terms of their abundance and damage potential. If the occurrence of pest / disease in time and space can be predicted in advance with reasonable accuracy on the basis of relevant weather parameters, appropriate and timely control measures can be programmed. Appropriate insecticide / fungicide interventions can certainly reduce the pesticide load in the environment and the related pollution and health hazards.

Effect of weather and climate on Soil Fertility

Red Soil (Alfisol)

Red soils are agriculturally important found in major portion of drylands. They are generally low in organic matter, available N and P. Soil pH ranges from 5.8 to 6.7. Rooting depth of crops is limited by the presence of compact subsoil. Many crops are susceptible to even moderate droughts. These soils are having the characteristics of rapidly sealing the surface after the rainfall. Water supply in the soil is reduced by limited infiltration due to lower conductivity. Problem of crusting affects the crop establishment. Soil moisture deficit is the major factor related to rainfall climatology affecting crop production. Soil erosion is the major factor reducing the fertility due to highly variable seasonal rainfall pattern.

Black Soil (Vertisol)

Black soils in India cover an area of about 72.9 m. ha which accounts for 22.2% of the total geographical area of the country. These are generally rainfed and experience considerable fluctuations in crop production due to climate variability. The clay content of the soil ranges from 40 to 60%, occasionally going to as high as 80%. Organic carbon content remains low ranging from 0.3 to 0.7%, pH of the soil normally ranges from 7.5 to 8.6%. The cation exchange capacity is 35-50 meq/100g. Inversion takes unique feature of the vertisols or deep black soils. Vertisols are necessarily deep. The soils invariably have wide and deep shrinkage cracks on the surface that changes with variation in the soil moisture regime. The cracks remain open depending on soil moisture and evaporation. The deep black soils, because of their high clay content, expanding nature of clay and depth, have a very high water holding capacity enabling crops to with stand drought at different stages of the crop.

Laterite Soils

The texture of the topsoil is loamy or clayey with many concretions. Laterite soils are generally associated with undulating topography in regions with a relatively high annual rainfall. These soils cover 13 m. ha in India. These soils are mostly dominated in hilly and high rainfall regions and slightly acidic in nature due to leaching of bases. They are rich in Iron and Alluminium.

Alluvial soil

They are generally loamy sands or sandy loams, very deep with moderate clay content. These soils are having firmly high water holding capacity. The drainage characteristics are highly varying. Water stagnation is the major problem affecting crop productivity during heavy rainfall seasons.

Sierozemic soil

These soils are sand, loamy sand and sandy loam in texture. Soil erosion through winds is common. Since these soils are light textured, water and nutrient holding capacity is less. Sub-soil Salinity is common due to extreme aridity. Kharif or Rabi cropping is possible in deep soils. But in loamy sands and sand, only the Kharif crops can be raised.

Submontane Soil

The soils are silty loam in texture and are medium to deep. Landslides and soil erosion are common. High rainfall lead to heavy soil erosion and major portion of top fertile soils are lost.

Lec 16. Climate change, climate variability - definition and causes of climate change.

Climate change: Any permanent change in weather phenomena from the normals of a long period average is referred as climate change. Eg. The global temperature has increased by 2.0 to 3.0 C and increase in CO2 from 180ppm to 350ppm.

Climate variability: The temporal changes in weather phenomena which is part of general circulation of atmosphere and occurs on a yearly basis on a global scale.

Climate change and climate variability are the concern of human kind in recent decades all over the world. The recurrent drought and desertification seriously threaten the livelihood of over 1-2 billion people who depend on the land for most of their needs. The weather related disasters viz. drought and floods, ice storms, dust storms, land slides, thunder clouds associated with lightening and forest fires are uncommon over one or other region of the world. The year 1998 was one of the recent weather related disaster years, which caused hurricane house in Central America and floods in China, India and Bangladesh. Canada and New England in the U.S. suffered heavily due to ice storm in January while Turkey, Argentina and Paraguay with floods in June 1998. Vast fires in Siberia burned over three million acres of forests. Human and crop losses are the worst phenomena in such weather disasters, affecting global economy to a considerable extent.

The 1997-'98 El-Nino events, the strongest of the last century is estimated to have affected 110 million people and cost the global economy nearly US $ 100 billion. Statistics compiled from insurance companies for the period 1950-1999. Show that major natural catastrophes which are mainly weather and climate related caused estimated economic losses of US $ 960 billion. Most of the losses were recorded in recent decades. Increase in aerosols due to emission of green house gases including black carbon and chlorofluorocarbons (CFCS), ozone depletion, UV-B filtered radiation, cold and heat waves, global cooling and warming and "human hand" in the form of deforestation and loss of wetlands in the process of imbalanced development for betterment of human kind may be caused factors for climate variability and climate change.

Causes of climatic variability

A. External causes

i) Solar output: An increase in solar output by 0.3%when compared to 1650 -1700AD data.
ii) Orbital variation: 1. Earth orbit varies form almost a complete circle to marked ellipse (Eccentricity). 2. Wobble of earth's axis (Precession of equinox) 3. Tilt of the earth's axis of rotation relative to the plane of the orbit varies between 21.8º and 24.4º.

B. Internal causes

i) Changes in the atmospheric composition. Change in the green house gases especially CO_2
ii) Land surface changes particularly the afforestation and deforestation
iii) The internal dynamics of southern oscillation – changes in the sea surface temperature in western tropical Pacific (El-Nino/La-Nina) coupled with Southern Oscillation Index, the Tahiti minus Darwin normalized pressure index leading to the ENSO phenomena
iv) Anthropogenic causes of climate variation in green house gases and aerosols.

Effects of climate change

1. The increase concentration of CO_2 and other green house gases are expected to increase the temperature of the earth.
2. Crop production is weather dependant and any change will have major effects on crop production and productivity.

3. Elevated CO_2 and temperature affects the biological process like respiration, photosynthesis, plant growth, reproduction, water use etc. Depending on the latitude the CO_2 may either offer beneficial effect or may behave otherwise also.

El-Nino and La-Nina

El-Nino is a Spanish word meaning "the boy child" ('Child Christ') because El-Nino occurs around Christmas time each year when the waters off the Peruvian coast warm slightly. In every three to six years, the waters become unusually warm. 'El Niño' is now used more widely to refer to this abnormal warming of the ocean and the resulting effects on weather. 'El Niño' is often coupled with 'Southern Oscillation' as the acronym ENSO. 'La Niña' is used popularly to signify the opposite of El Niño, occurring when the waters of the eastern Pacific are abnormally cold. La Niña episodes are associated with more rainfall over eastern Australia, and continuing drought in Peru. Peruvian meteorologists have objected to term La Niña-the Girl Child-because Christ is not known to have had a sister, and the term anti-ENSO is sometimes preferred.

The El-Nino event is due to decrease in atmospheric pressure over the South East Pacific Ocean. At the same time, the atmospheric pressure over Indonesia and North Australia increases. Once the El-Nino event is over, the atmospheric pressure over the above regions swings back. This sea-saw pattern of atmospheric pressure is called Southern Oscillation. Since El-Nino and Southern Oscillation are linked they often termed as ENSO.

It is most important one, which represents a tendency for high atmospheric pressure over the Pacific Ocean, represents to be associated with low pressure over the Indian Ocean and vice-versa. A measure of the monsoon low pressure is the Southern Oscillation Index (SOI) represented by the difference in sea level pressure over Tahiti, an Island in South central pacific and Darwin in North Australia, which represents the northern part of the Indian Ocean. The positive SOI denotes high pressure over the central pacific and low over Indonesia, North Australia and Northern Indian Ocean. Above average rainfall is expected over India, India and Indonesia and North Australia if the SOI is positive. Drought or deficit rainfall is expected in the above countries if the SOI is negative, indicating high atmospheric pressure over Indonesia and low in the central pacific.

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CAUSES & DETAILED IMPACT OF CLIMATE CHANGE ON AGRICULTURE-

Agriculture is situated at the interface between eco-systems and society. As such agriculture is affected by the changes in the global environmental conditions, but agriculture also contributes to about 20% of the emissions of greenhouse gases, notably methane and nitrous oxide. There is a growing concern about the potential effects of climate change on agriculture production in different regions of the world including India. Climate change could lead to sea level rise, increased weather variability, more droughts and the spread of infectious diseases.

Climate change is expected to affect agriculture very differently in different parts of the world. The resulting effects depend on current climatic and soil conditions, the direction of change and the available resources and infrastructure to cope with change.

Climate change and its impact

Mean sea level is expected to rise 15-95 cm by the year 2100, causing flooding of low-lying areas and other damage. Climatic zones could shift towards the poles by 150-550 km in the mid-latitude regions. Forests, deserts, rangelands and other unmanaged ecosystems would face new climatic stresses. As a result, many will decline or fragment and individual species will become extinct.

SEA-LEVEL RISE

Observations have detected sea-level rises over the last 100 years of between eight and 31 cm. The overall global mean value is estimated to be between 10 and 25 cm. This is related to the observed net melting of glaciers and ice fields and the net warming of the surface air temperature. Temperature increases in the ocean lead to change in sea level due to thermal expansion. Thermal expansion and ice melting are estimated to be of comparable importance for the sea-level changes.

Climate-change warming is expected to cause a sea-level rise of between 20 and 100 cm by the year 2100 as ocean warming and ice melting occurs at a faster rate than in the past 100 years.

The rise in sea level will have important impacts on coastal zones and small islands. Coastal erosion, flooding, saltwater intrusion and sedimentation changes will be factors that impact on human settlements, agriculture, freshwater supply and quality, fisheries, and human health. Millions of people will be affected in countries such as Bangladesh, Benin, China, Egypt, India, Japan, The Netherlands and Nigeria. In some, such as the Marshall Islands, entire countries will be affected.

Human society will face new risk and pressures. Food security is unlikely to be threatened at the global level, but some regions are likely to experience food shortages and hunger. Water resources will be affected as precipitation and evaporation patterns change around the world. Physical infrastructure will be damaged, particularly by sea-level rise and by extreme weather events. Economic activities, human settlements and human health will experience many direct and indirect effects. The poor and disadvantaged are the most vulnerable to the negative consequences of climate change.

Green house gases

i) Carbon di-oxide: The primary source of carbon dioxide build up is combustion of fossil fuel; gasoline, natural gas, coal, peat firewood and indiscriminate clearing the forests and extensive cultivation of land in crop production which result in irreversible oxidation of soil organic matter. The level has risen from 315 ppm in 1959 to more than 342 ppm in mid-1980. For India (1988-98), contributions from coal, petroleum natural gas and transportation emissions are around 85, 35, 5 and 15 million tonnes of carbon respectively giving a total of about 1.4×10^{14} g as against a global total of 5×10^5 g carbon / year. Thus, Indian contribution is 2.8 per cent of global CO_2 warming.

ii) Chlorofluorocarbons:

CFCs are responsible for destruction of the ozone layer, a protective shield that surrounds the earth at an altitude of 10-30 km. During the past two decades, ozone layer has been depleted 2 to 3 per cent at the global level.

The production of CFCs in India is around 5000 tonnes per year as against the global production of 7, 00,000 tonnes per year, thus India's contribution is less than one-hundredth of the total world production. The concentration of CFCs in the atmosphere is less than 1 ppm, but each molecule is 16,000 times more effective than carbon dioxide in absorbing heat and survives as long as 400 yeas.

iii) Methane

Methane contributes about 15 to 20 per cent towards green house effect. Most methane is generated by bacteria breaking down organic matter in the absence of oxygen, e.g. the flooded rice fields, guts of cattle, garbage dumps, leakage in the process of mining, transport, and use of coal and natural gas, and burning of biomass. Swamps and rice fields are thought to contribute about 25-30 per cent to the world methane production.

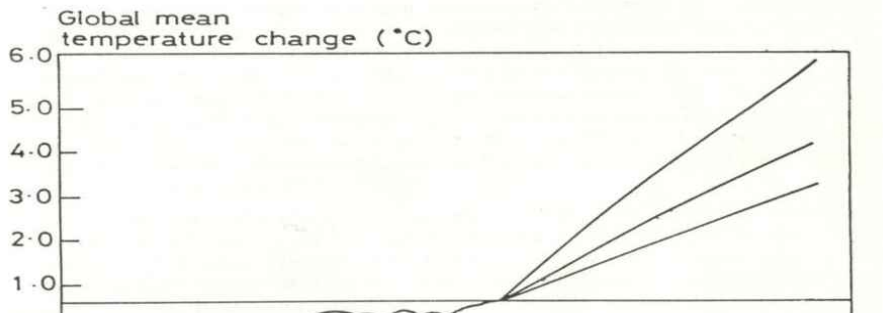
Almost 90 per cent of world's paddy fields exist in Asia and about 60 per cent of these are found in India and China. India produces 3.9×10^{12} g methane per year as against a global production of 110×10^{12} g methane per year i.e., about 2 per cent of the global paddy emissions. The contribution of Indian cattle is around 7×10^{12} g per year as against the world aver of 80×10^{12} g per year. Although methane lasts only 10 years, molecule for molecule it absorbs 20 to 30 times more heat than carbon dioxide.

iv) Other gases

Nitrous oxide is a green house gas and is also responsible for the destruction of ozone layer. Sources of N_2O include burning of forests, grassland and other biomass, natural soil and chemical fertilizers. N_2O lasts up to 180 years, is 200 times as heat absorbent as carbon di oxide and constitutes about 5 per cent of man-made emissions. Ozone in the stratosphere acts as a protective layer because it absorbs much of UV radiation from sun. However, in the troposphere its concentration is increasing and it acts as a greenhouse gas. It is responsible for about 10 to 20 per cent of the greenhouse effect.

Changes in global temperature

Projected global mean temperature changes relative to 1990 were calculated up to 2100. Climate models calculate that the global mean surface temperature could rise by about 1 to 4.5° C by 2100 (Fig 1).



The combined land-surface air and sea surface temperatures (°C) 1861 to 1998, relative to the average temperature between 1961 and 1990.The mean global surface temperature has increased by about 0.3 to 0.6°C since the late 19th century and by about 0.2 to 0.3°C over the last 40 years, which is the period with most reliable data.

LEC 17. IMPACT OF CLIMATE CHANGE ON AGRICULTURE, FORESTRY, HYDROLOGY, MARINE AND COASTAL ECOSYSTEM

CLIMATE:

Long-term regime of atmospheric variables of a given place or area is called as climate

Ex. Cold season, summer season

CLIMATE CHANGE:

Climate change is any long term change in weather statistics over a period of time that ranges from decades to millions of years. Climate change and agriculture are interrelated processes, both of which take place on a global scale. Global warming is projected to have significant impacts on conditions affecting agriculture, including temperature, carbon dioxide, glacial run-off, precipitation and the interaction of these elements.

PROJECTED CHANGES

- Temperature
- Rainfall
- Global warming
- Increased frequency of extreme climatic events such as flood , drought, etc…
- Enhanced carbon dioxide concenteration.
- Effect on soil fertility

Temperature effect:

For each plant variety, there is an optimal temperature for vegetative growth, with growth dropping off as temperatures increase or decrease.

If the temperature is increase, the drought occurs. If the temperature is decrease, the crops are affected by chilling injury(very low- negative).

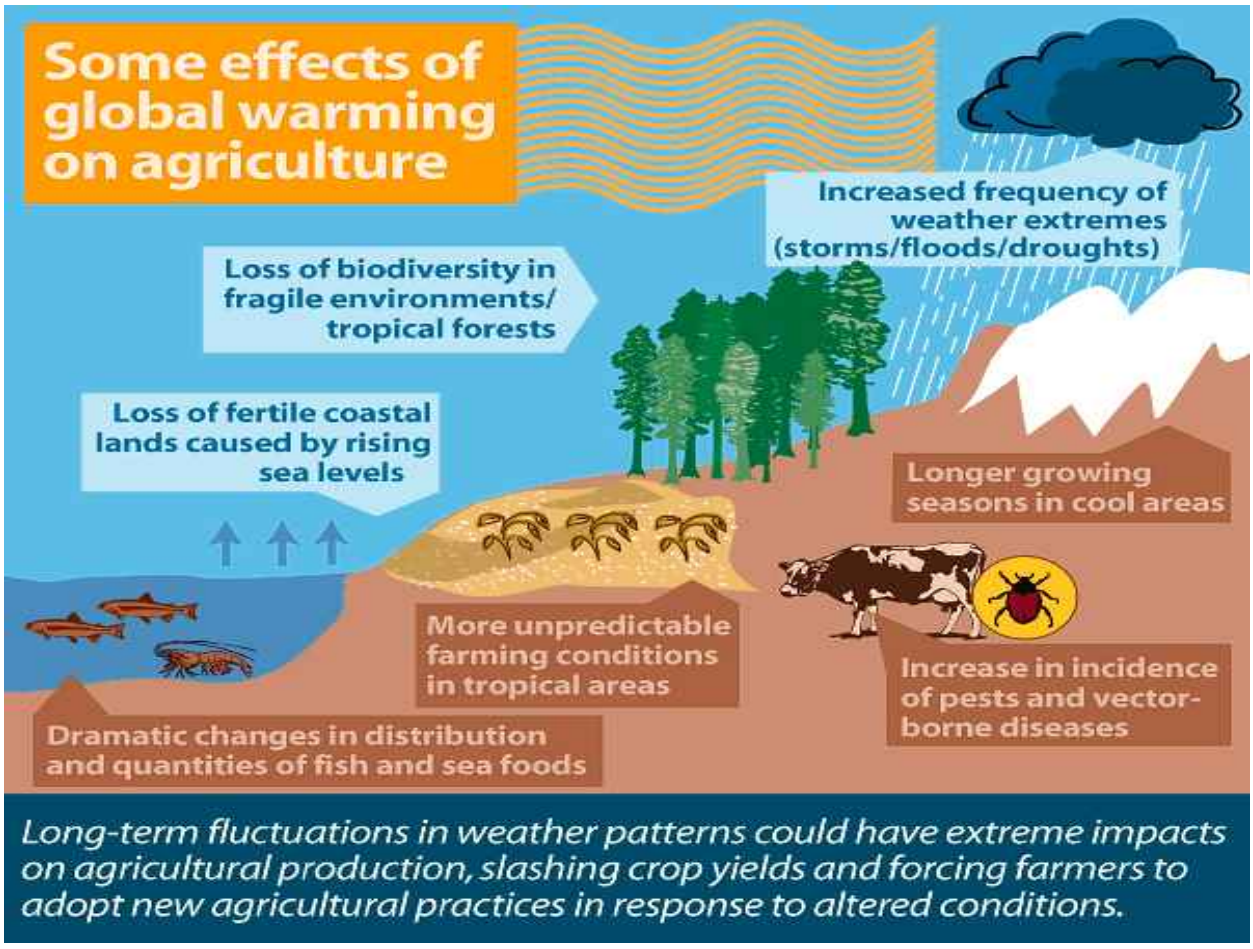
Similarly, there is a range of temperatures at which a plant will produce seed. Outside of this range, the plant will not reproduce. For example it is estimated that corn will fail to reproduce at temperatures above 95 °F and soybean above 102 °F.

Duration of crop growth cycles are above all, related to temperature. An increase in temperature will speed up development. In the case of an annual crop, the duration between sowing and harvesting will shorten. The shortening of such a cycle could have an adverse effect on productivity because senescence would occur sooner.



Rain fall:

It also affects the production of agriculture crops. If the rainfall is above the optimum level it will cause the flooding. So the crops are drowned by flood and also there is a reduction in yield.



EFFECT OF ELEVATED CO_2 ON CROPS

Carbon dioxide is essential to plant growth. Rising CO_2 concentration in the atmosphere can have both positive and negative consequences.

Positive:

- Beneficial effects of carbon dioxide are increases in crop growth and increase of efficiency in water use.
- Decrease in potential yields is likely to be caused by shortening of the growing period, decrease in water availability and poor vernalization
- Increased carbon dioxide

  Increases photosynthesis

  Fewer stomata – reduced water usage.

  C3 plants such as wheat have large productivity and yield.

Negative:

Most models study the relationship between CO_2 and productivity in isolation from other factors associated with climate change, such as an increased frequency of extreme weather events, seasonal shifts.

Erosion and fertility:

- The warmer atmospheric temperatures observed over the past decades are expected to lead to a more vigorous hydrological cycle, including more extreme rainfall events. Erosion and soil degradation is more likely to occur.

  - Soil fertility would also be affected by global warming. However, because the ratio of carbon to nitrogen is a constant, a doubling of carbon is likely to imply a higher storage of nitrogen in soils as nitrates, thus providing higher fertilizing elements for plants, providing better yields.

  - The average needs for nitrogen could decrease, and give the opportunity of changing often-costly fertilization strategies.

  - Due to the extremes of climate that would result, the increase in precipitations would probably result in greater risks of erosion, whilst at the same time providing soil with better hydration, according to the intensity of the rain.

  - The possible evolution of the organic matter in the soil is a highly contested issue: while the increase in the temperature would induce a greater rate in the production of minerals, lessening the soil organic matter content, the atmospheric CO_2 concentration would tend to increase it.

Potential Impacts on Indian Agriculture:

a. Productivity of most cereals would decrease due to increase in temperature, CO_2 anddecrease in water availability

b. A projected loss of 10-40% in crop production by 2100

c. 1°C increase in temperature may reduce yields of major food crops by 3-7%. Much greater losses at higher temperatures with longer duration. Greater loss expected in *rabi*

d. Length of growing period in rainfed areas is likely to reduce, especially in peninsular regions

e. Increased climatic extremes - likely to increase production variability

f. Increase in CO_2 to 550 ppm increases yields of rice, wheat, legumes and oilseeds by 10-20%.

g. Climate change induced by increasing greenhouse gases is likely to affect crops differently from region to region. Managing water to prevent water logging, erosion, and nutrient leaching where rainfall increases.

h. Altering the timing or location of cropping activities

i. Decrease in potential yields is likely to be caused by

✓ shortening of the growing period,
✓ decrease in water availability and
✓ poor vernalization.

IMPACTS OF CLIMATE VARIABILITY AND CHANGE IN ASIA:

- Mountain regions: impact on soil erosion and sedimentation
- Coastal regions: combined effects of subsidence and sea level rise on drainage and sedimentation problems
- Rangelands and dry lands: environmental and social stress
- Desert zones: negative impacts of temperature increases on natural vegetation
- Plants with surface root systems, which utilize mostly precipitation moisture will be vulnerable.
- Sheep breeding and lamb wool: negative impacts of climate Change
- Ttemperatures in winter: higher pathogen survival rates, more host plants for pathogens (extension of cropping area), Weakening of disease resistance due to heat-stress and more favourable growth conditions for pathogenic bacteria.
- Warmer and wetter climate*:* widespread incidence of wheat scab, rice blast, and sheath and culm blight of rice growth, reproduction, and spread of disease bacteria+

Possible effect of Climate Change on Rice:

- Increase in CO2 + Increase in Temperature + Monsoon variation
- Increasing glacier melt - Change in availability of irrigation water
- Change in crop water requirement
- Fertilizer Use Efficiency
- Green House Gas emission
- Latitudinal and Altitudinal effect
- Population dynamics of pest and disease
- Impact due to extreme weather events
- Quality of Agricultural Produces
- International Rice Research Institute forecast 20
- reduction in yields over the region per degree
- Celsius of temperature rise
- Consequences of increase in Sea level

Projected beneficial impacts of climate change on Indian agriculture:

- ❑ Reduced frequency of frost damage: less damage to potato, peas, mustard
- ❑ New 'flooded' areas may become available for fisheries in coastal regions
- ❑ Other potential benefits, if any, need to be characterized

- ❑ Diversifying income through altering integration with other farming activities such as livestock raising.
- ❑ Improving the effectiveness of pest, disease, and weed management practices through wider use of integrated pest and pathogen management, development.
- ❑ And also use of varieties and species resistant to pests and diseases and maintaining or improving quarantine capabilities and monitoring programs.
- ❑ Using climate forecasting to reduce production risk.

Climate projections for India	
Maximum temperature	increase by 2-4°C during 2050s in regions above 25°N
Minimum temperature	Increase up to 4°C all over the country. May exceed 4°C over southern peninsula, northeast India and some parts of Punjab, Haryana and Bihar
Monsoon Rainfall	marginal changes in monsoon months (JJAS) Large changes during non-monsoon months
Number of rainy days	Decrease in the number of rainy days over a major part of the country
Extreme Rainfall events	Overall increase in the rainy day intensity by 1-4 mm/day except for small areas in northwest India
Cyclonic storms	Increase in frequency and intensity of cyclonic storms

Conclusion:

- Climate change is a reality
- Indian agriculture is likely to suffer losses due to heat, erratic weather, and decreased irrigation availability
- Adaptation strategies can help minimize negative impacts
- These need research and policy support
- Costs of adaptation and mitigation are unknown but likely to be high; costs of inaction could be even higher
- Start with 'no-regrets' adaptation op

Effects of climate change:

GLOBAL WARMING:

The effects of global warming are the ecological and social changes caused by the rise in global temperatures. There is a scientific consensus that climate change is occurring, and that human activities are the primary driver. Global warming is projected to have a number of effects on the oceans. Ongoing effects include rising sea levels due to thermal expansion and melting of glaciers and ice sheets, and warming of the ocean surface, leading to increased temperature stratification.

Rising Sea Levels:

Sea levels around the world are rising faster because of the melting of land-based ice and the thermal expansion of oceans. According to the Arctic Climate Impact Assessment (ACIA, an intergovernmental forum composed of the eight arctic nations) in the past two decades the global sea level has risen by an average of 8 centimeters (3 inches), and projections show an additional rise of 10 to 90 centimeters (4 inches to 3 feet) during the twenty-first century. This rise in sea level, coupled with heavier than usual precipitation, may increase the risk of flooding for tens of millions of people in coastal areas across the world.

Ocean temperature rise:

From 1961 to 2003, the global ocean temperature has risen by 0.10 °C from the surface to a depth of 700 m. There is variability both year-to-year and over longer time scales, with global ocean heat content observations showing high rates of warming for 1991–2003, but some cooling from 2003 to 2007.The temperature of the Antarctic Southern Ocean rose by 0.17 °C (0.31 °F) between the 1950s and the 1980s, nearly twice the rate for the world's oceans as a whole. As well as having effects on ecosystems (e.g. by melting sea ice, affecting algae that grow on its underside), warming reduces the ocean's ability to absorb CO_2

It is likely (greater than 66% probability, based on expert judgment) that anthropogenic forcing contributed to the general warming observed in the upper several hundred meters of the ocean during the latter half of the 20th century.

Oxygen depletion:

The amount of oxygen dissolved in the oceans may decline, with adverse consequences for ocean life.

Acidification:

The oceans have already taken up about one-third of the carbon dioxide emitted by human activity. As carbon dioxide dissolves in sea water, carbonic acid is formed, which has the effect of acidifying the ocean, measured as a change in pH. The uptake of human carbon emissions since the year 1750 has led to an average decrease in pH of 0.1 units. A further reduction in average global surface ocean pH of between 0.14 and 0.35 units over the 21st century.

FISHERIES:

Fish has been an important part of the human diet in almost all countries of the world. It is highly nutritious; it can provide vital nutrients absent in typical starchy staples which dominate poor people’s diets. Fish provides about 20 % of animal protein intake and is one of the cheapest sources of animal proteins as far as availability and affordability is concerned.

While it serves as a health food for the affluent world owing to the fish oils rich in polyunsaturated fatty acids (PUFAs), for the people in the other extreme of the nutrition scale, fish is a health food owing to its proteins, oils, vitamins and minerals and the benefits associated with the consumption of small indigenous fishes . Although aquaculture has been contributing an increasingly significant proportion of fish over recent decades, approximately two-thirds of fish are still caught in capture fisheries.

The number of people directly employed in fisheries and aquaculture is estimated at 43.5 million, of which over 90 % are small-scale fishers. In addition to those directly employed in fishing, over 200 million people are thought to be dependent on small scale fishing in developing countries, in terms of other economic activities generated by the supply of fish (trade, processing, transport, retail, etc.) and supporting activities (boat building, net making, engine manufacture and repair, supply of services to fisherman and fuel to fishing boats etc.) in addition to millions for whom fisheries provide a supplemental income. Fisheries are often available in remote and rural areas where other economic activities are limited and can thus be important sources for economic growth and livelihoods in rural areas with few other economic activities.

IMPACTS OF CLIMATE CHANGE ON FISHERIES:

Fisheries and fisher folk may have the impact in a wide range of ways due to climate change. The distribution or productivity of marine and fresh water fish stocks might be affected owing to the processes such as ocean acidification, habitat damage, changes in oceanography, disruption to precipitation and freshwater availability .

Climate change, in particular, rising temperatures, can have both direct and indirect effects on global fish production. With increased global temperature, the spatial distribution of fish stocks might change due to the migration of fishes from one region to another in search of suitable conditions.

The effects of increasing temperature on marine and freshwater ecosystems are already evident, with rapid pole ward shifts in distributions of fish and plankton in regions such as North East Atlantic, where temperature change has been rapid.

Climate change has been implicated in mass mortalities of many aquatic species, including plants, fish, corals, and mammals. According to a study conducted by Prowse , the northward migration of species and the disruption and competition from invading species are already occurring and will continue to affect marine, terrestrial, and freshwater communities. This will have implications for the protection and management of wildlife, fish, and fisheries resources; protected areas; and forests.

Stress on populations of iconic wildlife species, such as the polar bear, ringed seals, and whales, will continue as a result of changes in critical sea-ice habitat interactions.

What Is the Coastal Zone?

The coastal zone, where land meets ocean, is one of the most dynamic natural systems. Here, the three main components of our planet—the hydrosphere, the lithosphere, and the atmosphere—meet and interact, forming interconnected systems. Coastlines are formed by morphological changes governed by climatic and geological processes. They constitute a transition zone where land and freshwater meet saline water, and across which the effects of land on the ocean, and vice versa, are transferred and modified. Coastal zones are a crucial battleground in the current fight against climate change.

Coastal areas are dynamic regions where terrestrial and marine ecosystems intersect and interact with human socioeconomic systems. Coastal systems also form a part of other larger marine ecosystems that include coasts and open ocean areas.Coasts are of great ecological and socioeconomic importance.Coasts are affected by two main types of influences, terrestrial and marine, that are considered external to the coastal zone. *Terrestrial influences* are mostly anthropogenic in nature. They include land use changes and all the consequences of changing hydrological regimes, nutrient loading from sediment transport, runoff, and reduction of sediments through rivers (for example, from dam and channel construction and extraction of

river sand upstream). *Marine influences* are mostly natural phenomena such as weather events (storms and cyclones), tsunamis, and wave patterns and coastal and ocean currents that affect the processes of nutrient, material, and heat transfer and mediate geomorphological changes.

Effect of climate change

1)**TEMPERATURE**:-Warmer temperatures are also predictedto change the ocean's areas of great primaryProductivity.The warmer or cooler a region, the more or less suited it is for growing certain types of plants.

GLOBAL WARMING:-An increase in earth temperature due to industrial processes leading to build up of green house gases in the atmosphere. These effects are increase temperature range, due to this rising of sea level. These affect coastal agriculture. Rising sea level decreases coastal land. This is problematic because the majority of the world's population resides on our near the coast.

2) **RAINFALL:** - Shifting rainfall patterns will change the growing locations of various crops. Some regions will be better suited for agriculture, while others will experience decreased yields. Rainfall data for the IMD stations of the State of Kerala for the period from 1871 to 2008(140 years) revealed a declining trend in annual and southwest monsoon rainfall during the past 60 years and an increasing trend in post monsoon rainfall, indicating likely shifts in rainfall patterns. The untimely rain in Kerala, which hit the entire region since March 14, 2008 has caused crop damage and flooding. The agriculture sector in Kerala was badly affected due to continuous rain.
3) **FLOOD:-** Flood affects agricultural production . The flood in1988 caused reduction of agricultural production by some 45 percent in the South-West coastal region of Bangladesh. Prolonged floods would tend to delay *Aman(rice var.)* plantation, resulting in significant loss of potential *Aman* production.
4) **CYCLONES- STORM SURGES**:-In meteorology, a cyclone is a closed, circular or elliptical area in which fluid is rotate. Pressure is lowest at center.Principal dangers from cyclones are: (i) very strong winds, (ii) rain.Cyclone :- environmental impact was quite large because: litter was in the ocean, destroying animal's habitats, trees were uprooted and turned over, metal, glass and other debris was thrown all over town. In terms of CC, 1°C increase in sea surface temperatures could increase tropical cyclone intensity by 10 percent .Cyclones bring severe winds, storm surges, and flood that impact on lives, crops and property.
5) **SALINE WATER:-**1) Stunted Growth and Production:- When plants are watered with salty water, they end up having to fight the osmosis process. They expend much more energy than normal, and because they cannot put their energy into flowering or making leaves, they may not grow as much or produce as well.2) Nutrition:-Salt water affects the ability of a plant to take up essential nutrients. This causes nutritional imbalances in the plant which may stunt the plant's growth or cause yellowing of the leaves. The adverse effects of saline water will be significant on coastal agriculture and the availability of fresh water for public water supply will fall. *Increased Soil Salinity from Saltwater Intrusion* -Sea level rise is expected to exacerbateintrusion of saline water into the fresh groundwater aquifers in the coastal zone. Increasing temperature is also expected to enhance soil evaporation, hence increasing soil salinity.

The potential Impacts of Climate Change on Indian coastal Agriculture

India has a distinct geographical entity since it is marked off from the rest of Asia by mountain and sea. The geologic and tectonic history of the region is complex. India has 667

Islands in Bay of Bengal and 508 in Arabian Sea. Length of coastline of mainland and Islands is 7516.6 KM.

1)Kerala and Climate Change:- a) Pepper output likely to fall in 2009:- Heavy pre-monsoon showers (and a lethal attack by wasps) may hit pepper production in Kerala, the main producer of the commodity in India. b) Production declines:- Increase in maximum temperature of 1-3° Cduring summer 2004 adversely affected thermo-sensitive crops like black pepper and cocoa in Kerala. c) Cashew production declines:- Though Kerala stood first in cashew production a decade ago, at present it occupies only fourth position and likely to go down further. It was due to steady decline in cashew area and also occurrence of weather aberrations during the reproductive phase of cashew. d)Climate change hits mango product:- Climate change and unseasonal rain in November and January over the last two years(2009 and 2010) have dampened the prospects of mango farmers in Muthalamada, known for its early mango harvest and large-scale export of the fruit. Late flowering of the fruit and the resultant delay in harvest has hit the annual export market. e) Coastal Coconut:- As per INCCA report(2010), yields of coconut are projected to increase in the west coast of India (includes Kerala) by up to 30% due to temperature increase.

2) MAHARASHTRA

Temperature rise by 2.3 to 3.8 C and increased rainfall by 2050 will affect: >30% reduction in sugarcane yield in Maharashtra.

3) ORISSA

"super cyclone" which struck the Indian state of Orissa in1999 generated a storm surge of 8 meters traveling up to 20 kilometers inland. The Orissa cyclone killed 10,000 people, destroyed 275,000 homes and 17,110 square kilometers of crops, and uprooted or snapped 90 million trees. Temperature rise by 2.3 to 3.8 C and increased rainfall by 2050 will affect >12% decline in rice production in Orissa.

4)KOLKATA

Compounding the effects of saltwater intrusion, overexploitation of ground water in and around Kolkata has led to a drop in its level, leading to further intrusion of seawater, thus making the subsurface ground water saline.

The country's Sundarban region, the largest continuous mangrove area in the world, provides direct employment to some 500,000 to 600,000 people for at least half the year, many of whom are engaged in harvesting the region's natural resources through fishing; woodcutting; and collecting thatching materials, honey, beeswax etc. By some estimates, just 10 centimeters of sea level rise would submerge 15 percent of the region, and the Sundarban would disappear altogether if the sea level rises up to 60 centimeter.

5) SMALL ISLAND DEVELOPING STATES

Sea level rise is expected to exacerbate coastal inundation, increase erosion, and magnify the effect of storm surges, thus threatening vital infrastructure, settlements, and facilities that support the livelihoods of island communities. Changes in precipitate on will alter water resource availability, which would in turn affect agriculture, biodiversity, and natural ecosystems dependent upon water supplies. Elevated sea surface temperatures and ocean acidification are already causing coral bleaching and may cause changes in fisheries distribution that would critically undermine commercial and subsistence fisheries in many SIDS.

IMPACT OF CLIMATE CHANGE

Coral reefs are among the most vulnerable ecosystems to climate change. Corals are especially sensitive to elevated sea surface temperatures. When physiologically stressed, corals may lose much symbiotic algae, which supply nutrients and colors. In this stage, corals appear white and are considered "bleached." Corals can recover from short-term bleaching, but prolonged bleaching can cause irreversible damage and subsequent mortality.

Mangroves and Coastal Wetlands

Asian wetlands have been increasingly threatened by a warmer climate in recent decades. Droughts and declining precipitation in most delta regions of Bangladesh, China, India, and Pakistan have caused wetlands to dry up and have severely degraded their ecosystems. Reduction of freshwater flows and saltwater intrusion in the Indus Delta and Bangladesh has seriously compromised these regions. About one-third of the world's mangroves have reportedly been lost during the last 50 years of the 20th century, likely as a result of human activities.

Sea grass Beds

Sea grass beds are formed by unique groups of species of flowering plants that grow completely submerged in shallow coastal waters. These beds are very productive and provide a host of ecosystem services, ranging from filtering and trapping sediments to nutrient cycling. They are an important nursery habitat for fish and invertebrates and a source of food for many coastal organisms in tropical areas. Climate change factors such as increased coastal water temperatures, sea level rise, and extreme events such as storms and storm surges can contribute to increasing stresses on seagrass meadows.

TERRESTRIAL ECOSYSTEMS

There are a number of ways in which climate change will affect terrestrial ecosystems. Terrestrial ecosystems depend directly on temperature and precipitation climatology. Changes in temperature and precipitation, including their extremes and seasonal or daily variations, will influence the distribution of biomes in the world, including those in agriculture. Rising temperatures alone would be expected to foster the pole ward migration of biome species. For example, a warming could be expected to improve options for agriculture in subarctic regions. Changes in climate will influence other determinants of the ecosystem condition such as disease, pest cycles and the incidence of fires. The increase in atmospheric carbon dioxide concentrations is expected to increase the primary productivity of plants, i.e. make them grow faster. This could change the balance among plants competing for the same space. The largest adverse impacts on the ecosystem are anticipated to be from direct human activity itself. The clearing of land for agriculture and urbanization and the segmentation of ecosystems will be important factors. More unfavorable impacts are likely in tropical and subtropical developing countries compared to developed countries because of population pressures and lack of resources for adaptation and mitigation responses to climate change impacts. Climate change will affect crops in a number of ways. These include the growth process of crop plants and those of insects, weeds and diseases as discussed above. Growth will be affected by changes in CO_2 concentration as well as those in temperature, moisture supply and severe weather. CO_2 increases alone are expected to increase the productivity of annual crops, particularly those that may be limited by existing concentrations of carbon dioxide (i.e. most crops including wheat, rice, barley, cassava and potato, and most trees). For these crops, increases on the order of 30 per cent would be expected for a doubling in CO_2 if there were no other changes in conditions. The increases would be less for plants that

have a special CO_2-concentrating mechanism (such crops as maize, millet, sugar cane, sorghum, and many tropical grasses).

Temperature and moisture supply have a dominant influence on crop growth. Each crop type has an optimal temperature range for growth. Moisture supply is critical throughout the growth period of the crop. The moisture supply depends on both precipitation and evaporation (evapotranspiration from the plant). The diurnal and day-to-day variability for temperature and moisture supply along with mean values are important climatic factors for plants because of the negative impact of extremes. Very preliminary impact assessments have been made for a selection of crops using simple estimates of climate change. The estimates were derived from climate-model simulations, historical data, or just an outright specification of a temperature change. The impacts show extreme variability, often ranging from increases to decreases of yields for a given crop. One may find comparable summaries for all other major regions of the Earth in chapter 13 in the 1995 IPCC Working Group II report. It is clear from these tables that there is great uncertainty in the quantitative impact assessments of climate change on agricultural crops. It is strongly suggested that crop yields and productivity will vary a great deal from one region to another. Some areas will see improvement in yields, others will see decreases. Agricultural patterns will change. It is likely that overall impacts will be significant. The overall effects for a region will depend on many factors in that region such as irrigation options, current agricultural infrastructure, adaptations in farming practices and so forth. It will take serious planning efforts to deal most constructively with the changes.

Forest Ecosystem

As the projected temperature increases are smaller in tropical latitudes, tropical forests will be affected less than those in other latitudes. However, climate change in terms of the amount and seasonality of rainfall could have larger impacts. Even so, other human impacts will likely affect tropical forests more than does climate change. Temperate forests will be impacted by temperature, precipitation and CO2 changes differently from region to region. However, the negative aspects of such changes on temperate forests will be minimized by reforestation and forest management programmes, since most temperate forests are located in developed countries. Boreal forests will be most affected by climate change as warming is expected to be largest at high latitudes. Increased fire and pest outbreaks will negatively impact the southern regions of boreal forests, whereas the increased temperature and moisture supply in the northern regions of the boreal forests will enhance the forest, which is expected to advance northward into the tundra. Forest fires are a matter of special interest. Forest fires occur commonly in seasonally dry forest areas due to human and natural causes. They play an important role in the ecosystem dynamics in tropical, temperate and boreal zones. Handling them is an important part of forest management. In some cases of major fires there can be extreme danger to human life and property locally, as well as region-wide health effects from air pollution. Temperature increases due to climate change are expected to increase drought conditions, which would lead to more favorable conditions for forest fires in seasonally dry areas. This would be of particular concern to areas that do not have integrated fire, pest and disease management.

DESERTS, LAND DEGRADATION, AND DESERTIFICATION

Roughly 30 per cent of the earth's land surface is desert or semi-desert. These are areas where the lack of moisture is a serious impediment to the growth of plants. Adjacent to these areas and elsewhere are regions, estimated at 17 per cent of the total earth land surface, where desertification' ascribed to human activity is occurring. This process is an ecological degradation which causes economically-productive land to become less productive, more desert-like and

incapable of continuing to sustain an existing community of people. Currently about one sixth of the earth's population lives in regions where such desertification is occurring. Desertification involves a number of factors. Soil that is cultivated for agriculture can be eroded by water flow and wind. Salinization of soils can occur near coastlines due to sea-level rises. Inland salinization can occur due to salt accumulation related to erosion, seepage and wind deposition. Overall significant causes of desertification can be traced to over cultivation, overstocking, fuel and wood collection, salinization and urbanization. It is difficult to define the effects of global climate change in these arid areas, especially where direct effects due to human activity are already taking place. Generally climate-change temperature increases would be expected to increase stresses on plants. This would tend to make desert conditions more severe and to accentuate desertification processes. The increase in CO_2 would be expected to reduce plant transpiration and to increase the water use efficiency of plants. Climate model projections for changes in precipitation are very uncertain. Impacts of precipitation changes depend critically on changes in distribution throughout the year and in extreme events, aspects which are not reliably handled by the models. Most model simulations to date do not suggest significantly wetter conditions in arid regions. In summary, local environmental change due to human activity (desertification) is significant in many arid areas and is independent of global climate change. It is not expected that climate change will offset the desertification process, but rather that climate-change-related factors, such as increased drought conditions resulting from rising temperatures, will increase the vulnerability of land to desertification. Many arid and semi-desertareas are in developing countries where the negative impacts would be most severe.

FRESHWATER RESOURCES MANAGEMENT

As discussed earlier, climate-change estimates for precipitation and evaporation from climate models have considerable uncertainty. Predictions for any given country range from increases to decreases in most cases. Clearly there is great uncertainty in the results. However, overall the number of countries with shortages in water supply is expected to increase by the year 2050. This is due to population increases which are not offset by precipitation increases (actually precipitation minus evaporation increases) even for the most favourable of the three model predictions. A value of 1000 m3/yr/person has been used as the baseline for minimum water requirement. Based on this number, only two of the 21 countries listed had a water shortage in 1990, eight of these countries would have a water shortage in 2050 due to population increase alone, and anywhere from seven to eleven would have a water shortage in 2050 based on both population and climate change projections. Freshwater demands are expected to increase with the warming associated with climate change. However, this impact will be combined with population growth, economic factors, and changes in agricultural, industrial and domestic practices, acting at the same time. Irrigation for agriculture is the biggest user for water extracted from the natural reservoirs (rivers, lakes and ground water). Temperature increases are expected to increase irrigation requirements more than precipitation enhancement would decrease such requirements. Impacts of climate change on municipal uses for domestic and industrial purposes are not well defined.

Indian monsoon

climate refers to the sum total of weather conditions and variations over a large area for a long period of time (more than thirty years). **Weather** refers to the state of the atmosphere over an area at any point of time. The elements of weather and climate are the same, i.e. temperature, atmospheric pressure, wind, humidity and precipitation. You may have observed that the weather conditions fluctuate very often even within a day. But there is some common pattern over a few weeks or months, i.e. days are cool or hot, windy or calm, cloudy or bright, and wet or dry. On the basis of the generalised monthly atmospheric conditions, the year is divided into seasons such as winter, summer or rainy seasons.

The climate of India is described as the **'monsoon'** type. In Asia, this type of climate is found mainly in the south and the southeast. Despite an overall unity in the general pattern, there are perceptible regional variations in climatic conditions within the country. Let us take two important elements – temperature and precipitation, and examine how they vary from place to place and season to season.

In summer, the mercury occasionally touches 50°C in some parts of the Rajasthan desert, whereas it may be around 20°C in Pahalgam in Jammu and Kashmir. On a winter night, temperature at Drass in Jammu and Kashmir may be as low as minus 45°C. Thiruvananthapuram, on the other hand, may have a temperature of 22°C.

DO YOU Know? In certain places there is a wide difference between day and night temperatures. In the Thar Desert the day temperature may rise to 50°C, and drop down to near 15°C the same night. On the other hand, there is hardly any difference in day and night temperatures in the Andaman and Nicobar islands or in Kerala.

DO YOU Know? The word monsoon is derived from the Arabic word 'mausim' which literally means season.

- 'Monsoon' refers to the seasonal reversal in the wind direction during a year.

Climatic Controls

There are six major controls of the climate of any place. They are: **latitude**, **altitude pressure and wind system, distance from the sea** (continentality), **ocean currents and relief features**.

Due to the curvature of the earth, the amount of solar energy received varies according to **latitude**. As a result, air temperature generally decreases from the equator towards the poles. As one goes from the surface of the earth to higher **altitudes**, the atmosphere becomes less dense and temperature decreases. The hills are therefore cooler during summers. The **pressure and wind** system of any area depend on the latitude and altitude of the place. Thus it influences the temperature and rainfall pattern. The sea exerts a moderating influence on climate: As the **distance from the sea** increases, its moderating influence decreases and the people experience extreme weather conditions. This condition is known as continentality (i.e. very hot during summers and very cold during winters). **Ocean currents** along with onshore winds affect the climate of the coastal areas, For example, any coastal area with warm or cold currents flowing past it, will be warmed or cooled if the winds are onshore.

Finally, **relief** too plays a major role in determining the climate of a place. High mountains act as barriers for cold or hot winds; they may also cause precipitation if they are high enough and lie in the path of rain-bearing winds. The leeward side of mountains remains relatively dry.

Factors Affecting India's Climate

Latitude

The Tropic of Cancer passes through the middle of the country from the Rann of Kuchchh in the west to Mizoram in the east. Almost half of the country, lying south of the Tropic of Cancer, belongs to the tropical area. All the remaining area, north of the Tropic, lies in the sub-tropics.

Altitude

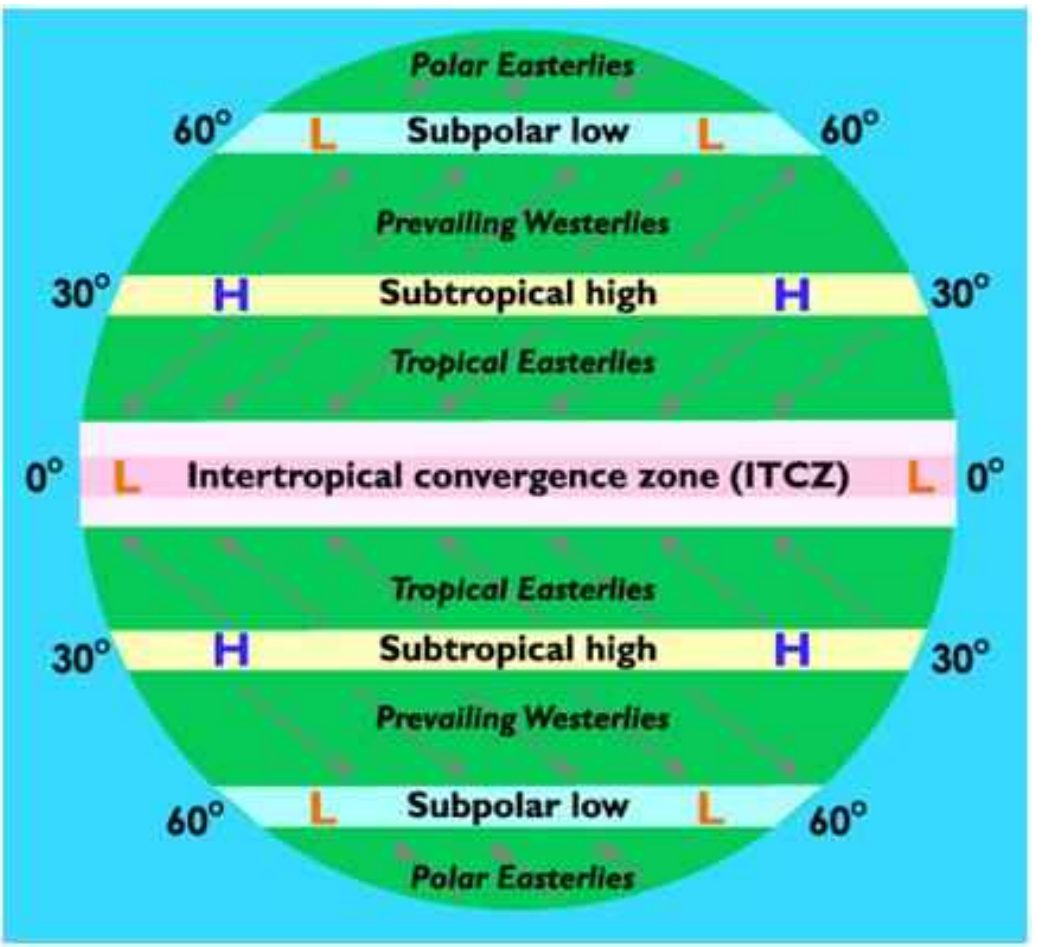
India has mountains to the north, which have an average height of about 6,000 metres. India also has a vast coastal area where the maximum elevation is about 30 metres. The Himalayas prevent the cold winds from Central Asia from entering the subcontinent. It is because of these mountains that this subcontinent experiences comparatively milder winters as compared to central Asia.

Pressure and Winds

The climate and associated weather conditions in India are governed by the following atmospheric conditions:

- ***Pressure and surface winds;***
- ***Upper air circulation; and***
- ***Western cyclonic disturbances and tropical cyclones.***

India lies in the region of north easterly winds. These winds originate from the subtropical high-pressure belt of the northern



hemisphere. They blow southwards, get deflected to the right due to the Coriolis force, and move towards the equatorial low-pressure area. Generally, these winds carry little moisture as they originate and blow over land. Therefore, they bring little or no rain. Hence, India should have been an arid land, but it is not so. Let us see why?

Coriolis force: An apparent force caused by the earth's rotation. The Coriolis force is responsible for deflecting winds towards the right in the northern hemisphere and towards the left in the southern hemisphere. This is also known as 'Ferrel's Law'.

The pressure and wind conditions over India are unique. During winter, there is a high-pressure area north of the Himalayas. Cold dry winds blow from this region to the low-pressure areas over the oceans to the south. In summer, a low-pressure area develops over interior Asia, as well as, over northwestern India. This causes a complete reversal of the direction of winds during summer. Air moves from the high-pressure area over the southern Indian Ocean, in a south-easterly direction, crosses the equator, and turns right towards the low-pressure areas over the Indian subcontinent. These are known as the Southwest Monsoon winds. These winds blow over the warm oceans, gather moisture and bring widespread rainfall over the mainland of India.

The upper air circulation in this region is dominated by a westerly flow. An important component of this flow is the **jet stream**.

These jet streams are located approximately over 27°-30° north latitude, therefore, they are known as *subtropical westerly jet streams*. Over India, these jet streams blow south of the Himalayas, all through the year except in summer. The western cyclonic disturbances experienced in the north and north-western parts of the country are brought in by this westerly flow. In summer, the subtropical westerly jet stream moves north of the Himalayas with the apparent movement of the sun. An easterly jet stream, called the *sub-tropical easterly jet* stream blows over peninsular India, approximately over 14°N during the summer months.

Jet stream: These are a narrow belt of high altitude (above 12,000 m) westerly winds in the troposphere. Their speed varies from about 110 km/h in summer to about 184 km/h in winter. A number of separate jet streams have been identified. The most constant are the mid-latitude and the sub-tropical jet stream.

Western Cyclonic Disturbances

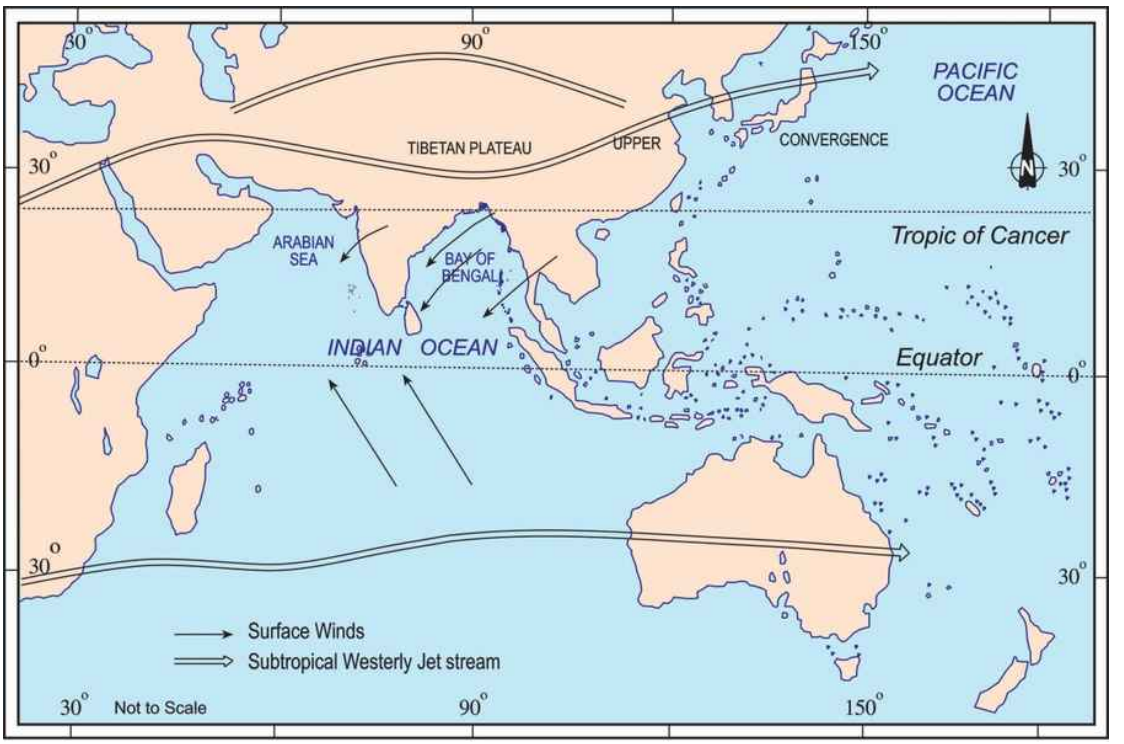
The western cyclonic disturbances are weather phenomena of the winter months brought in by the westerly flow from the Mediterranean region. They usually influence the weather of the north and north-western regions of India. Tropical cyclones occur during the monsoon, as well as, in October – November, and are part of the easterly flow. These disturbances affect the coastal regions of the country. Have you read or heard about the disasters caused by them on Odisha and Andhra Pradesh coast?

The Indian Monsoon

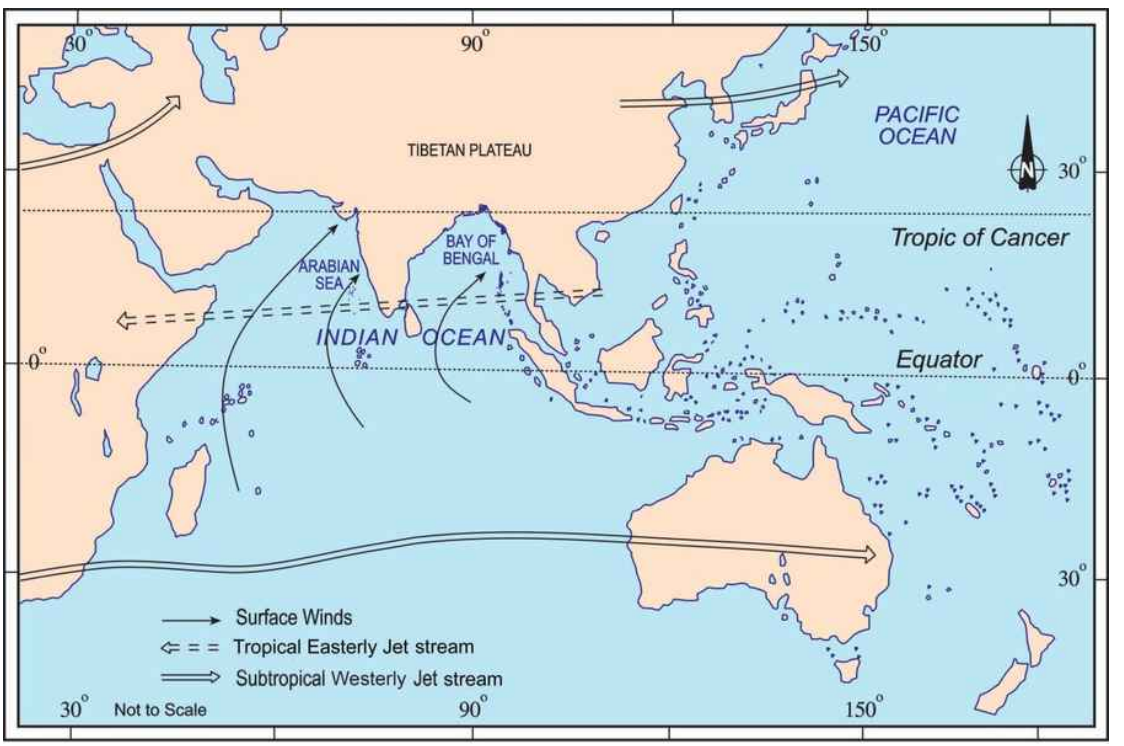
The climate of India is strongly influenced by monsoon winds. The sailors who came to India in historic times were one of the first to have noticed the phenomenon of the monsoon. They benefited from the reversal of the wind system as they came by sailing ships at the mercy of winds. The Arabs, who had also come to India as traders named this seasonal reversal of the wind system 'monsoon'.



Arrival of Monsoon



Atmospheric Conditions over the Indian Subcontinent in the Month of January



Atmospheric Conditions over the Indian Subcontinent in the Month of June

The monsoons are experienced in the tropical area roughly between 20° N and 20° S. To understand the mechanism of the monsoons, the following facts are important.

(a) **The differential heating and cooling of land and water** creates low pressure on the landmass of India while the seas around experience comparatively high pressure.
(b) **The shift of the position of Inter Tropical Convergence Zone (ITCZ)** in summer, over the Ganga plain (this is the equatorial trough normally positioned about 5°N of the equator. It is also known as the monsoon-trough during the monsoon season).
(c) The presence of the **high-pressure area, east of Madagascar**, approximately at 20°S over the Indian Ocean. The intensity and position of this high-pressure area affects the Indian Monsoon.
(d) The **Tibetan plateau gets intensely heated** during summer, which results in strong vertical air currents and the formation of low pressure over the plateau at about 9 km above sea level.
(e) The **movement of the westerly jet stream to the north of the Himalayas** and the presence of the **tropical easterly jet stream over the Indian peninsula** during summer.

Inter Tropical Convergence Zone

The Inter Tropical Convergence Zone (ITCZ,) is a broad trough of low pressure in equatorial latitudes. This is where the northeast and the southeast trade winds converge. This convergence zone lies more or less parallel to the equator but moves north or south with the apparent movement of the sun.

Apart from this, it has also been noticed that changes in the pressure conditions over the southern oceans also affect the monsoons. Normally when the tropical eastern south Pacific Ocean experiences high pressure, the tropical eastern Indian Ocean experiences low pressure. But in certain years, there is a reversal in the pressure conditions and the eastern Pacific has lower pressure in comparison to the eastern Indian Ocean. This periodic change in pressure conditions is known as the **Southern Oscillation** or **SO**. The difference in pressure over Tahiti (Pacific Ocean, 18°S/149°W) and Darwin in northern Australia (Indian Ocean, 12°30'S/131°E) is computed to predict the intensity of the monsoons. If the pressure differences were negative, it would mean below average and late monsoons. A feature connected with the SO is the **El Nino** phenomenon in which a warm ocean current that flows past the Peruvian Coast, in place of the cold Peruvian current, every 2 to 5 years. The changes in pressure conditions are connected to the El Nino. Hence, the phenomenon is referred to as **ENSO** (El Nino Southern Oscillations).

El Nino: This is a name given to the periodic development of a warm ocean current along the coast of Peru as a temporary replacement of the cold Peruvian current. 'El Nino' is a Spanish word meaning 'the child', and refers to the baby Christ, as this current starts flowing during Christmas. The presence of the El Nino leads to an increase in sea-surface temperatures and weakening of the trade winds in the region.

The Onset of the Monsoon and Withdrawal

The Monsoon, unlike the trades, are not steady winds but are *pulsating* in nature, affected by different atmospheric conditions encountered by it, on its way over the warm tropical seas. The duration of the monsoon is between 100-120 days from early June to mid-September. Around the time of its arrival, the normal rainfall increases suddenly and continues constantly for several days. This is known as the **'burst'** of the monsoon, and can be distinguished from the pre-monsoon showers. The monsoon arrives at the southern tip of the Indian peninsula generally by the first week of June. Subsequently, it proceeds into two – the Arabian Sea branch and the Bay of Bengal branch. The Arabian Sea branch reaches Mumbai about ten days later on approximately the 10th of June. This is a fairly rapid advance. The Bay of Bengal branch also advances rapidly and arrives in Assam in the first week of June. The lofty mountains causes the monsoon winds to deflect towards the west

over the Ganga plains. By mid-June the Arabian Sea branch of the monsoon arrives over Saurashtra-Kuchchh and the central part of the country. The Arabian Sea and the Bay of Bengal branches of the monsoon merge over the northwestern part of the Ganga plains. Delhi generally receives the monsoon showers from the Bay of Bengal branch by the end of June (tentative date is 29th of June). By the first week of July, western Uttar Pradesh, Punjab, Haryana and eastern Rajasthan experience the monsoon. By mid-July, the monsoon reaches Himachal Pradesh and the rest of the country (Figure 4.3).

Withdrawal or the retreat of the monsoon is a more gradual process (Figure 4.4). The withdrawal of the monsoon begins in northwestern states of India by early September. By mid-October, it withdraws completely from the northern half of the peninsula. The withdrawal from the southern half of the peninsula is fairly rapid. By early December, the monsoon has withdrawn from the rest of the country.

The islands receive the very first monsoon showers, progressively from south to north, from the last week of April to the first week of May. The withdrawal, takes place progressively from north to south from the first week of December to the first week of January. By this time the rest of the country is already under the influence of the winter monsoon.

The Seasons

The monsoon type of climate is characterised by a distinct seasonal pattern. The weather conditions greatly change from one season to the other. These changes are particularly noticeable in the interior parts of the country. The coastal areas do not experience much variation in temperature though there is variation in rainfall pattern. How many seasons are experienced in your place? Four main seasons can be identified in India – the cold weather season, the hot weather season, the advancing monsoon and the retreating monsoon with some regional variations.

The Cold Weather Season (Winter)

The cold weather season begins from mid-November in northern India and stays till February. December and January are the coldest months in the northern part of India. The temperature decreases from south to the north. The average temperature of Chennai, on the eastern coast, is between 24° – 25° Celsius, while in the northern plains, it ranges between 10°C and 15° Celsius. Days are warm and nights are cold. Frost is common in the north and the higher slopes of the Himalayas experience snowfall.

During this season, the northeast trade winds prevail over the country. They blow from land to sea and hence, for most part of the country, it is a dry season. Some amount of rainfall occurs on the Tamil Nadu coast from these winds as, here they blow from sea to land.

In the northern part of the country, a feeble high-pressure region develops, with light winds moving outwards from this area. Influenced by the relief, these winds blow through the Ganga valley from the west and the northwest. The weather is normally marked by clear sky, low temperatures and low humidity and feeble, variable winds.

A characteristic feature of the cold weather season over the northern plains is the inflow of cyclonic disturbances from the west and the northwest. These low-pressure systems, originate over the Mediterranean Sea and western Asia and move into India, along with the westerly flow. They cause the much-needed winter rains over the plains and snowfall in the mountains. Although the total amount of winter rainfall locally known as *'mahawat'* is small, they are of immense importance for the cultivation of *'rabi'* crops.

The peninsular region does not have a well-defined cold season. There is hardly any noticeable seasonal change in temperature pattern during winters due to the moderating influence of the sea.

The Hot Weather Season (Summer)

Due to the apparent northward movement of the sun, the global heat belt shifts northwards. As such, from March to May, it is hot weather season



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in India. The influence of the shifting of the heat belt can be seen clearly from temperature recordings taken during March-May at different latitudes. In March, the highest temperature is about 38° Celsius, recorded on the Deccan plateau. In April, temperatures in Gujarat and Madhya Pradesh are around 42° Celsius. In May, temperature of 45° Celsius is common in the northwestern parts of the country. In peninsular India, temperatures remain lower due to the moderating influence of the oceans.

The summer months experience rising temperature and falling air pressure in the northern part of the country. Towards the end of May, an elongated low-pressure area develops in the region extending from the Thar Desert in the northwest to Patna and Chotanagpur plateau in the east and southeast. Circulation of air begins to set in around this trough.

A striking feature of the hot weather season is the *'loo'*. These are strong, gusty, hot, dry winds blowing during the day over the north and northwestern India. Sometimes they even continue until late in the evening. Direct exposure to these winds may even prove to be fatal. Dust storms are very common during the month of May in northern India. These storms bring temporary relief as they lower the temperature and may bring light rain and cool breeze. This is also the season for localised thunderstorms, associated with violent winds, torrential downpours, often accompanied by hail. In West Bengal, these storms are known as the '*Kaal Baisakhi'*.

Towards the close of the summer season, pre-monsoon showers are common especially, in Kerala and Karnataka. They help in the early ripening of mangoes, and are often referred to as '*mango showers'*.

Advancing Monsoon (The Rainy Season)

By early June, the low-pressure condition over the northern plains intensifies. It attracts, the trade winds of the southern hemisphere. These south-east trade winds originate over the warm subtropical areas of the southern oceans. They cross the equator and blow in a south-westerly direction entering the Indian peninsula as the south-west monsoon. As these winds blow over warm oceans, they bring abundant moisture to the subcontinent. These winds are strong and blow at an average velocity of 30 km per hour. With the exception of the extreme north-west, the monsoon winds cover the country in about a month.

The inflow of the south-west monsoon into India brings about a total change in the weather. Early in the season, the windward side of the Western Ghats receives very heavy rainfall, more than 250 cm. The Deccan Plateau and parts of Madhya Pradesh also receive some amount of rain in spite of lying in the rain shadow area. The maximum rainfall of this season is received in the north-eastern part of the country. Mawsynram in the southern ranges of the Khasi Hills receives the highest average rainfall in the world. Rainfall in the Ganga valley decreases from the east to the west. Rajasthan and parts of Gujarat get scanty rainfall.

Another phenomenon associated with the monsoon is its tendency to have 'breaks' in rainfall. Thus, it has wet and dry spells. In other words, the monsoon rains take place only for a few days at a time. They are interspersed with rainless intervals. These breaks in monsoon are related to the movement of the monsoon trough. For various reasons, the trough and its axis keep on moving northward or southward, which determines the spatial distribution of rainfall. When the axis of the monsoon trough lies over the plains, rainfall is good in these parts. On the other hand, whenever the axis shifts closer to the Himalayas, there are longer dry spells in the plains, and widespread rain occur in the mountainous catchment areas of the Himalayan rivers. These heavy rains bring in their wake, devastating floods causing damage to life and property in the plains. The frequency and intensity of tropical depressions too, determine the amount and duration of monsoon rains. These depressions form at the head of the Bay of Bengal and cross over to the mainland. The depressions follow the axis of the "monsoon



Retreating Monsoon

trough of low pressure". The monsoon is known for its uncertainties. The alternation of dry and wet spells vary in intensity, frequency and duration. While it causes heavy floods in one part, it may be responsible for droughts in the other. It is often irregular in its arrival and its retreat. Hence, it sometimes disturbs the farming schedule of millions of farmers all over the country.

Retreating/Post Monsoons (The Transition Season)

During October-November, with the apparent movement of the sun towards the south, the monsoon trough or the low-pressure trough over the northern plains becomes weaker. This is gradually replaced by a high-pressure system. The south-west monsoon winds weaken and start withdrawing gradually. By the beginning of October, the monsoon withdraws from the Northern Plains.

The months of October-November form a period of transition from hot rainy season to dry winter conditions. The retreat of the monsoon is marked by clear skies and rise in temperature. While day temperatures are high, nights are cool and pleasant. The land is still moist. Owing to the conditions of high temperature and humidity, the weather becomes rather oppressive during the day. This is commonly known as 'October heat'. In the second half of October, the mercury begins to fall rapidly in northern India.

Do You Know? Mawsynram, the wettest place on the earth is also reputed for its stalagmite and stalactite caves.

The low-pressure conditions, over north-western India, get transferred to the Bay of Bengal by early November. This shift is associated with the occurrence of cyclonic depressions, which originate over the Andaman Sea. These cyclones generally cross the eastern coasts of India cause heavy and widespread rain. These tropical cyclones are often very destructive. The thickly populated deltas of the Godavari, the Krishna and the Kaveri are frequently struck by cyclones, which cause great damage to life and property. Sometimes, these cyclones arrive at the coasts of Odisha, West Bengal and Bangladesh. The bulk of the rainfall of the Coromandel Coast is derived from depressions and cyclones.

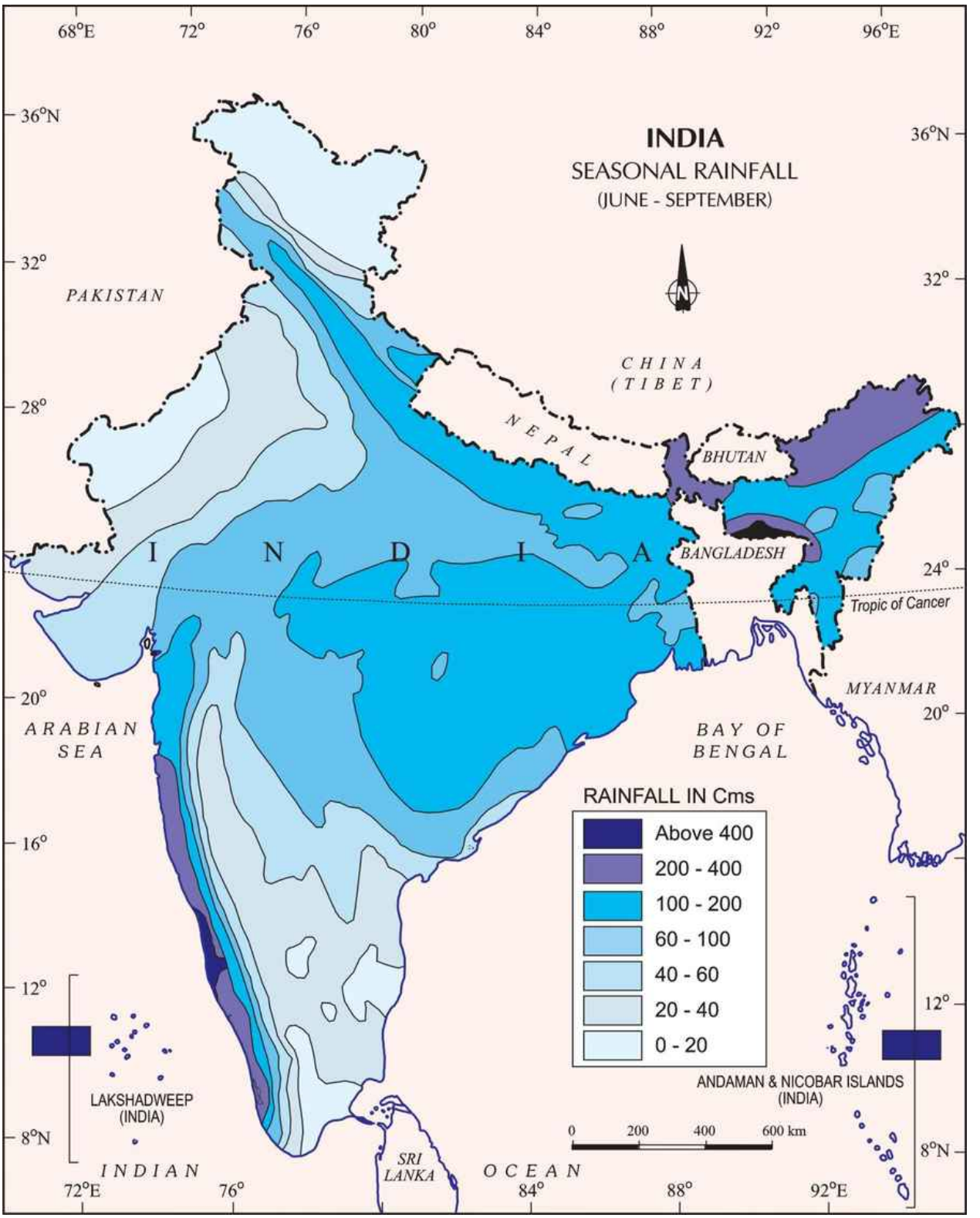
Distribution of Rainfall

Parts of western coast and northeastern India receive over about 400 cm of rainfall annually. However, it is less than 60 cm in western Rajasthan and adjoining parts of Gujarat, Haryana and Punjab. Rainfall is equally low in the interior of the Deccan plateau, and east of the Sahyadris. Why do these regions receive low rainfall? A third area of low precipitation is around Leh in Jammu and Kashmir. The rest of the country receives moderate rainfall. Snowfall is restricted to the Himalayan region.

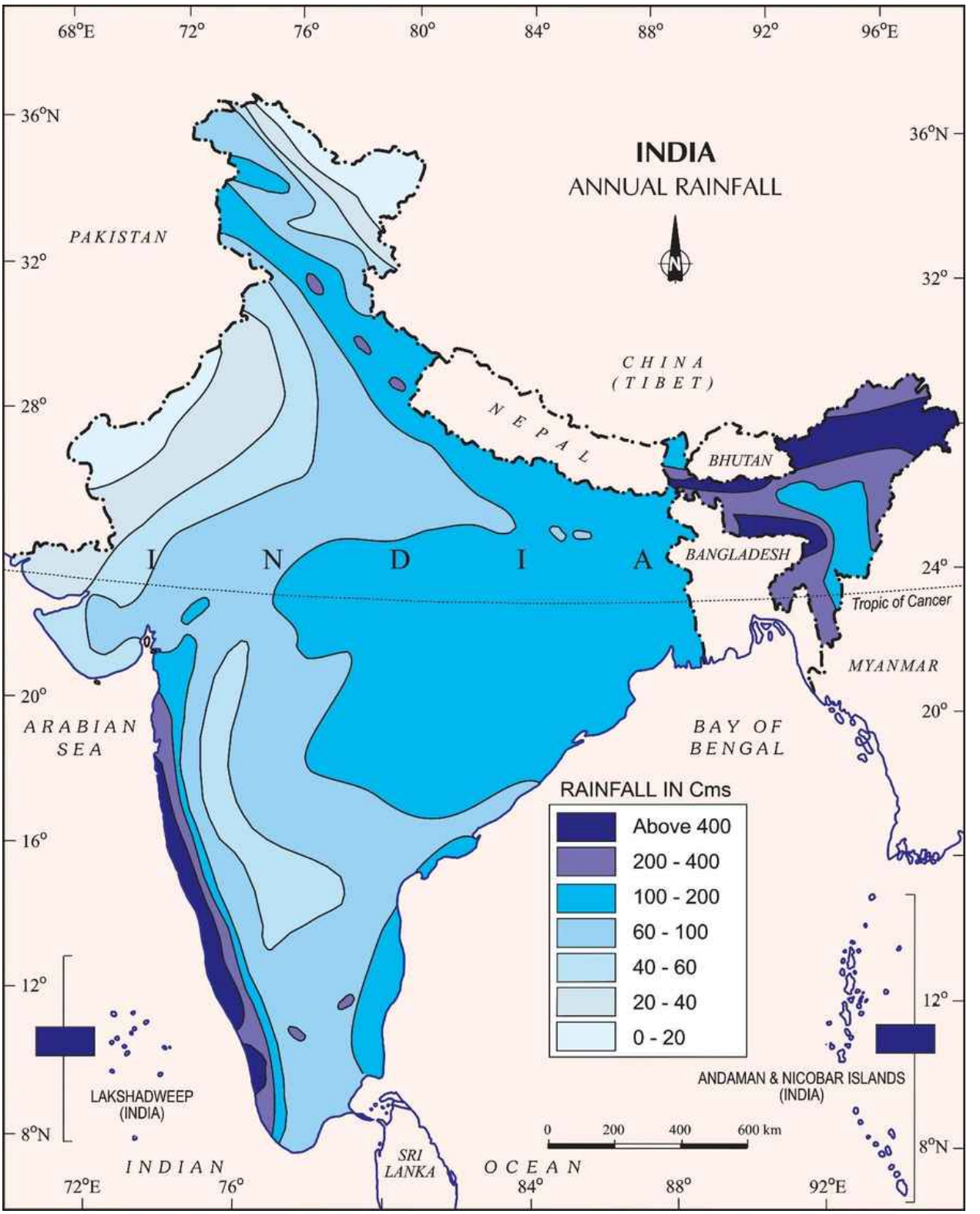
Owing to the nature of monsoons, the annual rainfall is highly variable from year to year. Variability is high in the regions of low rainfall, such as parts of Rajasthan, Gujarat and the leeward side of the Western Ghats. As such, while areas of high rainfall are liable to be affected by floods, areas of low rainfall are drought-prone (Figure 4.6 and 4.7).

Monsoon as a Unifying Bond

You have already known the way the Himalayas protect the subcontinent from extremely cold winds from central Asia. This enables northern India to have uniformly higher temperatures compared to other areas on the same latitudes. Similarly, the Peninsular plateau, under the influence of the sea from three sides, has moderate temperatures. Despite such moderating influences, there are great variations in the temperature conditions. Nevertheless, the unifying influence of the monsoon on the Indian subcontinent is quite perceptible. The seasonal alteration of the wind systems and the associated weather conditions provide a rhythmic cycle of seasons. Even the uncertainties of rain and uneven distribution



Seasonal Rainfall (June-September)


INDIA
ANNUAL RAINFALL
68°E
72°
76°
80°
84°
88°
92°
96°E
36°N
32°
28°
24°
20°
16°
12°
8°N
PAKISTAN
CHINA (TIBET)
NEPAL
BHUTAN
BANGLADESH
MYANMAR
I N D I A
Tropic of Cancer
ARABIAN SEA
BAY OF BENGAL
INDIAN OCEAN
SRI LANKA
LAKSHADWEEP (INDIA)
ANDAMAN & NICOBAR ISLANDS (INDIA)
RAINFALL IN Cms
Above 400
200 - 400
100 - 200
60 - 100
40 - 60
20 - 40
0 - 20
0 200 400 600 km